

2.1. Carbohydrate Metabolism

Introduction to Metabolism

We eat, we digest, we absorb, then what?

Three fates for nutrients

- 1) Most are used to **supply energy** for life
- 2) Some are used to **synthesize** structural or functional molecules
- 3) The rest are **stored** for future use



Biochemical Reactions

- **Metabolism:** total sum of the chemical reaction happening in a living organism (highly coordinated and purposeful activity)
 - a. **Anabolism-** energy requiring biosynthetic pathways
 - where small molecules joined to form larger molecules.
 - E,g. *Gluconeogenesis* and *Glycogenesis*
 - Such reactions are accelerated when energy is readily available, when precursor molecules are in abundance, or during periods of **growth** or **regeneration of cellular material**.
 - b. **Catabolism-** degradation of fuel molecules and the production of energy for cellular function
 - where large molecules broken down to smaller.
 - E.g *Glycolysis*, *Pentose shunt*, *glycogenolysis* and *Pyruvate oxidation*.
 - Such reactions are accelerated during periods of **fuel deprivation** or **stress** to an organism.
- All reactions are catalyzed by **enzymes**

Catabolic pathways

- Energy is extracted from food via oxidation, resulting in the end products **carbon dioxide and water**.
- Serve to capture chemical energy in the form of **ATP** from the degradation of energy-rich fuel molecules.
- Energy generation by degradation of complex molecules.
- **NB** Catabolic pathways are typically oxidative, and require coenzymes such as **NAD⁺** .

Anabolic pathways

- Combine small molecules, such as amino acids, *to form complex molecules*, such as proteins.
- Anabolic reactions **require energy** (are endergonic), which is generally provided by the breakdown of ATP.
- Often *involve chemical reductions* in which the reducing power is most frequently provided by the electron donor NADPH.

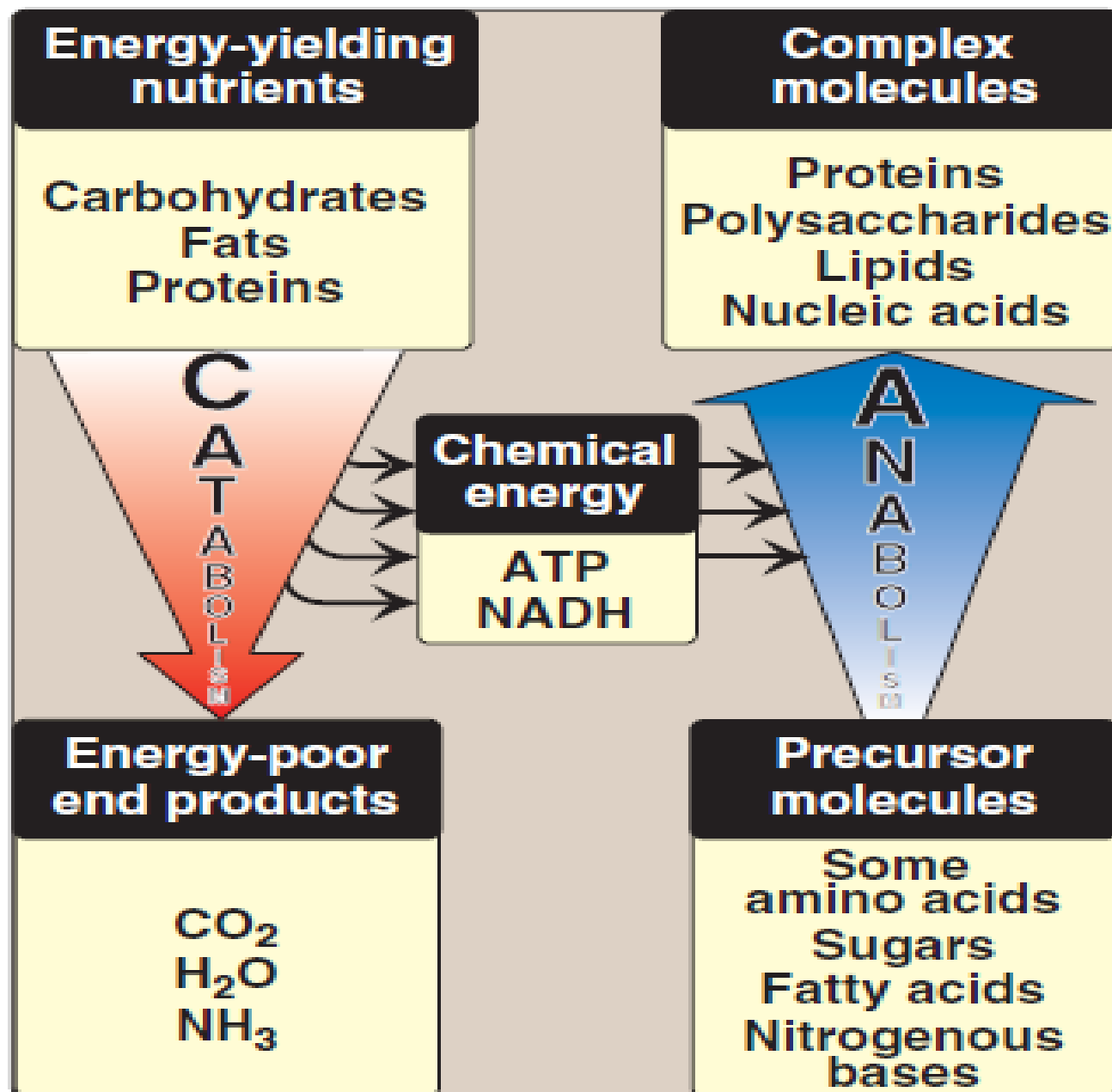


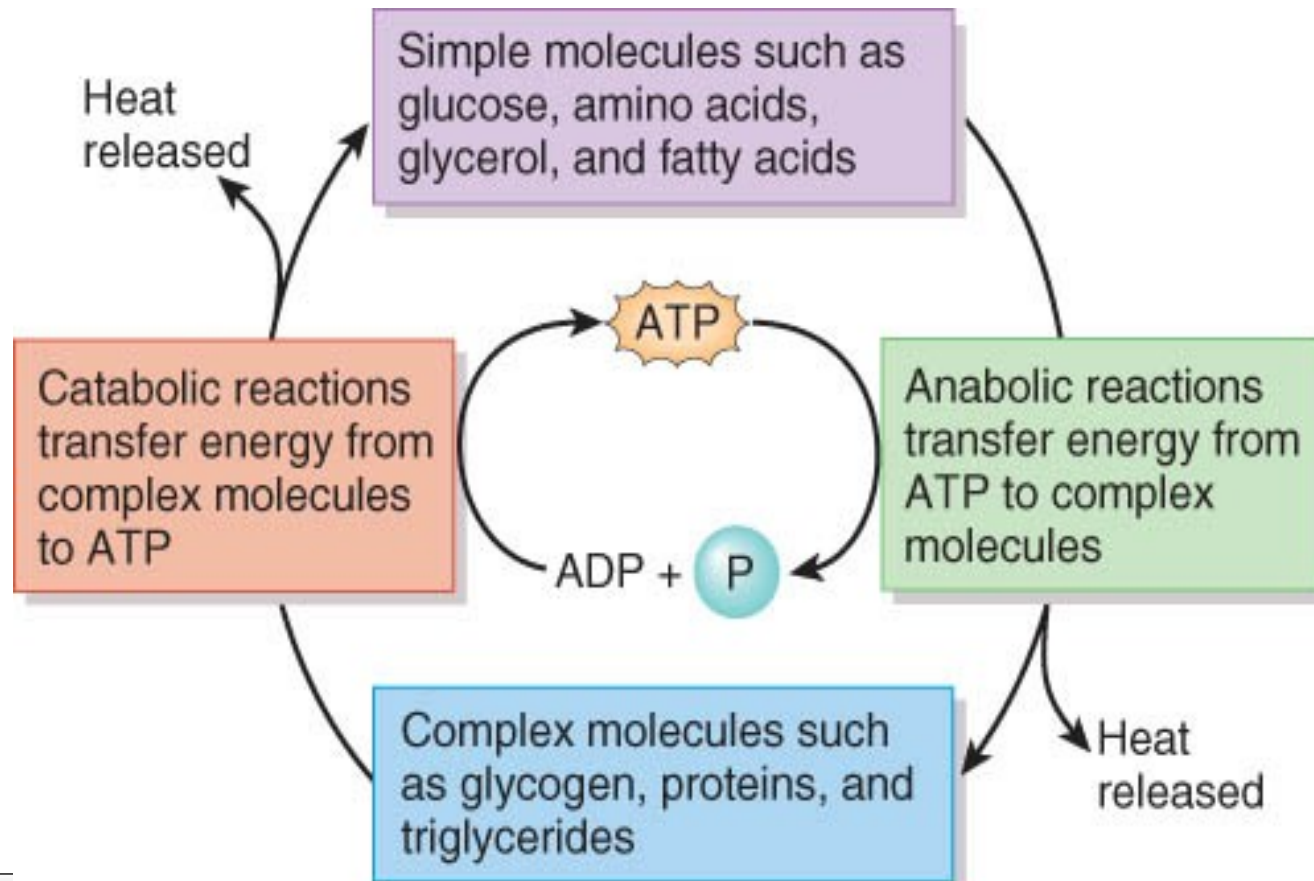
Fig. Comparison of catabolic and anabolic pathways.

The primary functions of metabolism are:

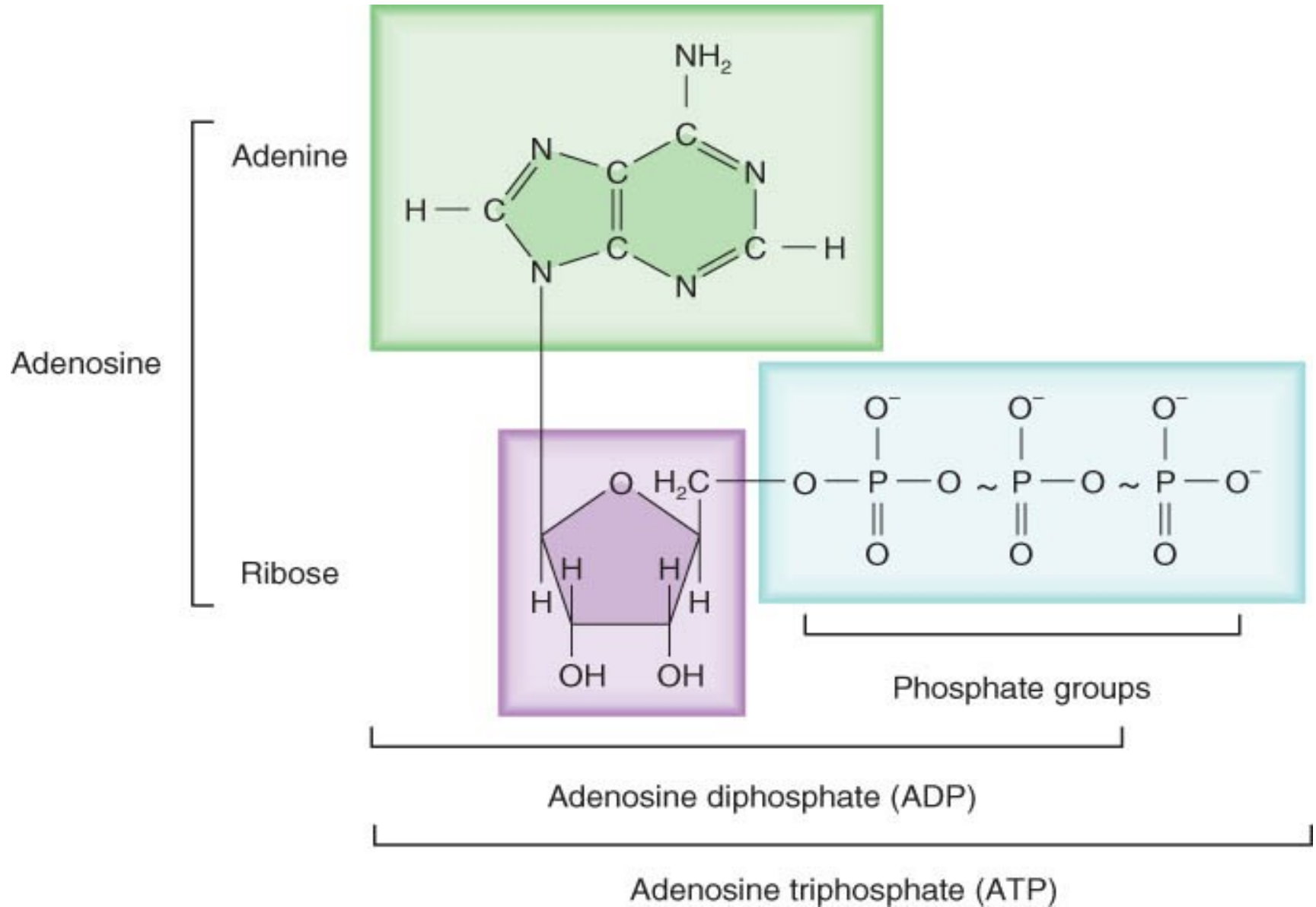
- a. Acquisition & utilization of **energy**
 - b. **Synthesis** of molecules needed for cell structure and functioning (i.e. proteins, nucleic acids, lipids, & CHO)
 - c. Removal of **waste** products
-
- Energy in carbohydrates, lipids and proteins is used to produce ATP through **oxidation-reduction reactions**.

METABOLIC REACTIONS

- The chemical reactions of living systems depend on transfer of manageable amounts of **energy** from one molecule to another. This transfer is usually performed by **ATP**.

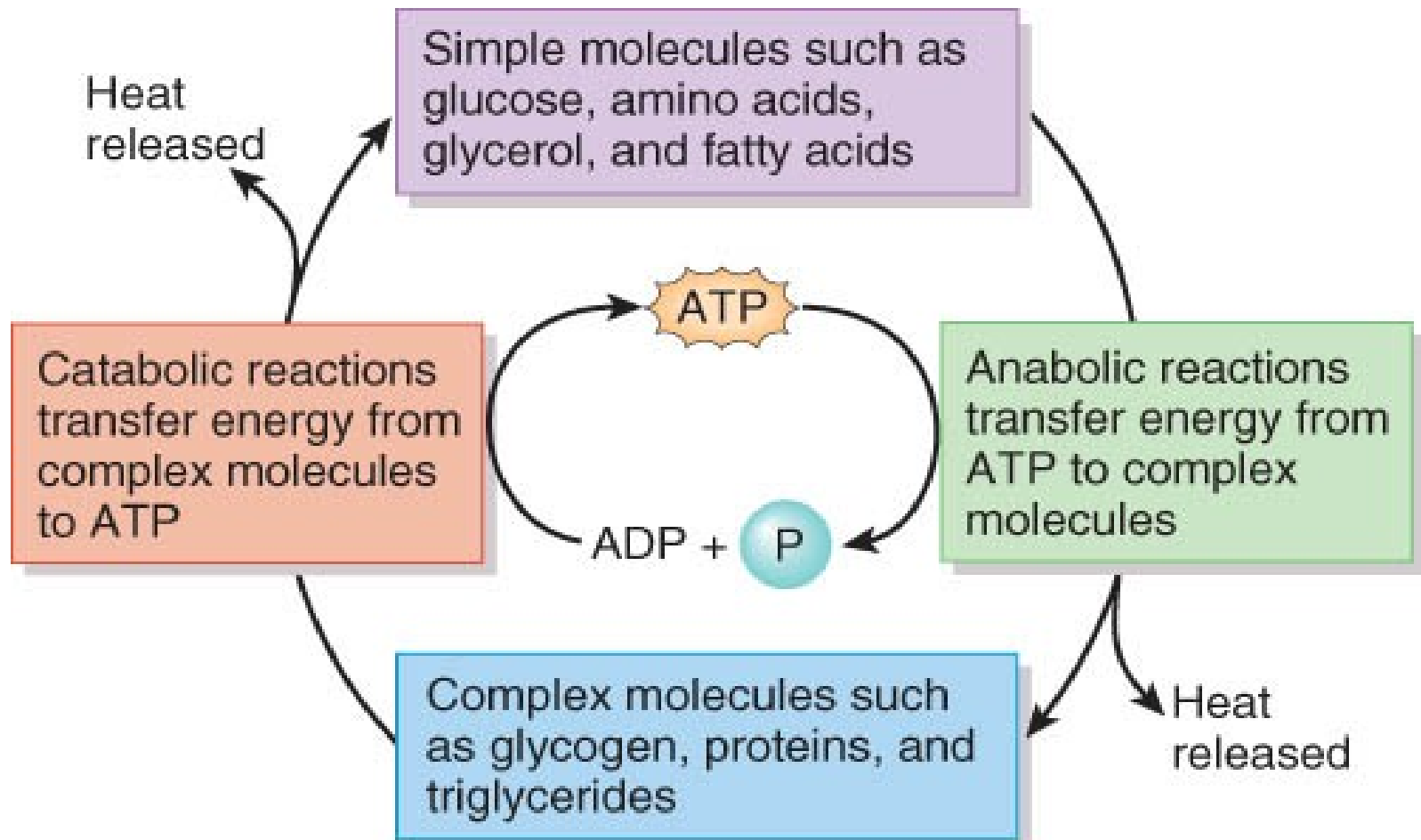


ATP



ENERGY TRANSFER

- All molecules (nutrient molecules included) have **energy stored** in the bonds between their atoms.



Oxidation-Reduction Reactions

- **Oxidation** is the **removal of electrons** from a molecule and results in a **decrease in the energy content** of the molecule. Because most biological oxidations involve the **loss of hydrogen** atoms, they are called **dehydrogenation reactions**.
 - When a substance is **oxidized**, the liberated hydrogen atoms **do not remain free in the cell** but are transferred immediately by **coenzymes** to another compound.
- **Reduction** is the opposite of oxidation.
 - i.e. the **addition of electrons** to a molecule and results in an **increase in the energy** content of the molecule.
- Oxidation and Reduction reactions are always coupled.
 - Redox reaction.

Coenzymes

- Two coenzymes are commonly used by living cells to **carry hydrogen atoms**:
 - *nicotinamide adenine dinucleotide (NAD)* and
 - Made from vitamin B (**niacin**)
 - *flavin adenine dinucleotide (FAD)*.
 - Made from Vitamin B₂ (**riboflavin**)
- An important point to remember about oxidation-reduction reactions is that **oxidation** is usually an **energy-releasing reaction**.

Mechanisms of ATP Generation

- **Phosphorylation** is **ADP + P = ATP**
 - bond attaching 3rd phosphate group contains stored energy.
- Mechanisms of phosphorylation:
 - **within animals**
 - **substrate-level phosphorylation** in cytosol/mitochondria
 - **oxidative phosphorylation** in mitochondria
 - in chlorophyll-containing plants or bacteria
 - **photophosphorylation**.

Substrate-level phosphorylation Vs. Oxidative phosphorylation

- **Phosphorylation** is the metabolic reaction of introducing a phosphate group into an organic molecule.
- **Oxidative phosphorylation:**
 - The formation of high-energy phosphate bonds by phosphorylation of ADP to ATP coupled to the transfer of electrons from reduced coenzymes to molecular **oxygen** via the electron transport chain (ETC).
 - it occurs in the **mitochondria**.
- **Substrate-level phosphorylation:**
 - The formation of high-energy phosphate bonds by phosphorylation of ADP to ATP (or GDP to GTP) coupled to cleavage of a high-energy metabolic **intermediate (substrate)**.
 - It may occur in **cytosol or mitochondria**.

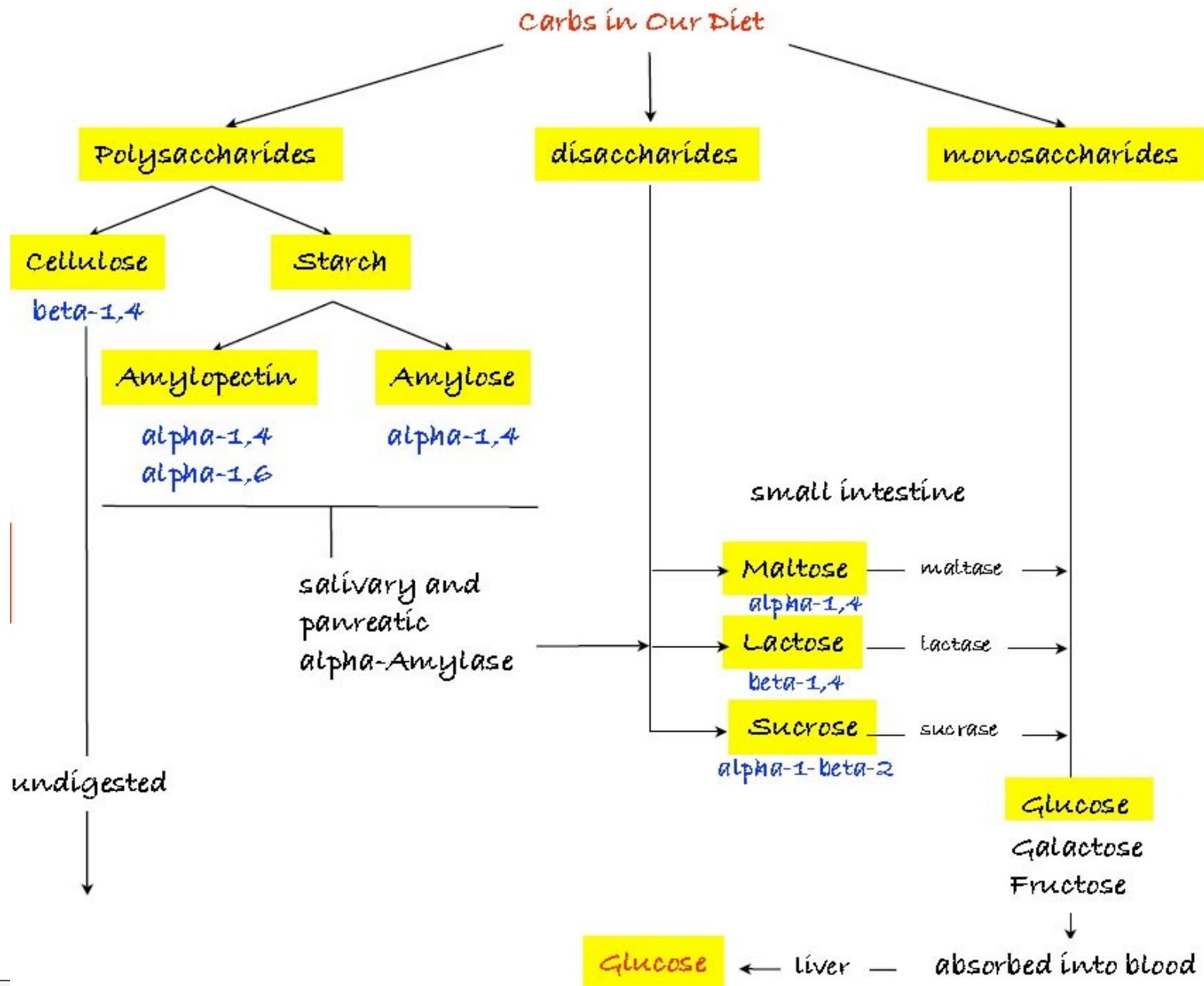
Preferred Fuels in the Well-Fed and Fasting States

Organ	Well-Fed	Fasting
Liver	Glucose and amino acids	Fatty acids
Skeletal muscle	Glucose	Fatty acids, ketones
Cardiac muscle	Glucose	Fatty acids, ketones
Adipose tissue	Glucose	Fatty acids
Brain	Glucose	Glucose (ketones in prolonged fast)
Red blood cells	Glucose	Glucose

Carbohydrate Digestion and Absorption

- **Mouth**
 - Salivary amylase begins digestion of starch into maltose
- **Small intestine**
 - Pancreatic amylase completes starch digestion
 - Brush border enzymes digest the disaccharides- maltase, sucrase, lactase
- **End products of carbohydrate digestion**
 - Glucose, fructose, galactose
 - Absorbed into bloodstream
- **Fiber not digested, excreted in feces**

Carbohydrate Digestion and Absorption



Carbohydrate Metabolism

General

80% of carbohydrates ingested contain glucose; remainder: fructose, galactose,
glucose is the body's preferred carbohydrate energy source

Metabolism of carbohydrates

Glycolysis

Citric acid cycle

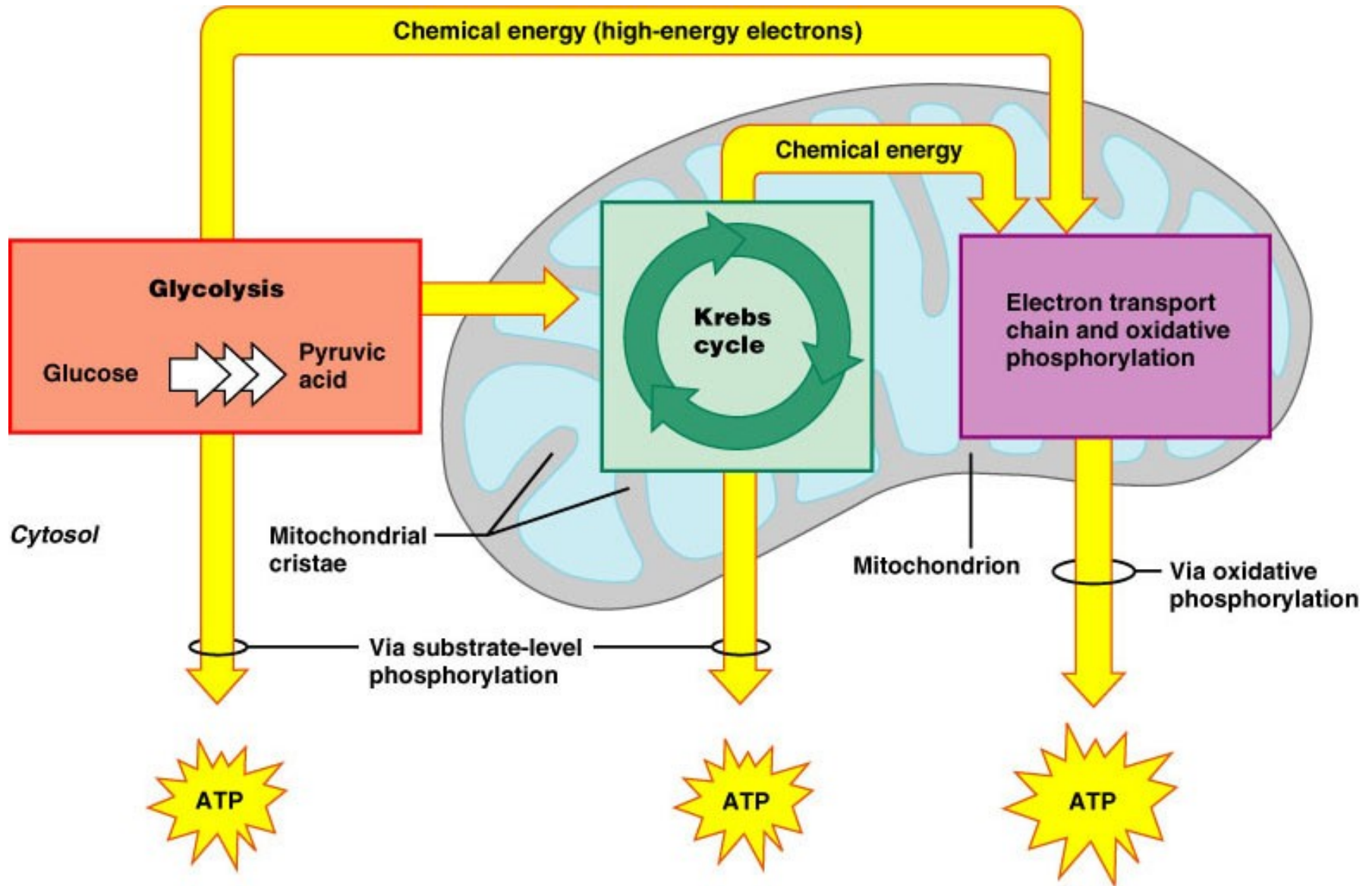
Pentose Phosphate Pathway

Glycogen metabolism

Gluconeogenesis

Control of blood glucose level

Carbohydrate Catabolism



Uptake of glucose

- The duodenum and upper jejunum absorb the bulk of the dietary sugars.
- **Insulin is not** required for the uptake of glucose by intestinal cells.
- However, different sugars have different mechanisms of absorption.
- E.g. galactose and glucose are transported into the mucosal cells by an active, energy-requiring process that requires a concurrent uptake of sodium ions; the transport protein is the **sodium-dependent glucose cotransporter 1 (SGLT-1)**.

- Fructose uptake requires a **sodium-independent monosaccharide transporter (GLUT-5)** for its absorption.
- All three monosaccharides are transported from the intestinal mucosal cell into the portal circulation by yet another transporter, **GLUT-2**.

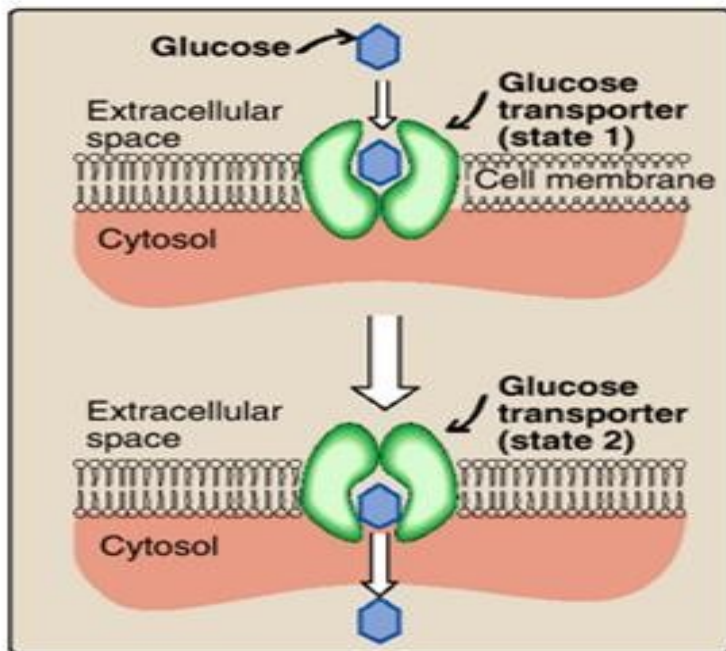
Transport of Glucose Into Cells

- Glucose can**not** diffuse directly into cells, but enters by one of two transport mechanisms:

	Active transport	Facilitated transport
Insulin-sensitive		Most tissues (for example, skeletal muscle and adipose tissue)
Insulin-insensitive	Epithelia of intestine Renal tubules Choroid plexus	Erythrocytes Leukocytes Lens of eye Cornea Liver Brain

A. Na^+ -independent facilitated diffusion transport

- mediated by a family of at least 14 glucose transporters in cell membranes.
- Designated as GLUT-1 to GLUT-14 (glucose transporter isoforms 1–14).
- These transporters exist in the membrane in two conformational states .



Extracellular glucose binds to the transporter, which then alters its conformation, transporting glucose across the cell membrane.

Fig. facilitated transport of glucose through a cell membrane.

B. Na⁺-monosaccharide cotransporter system

- This is an energy-requiring process that transports glucose “against” a concentration gradient—i.e. from low glucose concentrations outside the cell to higher concentrations within the cell.
- It is a carrier-mediated process in which the movement of glucose is coupled to the concentration gradient of Na⁺, which is transported into the cell at the same time.
- The carrier is a sodium-dependent–glucose transporter or SGLT.
- It occurs in the **epithelial cells of the intestine, renal tubules, and choroid plexus.**

Regulation of Cellular Glucose Uptake

■ Brain & RBC:

- GLUT-1 has high affinity (low K_m) for glucose and are always saturated.
- Insures that brain and RBC always have glucose.

■ Liver:

- GLUT-2 has low affinity (hi K_m) and high capacity.
- Uses glucose when fed at rate proportional to glucose concentration

■ Muscle & Adipose:

- GLUT-4 is sensitive to insulin

Major Glucose Transporters in Human Cells

Name	Tissues	K_m , glucose	Functions
GLUT 1	Most tissues (brain, red cells)	~ 1 mM	Basal uptake of glucose
GLUT 2	Liver Pancreatic β -cells	~ 15 mM	Uptake and release of glucose by the liver β -cell glucose sensor
GLUT 3	Most tissues	~ 1 mM	Basal uptake
GLUT 4	Skeletal muscle Adipose tissue	~ 5 mM	Insulin-stimulated glucose uptake

Normal blood glucose concentration is 4–8 mM.

Glycolysis: An Overview

- Glycolysis, the major pathway for glucose oxidation, occurs in the *cytosol of all cells*.
- It is **unique**, in that it can function either aerobically or anaerobically, depending on the availability of oxygen and intact mitochondria.
- It allows tissues to survive in presence or absence of oxygen, e.g., skeletal muscle.
- RBCs, which lack mitochondria, are completely reliant on glucose as their metabolic fuel, and metabolize it by anaerobic glycolysis.

GLYCOLYSIS

Either **Aerobic** → **Pyruvate**
Anaerobic → **Lactic Acid**

Occurs in the Cytosol

Overall Rxn:

$\text{Glucose} + 2 \text{ ADP} + 2 \text{ NAD}^+ + 2 \text{ Pi} \rightarrow 2 \text{ Pyruvate} + 2 \text{ ATP} + 2 \text{ NADH} + 2 \text{ H}^+$

NADH generated during glycolysis is used to fuel mitochondrial ATP synthesis via oxidative phosphorylation. Does **not** pass through mitochondrial membrane

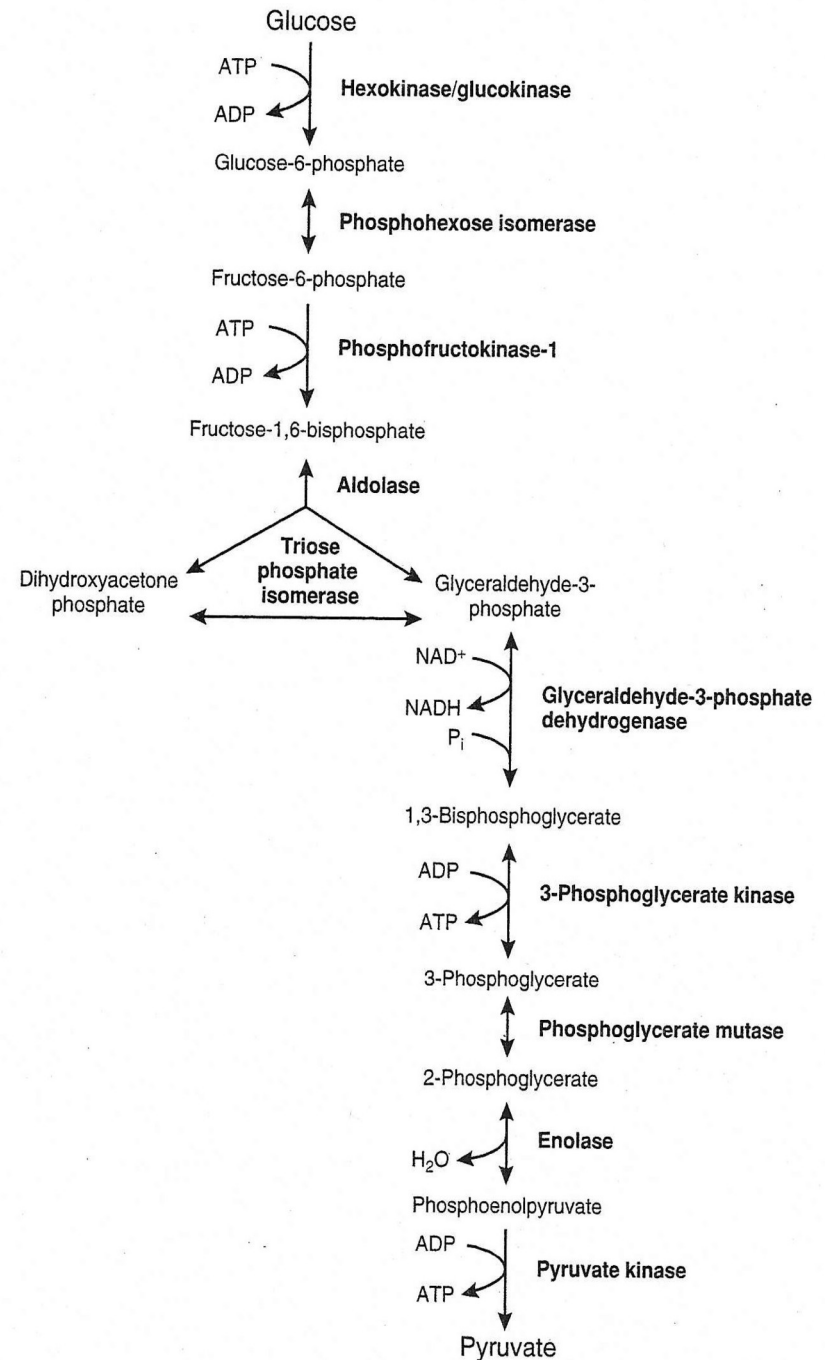
2 ATP generated → glycerol phosphate shuttle
3 ATP generated → malate-aspartate shuttle

If used to transport the electrons from cytoplasm NADH into the mitochondria.

Glycolysis:

Embden-Myerhof Pathway

- Oxidation of glucose
- Products:
 - 2 Pyruvate
 - 2 ATP
 - 2 NADH
- Cytosolic



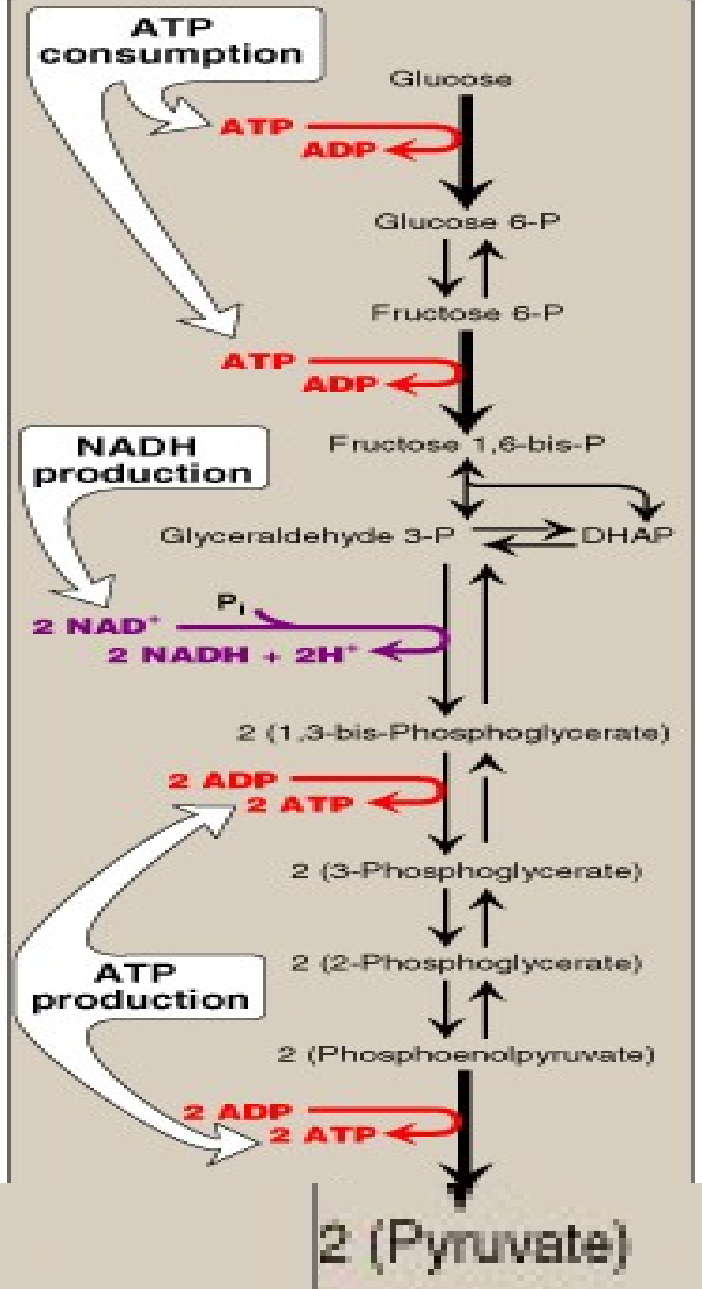
What are the different types of Glycolysis?

Anaerobic Glycolysis:

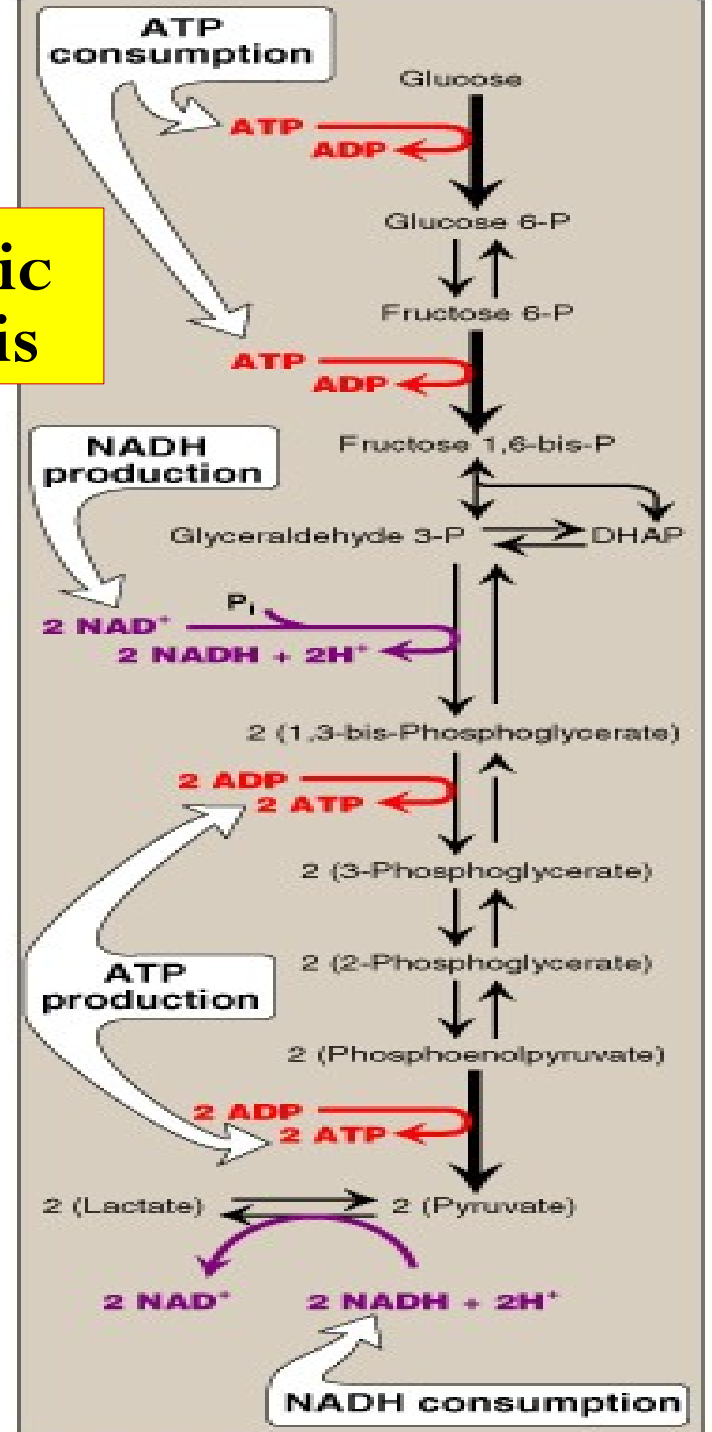
- Occurs in the absence of Oxygen,
- Produces 2 molecules of Lactate and a **Net of 2 ATP** per molecule of Glucose
- *Lactate* is the end product of Anaerobic Glycolysis

Aerobic Glycolysis:

- Occurs in the presence of Oxygen,
- Produces 2 molecules of Pyruvate and Net 2 ATP, 2 NADH per molecule of Glucose
- *Pyruvate* is end product of Aerobic Glycolysis



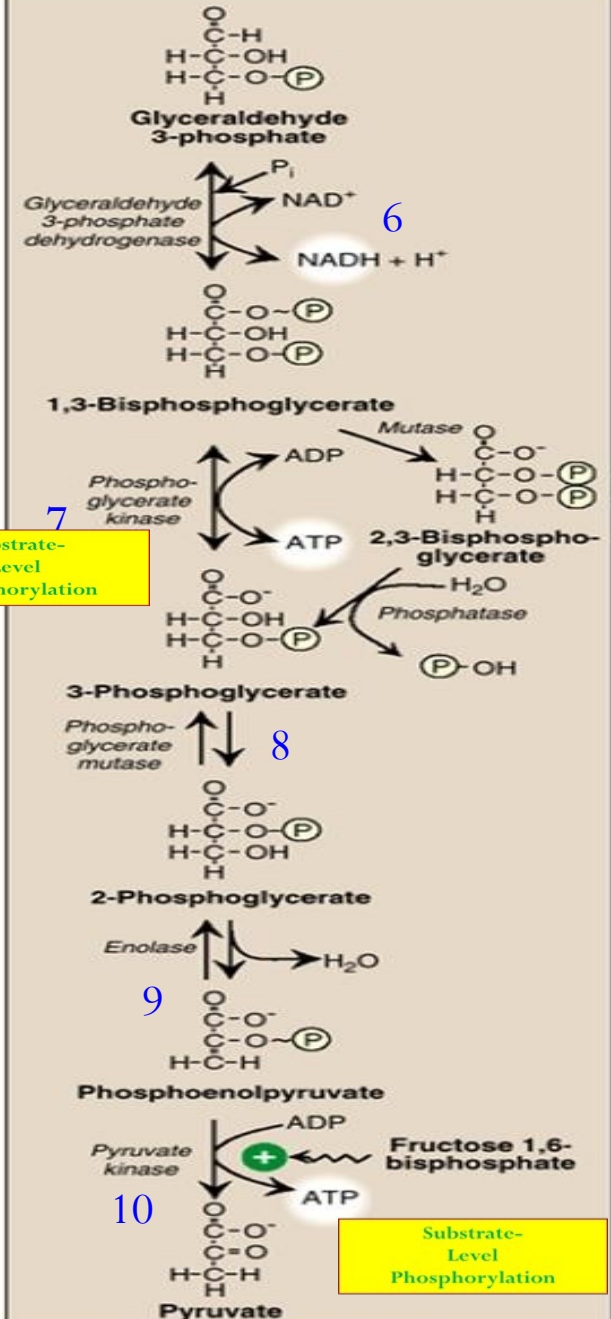
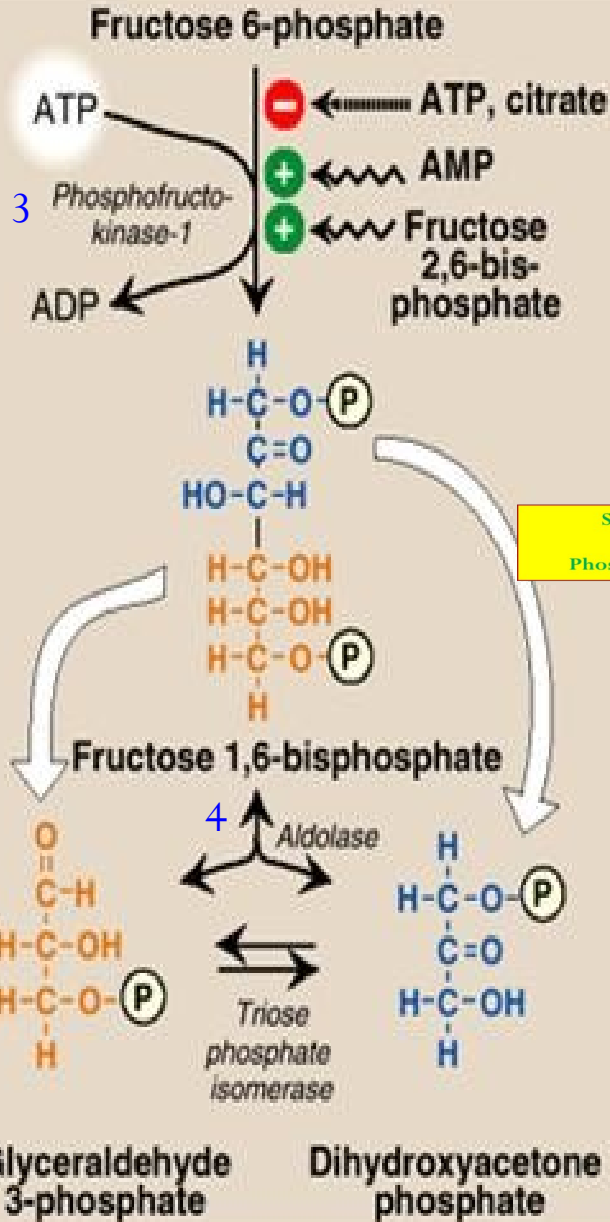
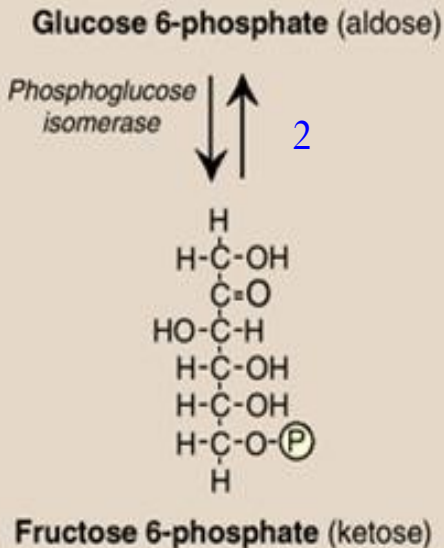
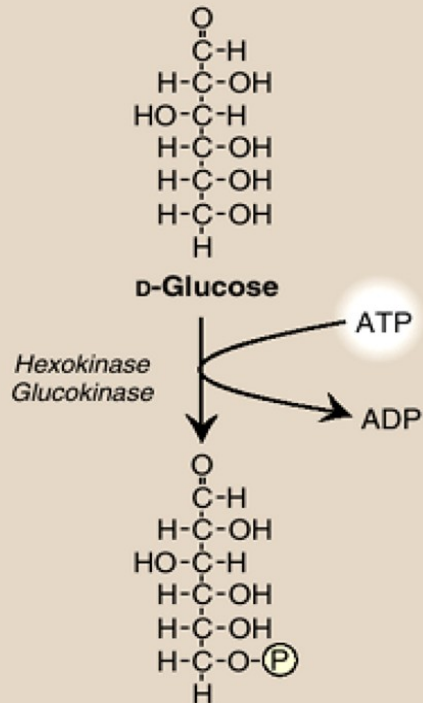
Anaerobic Glycolysis



Aerobic Glycolysis

Energy investment phase

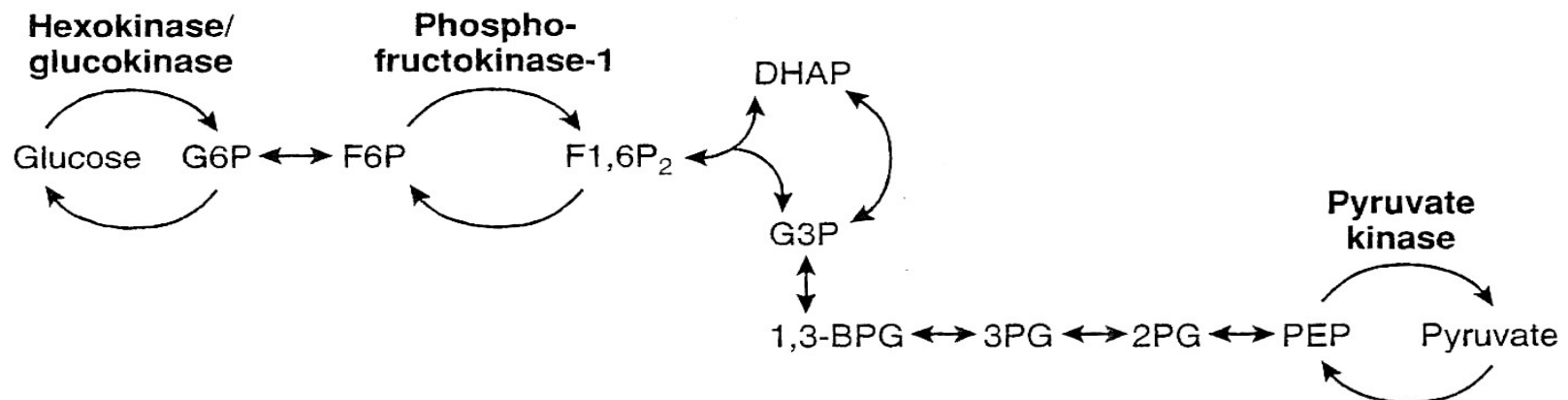
Energy generation phase



Glycolysis

Hexokinase

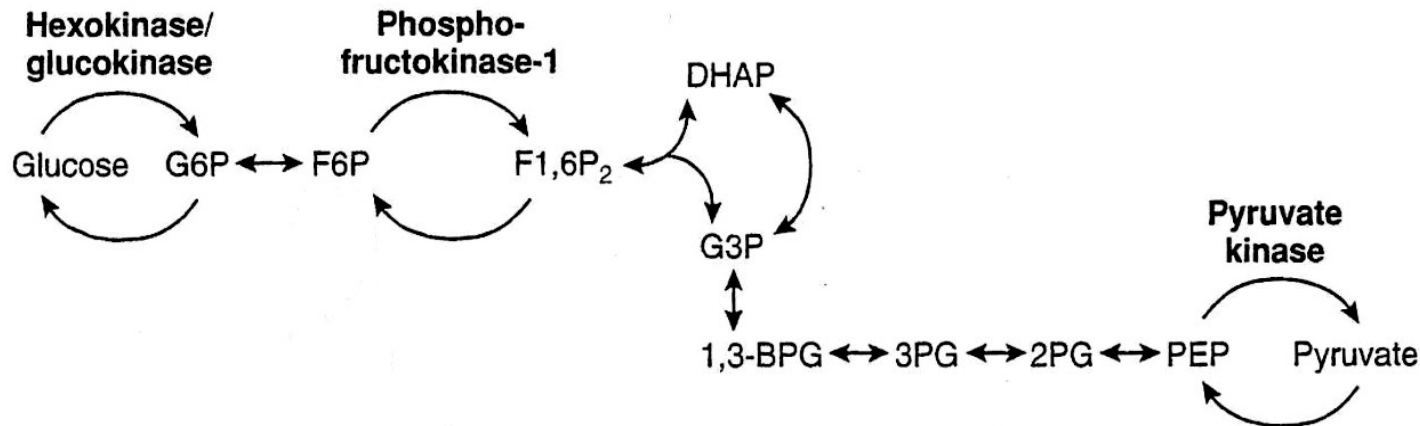
- ♦ **Inhibited** by product **glucose-6-phosphate**:
 - by competition at the active site
 - by allosteric interaction at a separate enzyme site.
- ♦ Cells trap glucose by phosphorylating it, preventing exit through glucose carriers.



Glucokinase

--a variant of hexokinase found in liver.

- ♦ Glucokinase has a higher K_M for glucose.
- ♦ It is active only at high [glucose].
- ♦ Glucokinase is not subject to product inhibition by glucose-6-phosphate.
- ♦ Liver will take up & phosphorylate glucose even when liver [glucose-6-phosphate] is high.



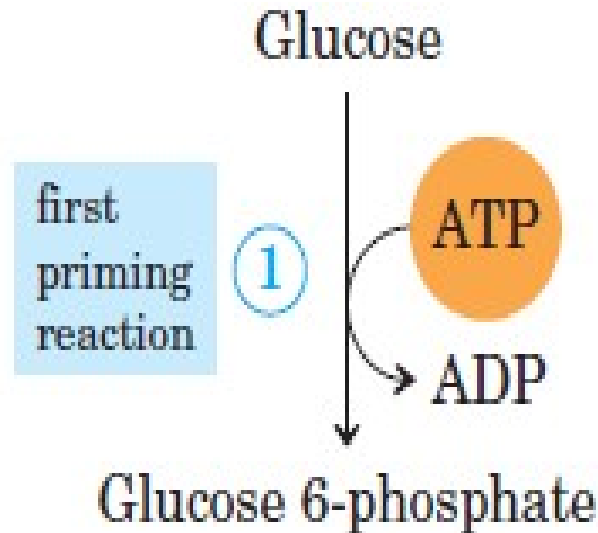
Comparison of Glucokinase and Hexokinase

	<i>Glucokinase</i>	<i>Hexokinase</i>
Kinetic parameters		
K_m	High (10 mM)	Low (<100 μ M)
V_{max}	High	Low
Tissue distribution	Liver Pancreatic beta cells	Most tissues
Regulation		
Short-term	Activity responds to changes in glucose concentration	Inhibited by glucose-6-phosphate
Long-term	Synthesis induced by insulin	Constitutive

Three irreversible reactions:

1- Glucose is **phosphorylated**.

- ✓ phosphorylated at the expense of an ATP. The investment of an ATP here is called “priming.”
- ✓ Enzymes = **hexokinase/ glucokinase**



Reaction:

- ◆ first energy investment
- ◆ highly exergonic, $\Delta G^{\circ'} = -16.7 \text{ kJ/mole}$, (essentially irreversible)

Hexokinase

- ◆ found in all cells of every organism
- ◆ one of three regulatory enzymes of glycolysis (see also phosphofructokinase and pyruvate kinase).
- ◆ relatively **high affinity** for glucose, $K_M = 0.1 \text{ mM}$
- ◆ **low specificity** for monosaccharides (simple sugars) i.e., other monosaccharides can be phosphorylated by hexokinase.
- ◆ **inhibited** by its product, glucose 6-phosphate

3. Phosphorylation of fructose 6-phosphate

- **Second phosphorylation**

- **Enzyme = phosphofructokinase-1 (PFK-1)**

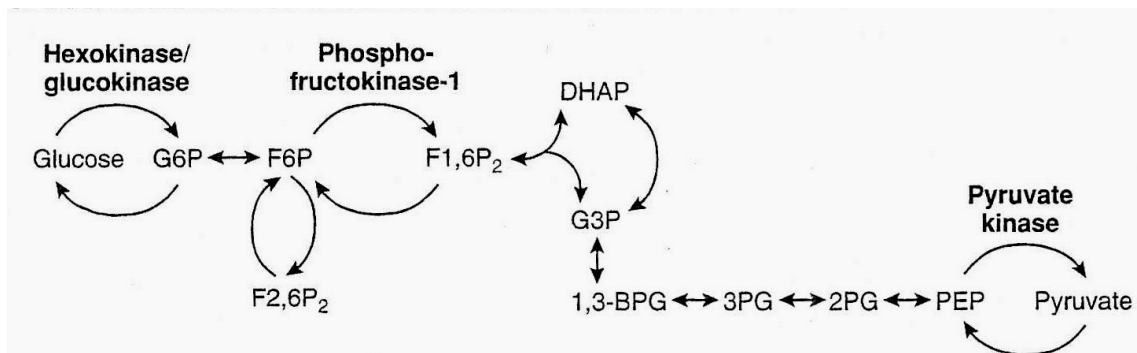
- second ATP investment

- highly **exergonic**, essentially irreversible,
 $\Delta G^{\circ'} = -14.2 \text{ kJ/mole}$

- ✓ is the most important control point and the **rate-limiting** and committed step of glycolysis.

Phosphofructokinase-1 (PFK-1)

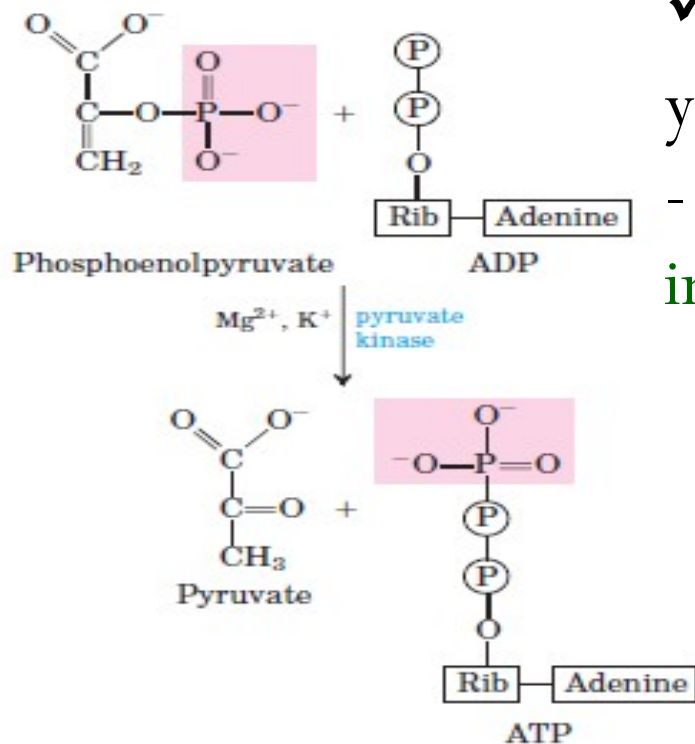
- The rate-limiting step of the Glycolysis pathway
- Inducible enzyme
 - **Induced** in feeding by **insulin**
 - **Repressed** in starvation by **glucagon**
- Allosteric regulation
 - Activated by AMP
 - Inhibited by ATP and Citrate
 - Activated by Fructose-2,6-bisphosphate



10. Formation of pyruvate producing ATP

➤ Final generation of ATP

- Enzyme = **pyruvate kinase**



✓ second substrate level phosphorylation yielding ATP

- highly exergonic reaction, **irreversible**, $\Delta G^{\circ'} = -31.4 \text{ kJ/mole}$.

✓ the third irreversible reaction of glycolysis.

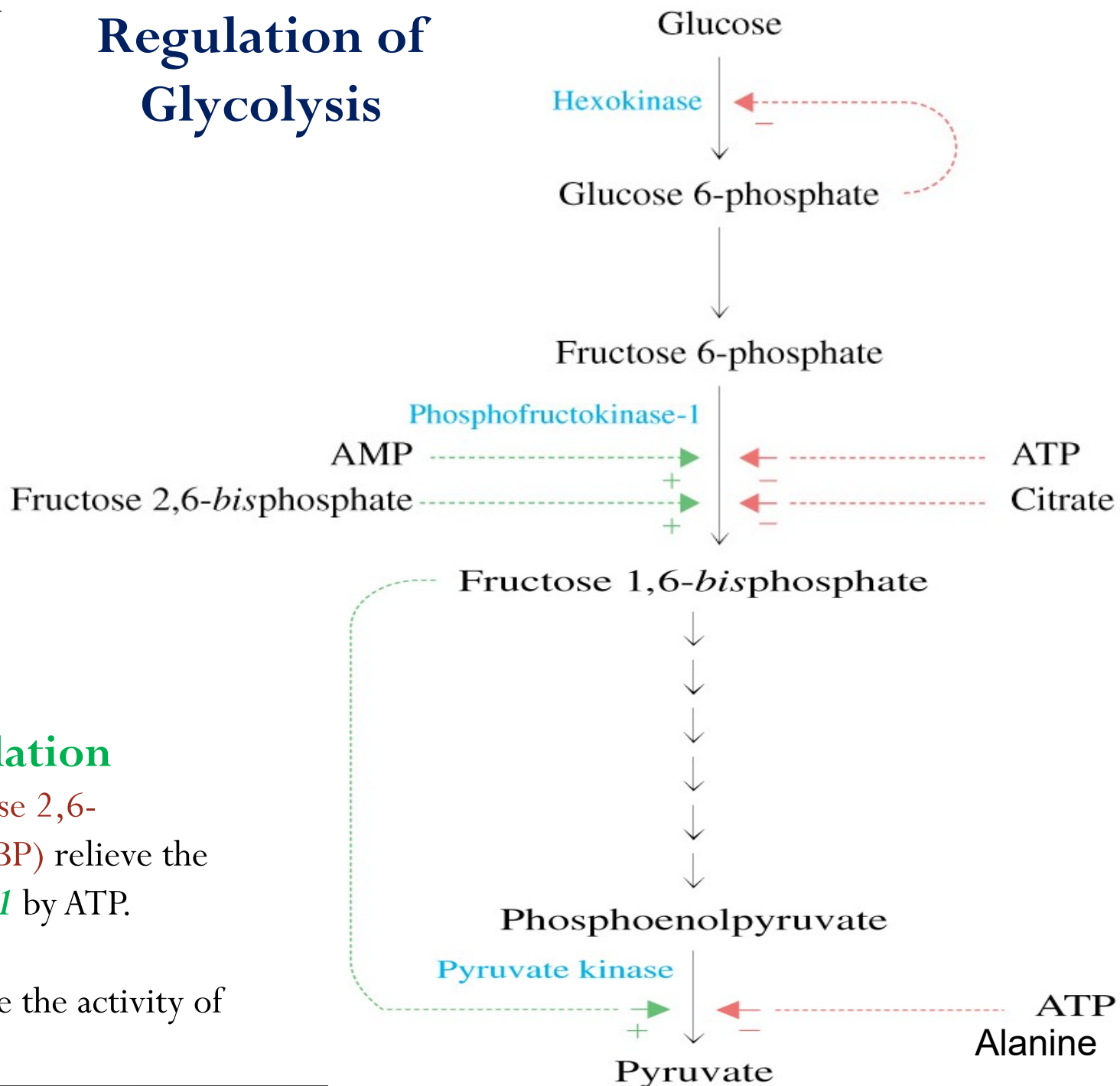
Regulation of Glycolysis

Inhibition

- 1) **PFK-1** is inhibited by **ATP** and **citrate**.
- 2) **Pyruvate kinase** is inhibited by **ATP** and **alanine**.
- 3) **Hexokinase** is inhibited by excess **glucose 6-phosphate**.

Stimulation

- 1) **AMP** and **fructose 2,6-bisphosphate (F2,6BP)** relieve the inhibition of **PFK-1** by **ATP**.
- 2) **F1,6BP** stimulate the activity of **pyruvate kinase**.



Long-Term Regulation of Glycolysis

Insulin: Induction

Glucagon: Repression

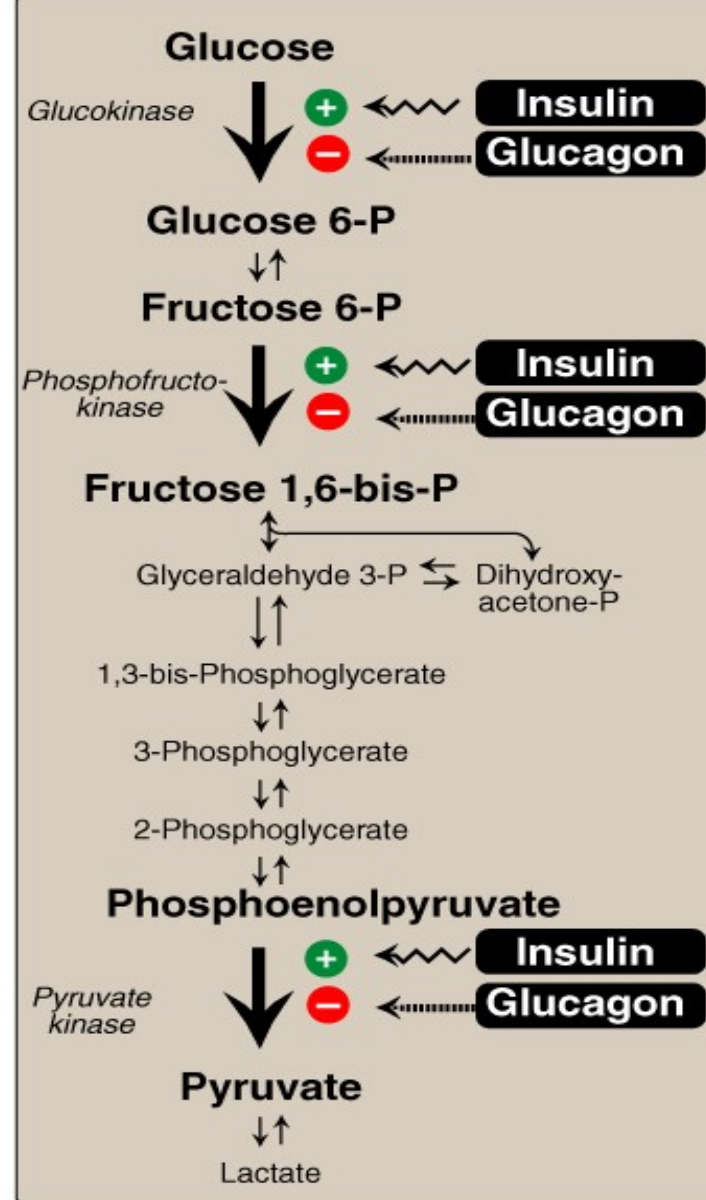


Fig. Effect of insulin & glucagon on the synthesis of key enzymes of glycolysis in liver.

Glycolysis

Key points about the Shuttle System:

- **Malate-Asparate shuttle** is the primary system
- By default **Glycerol shuttle** is secondary

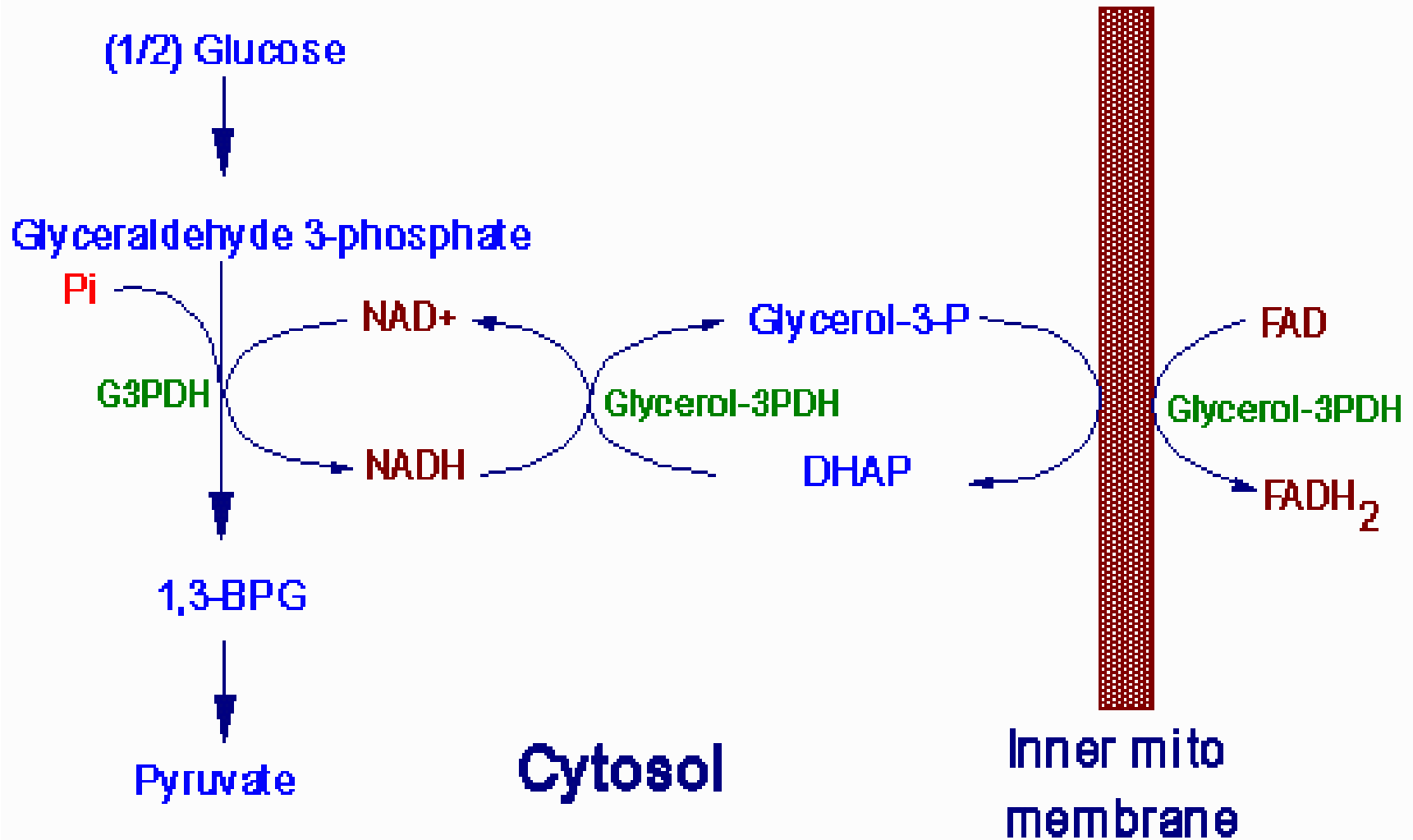
Two enzymes are involved in this shuttle:

1. **cytosolic version** of the enzyme *glycerol-3-phosphate dehydrogenase (glycerol-3-PDH)* which has as one substrate, NADH.

2. **mitochondrial form** of the enzyme which has as one of its' substrates, FAD^+ .

- Since the electrons from mitochondrial FADH_2 feed into the oxidative phosphorylation pathway at coenzyme Q (as opposed to *NADH-ubiquinone oxidoreductase* [complex I]) only 2 moles of ATP will be generated from glycolysis.

Glycerol Phosphate Shuttle



DHAP= Dihydroxyacetone phosphate

Malate -Asp Shuttle

The electrons are "carried" into the mitochondria in the form of **malate**.

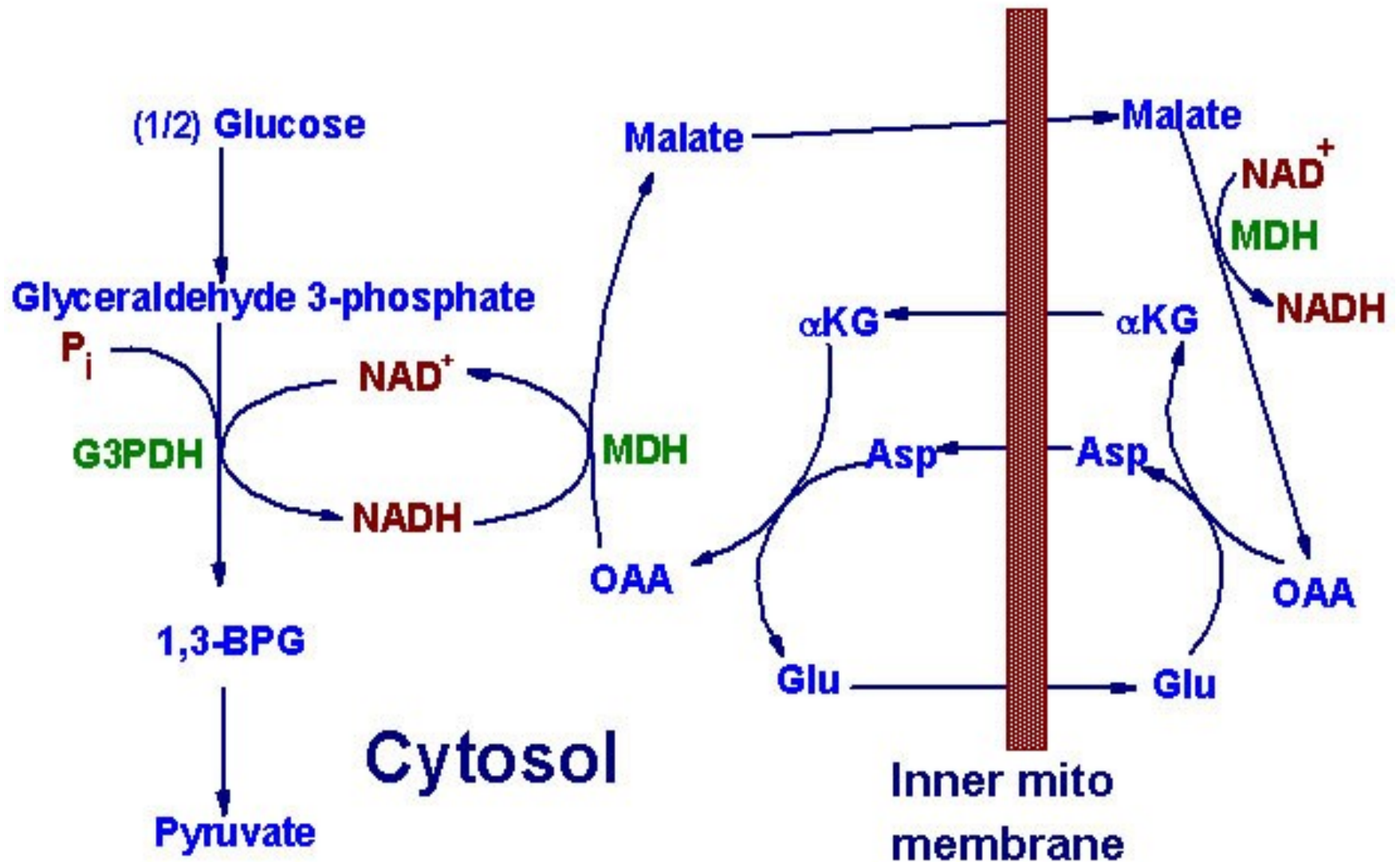
- **Cytoplasmic malate dehydrogenase** (MDH) reduces oxaloacetate (OAA) to malate while oxidizing NADH to NAD⁺.
- **Malate** then enters the mitochondria where the reverse reaction is carried out by **mitochondrial MDH**.

Mitochondrial OAA goes to the cytoplasm to maintain this cycle ;

- must be transaminated to aspartate (Asp) with the amino group being donated by glutamate (Glu).
- The Asp then leaves the mitochondria and enters the cytoplasm.
- The deamination of glutamate generates α -ketoglutarate (α -KG) which leaves the mitochondria for the cytoplasm.

When the **energy level** of the **cell rises**, **the rate of mitochondrial oxidation of NADH to NAD⁺ declines** and therefore, **the shuttle slows**.

Malate-Aspartate Shuttle



Aerobic Glycolysis: ATP Production

ATP Consumed:

2 ATP

ATP Produced:

Substrate-level 2 X 2 = 4 ATP

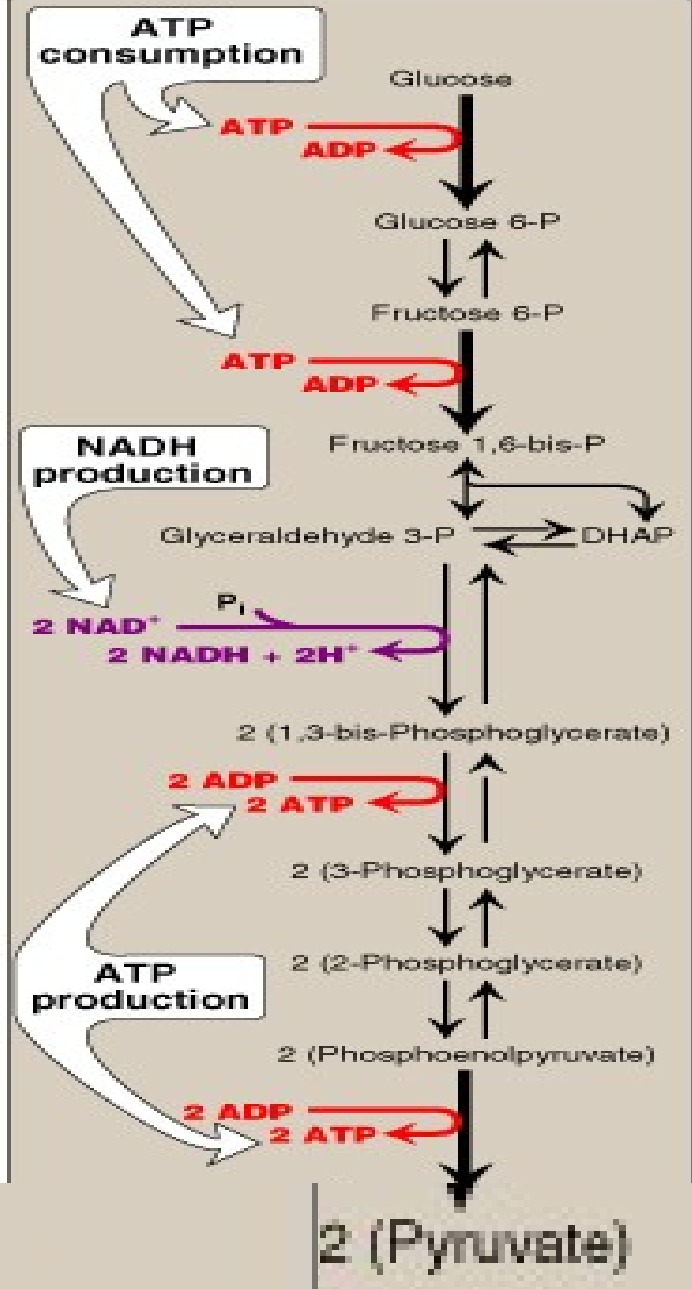
Oxidative-level 2 X 3 = 6 ATP

Total 10 ATP

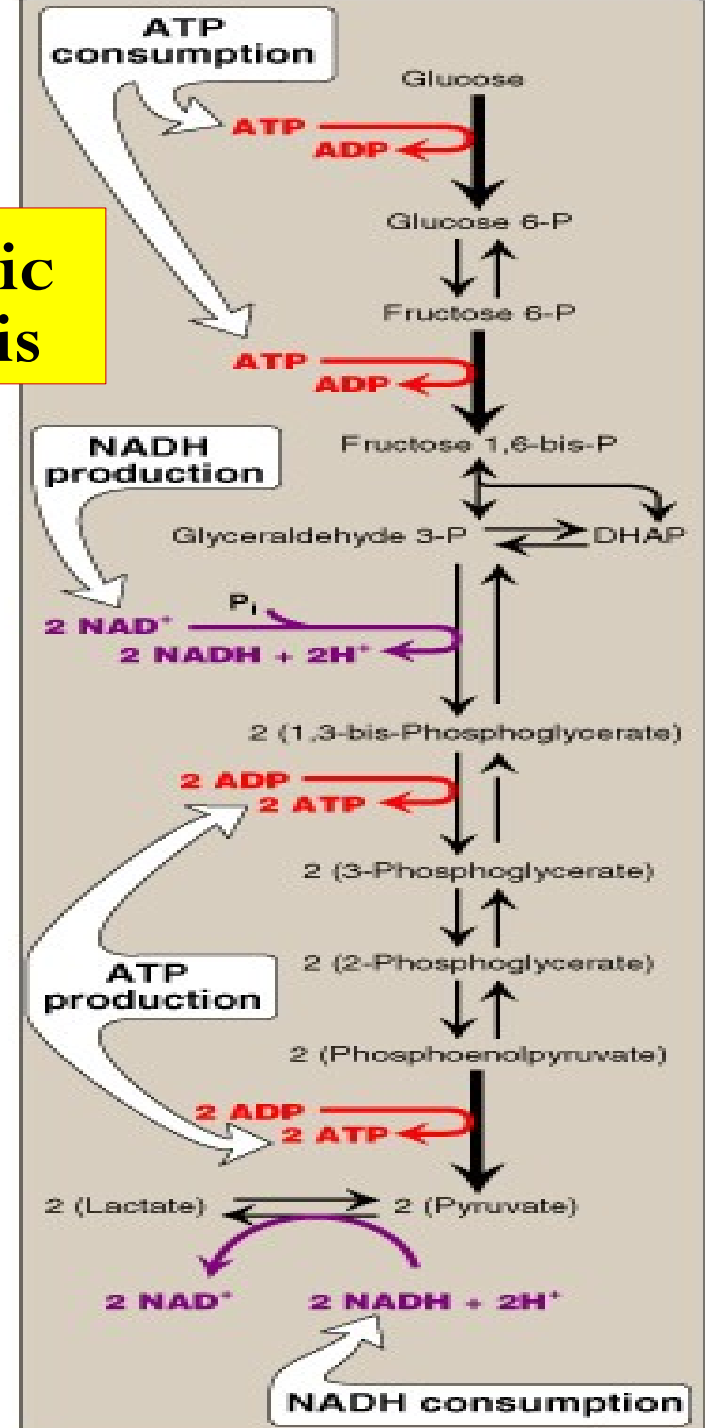
Net: 10 – 2 = 8 ATP

By Malate_Aspartate shuttle

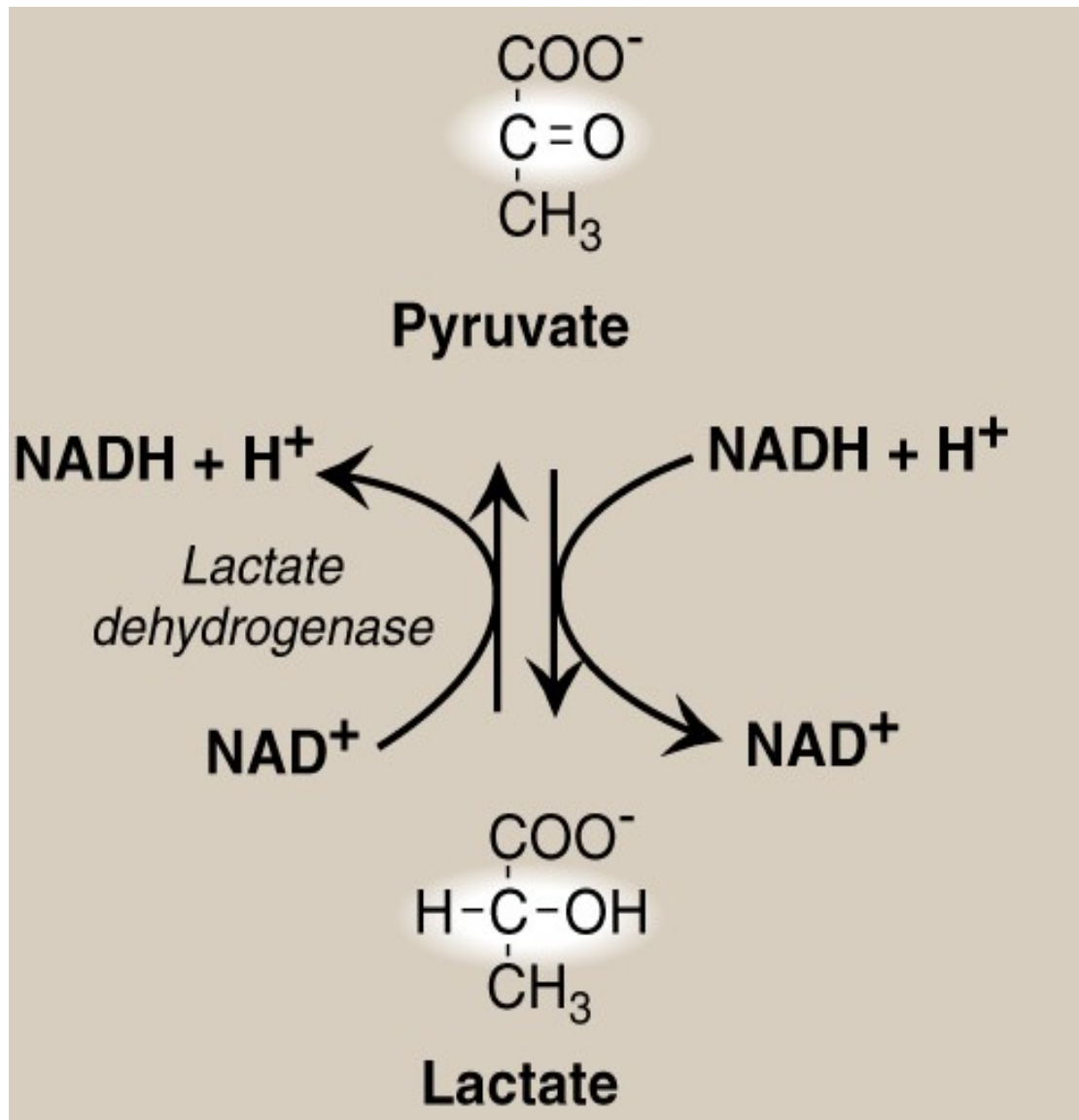
Anaerobic Glycolysis



Aerobic Glycolysis



Lactate Dehydrogenase



Anaerobic Glycolysis: ATP Production

ATP Consumed:

2 ATP

ATP Produced:

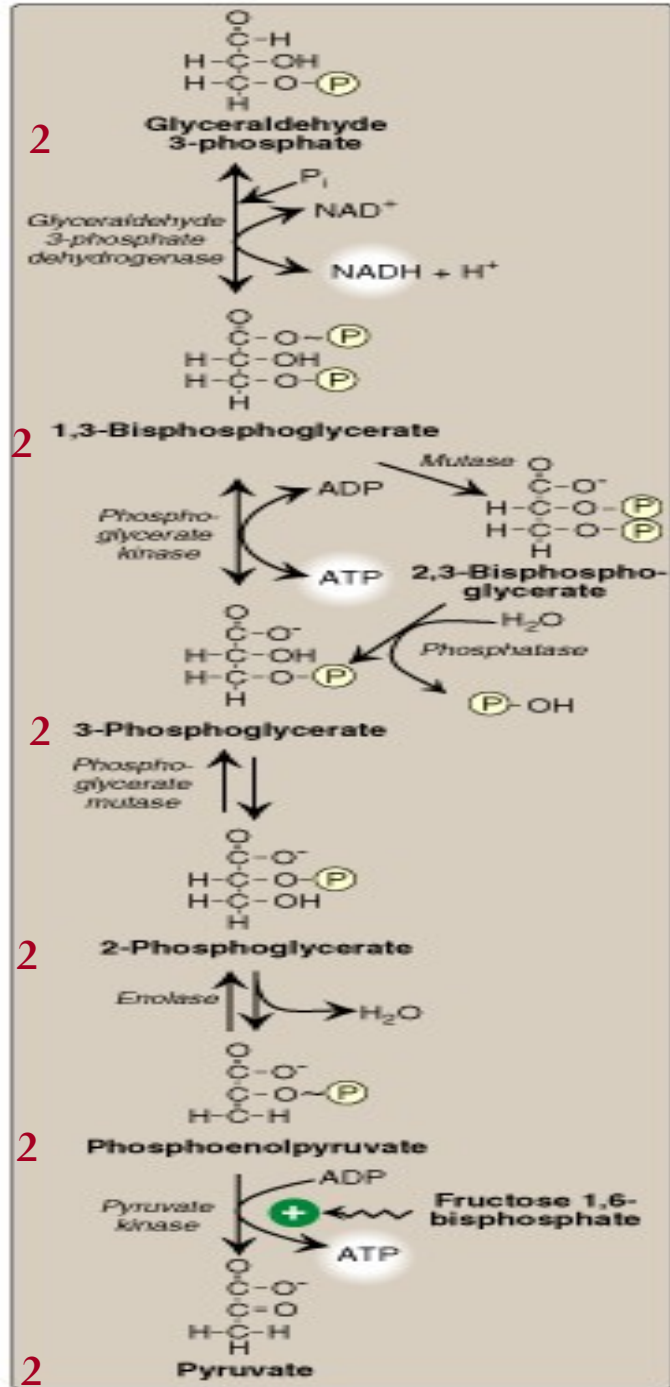
Substrate-level **2 X 2 = 4** **ATP**

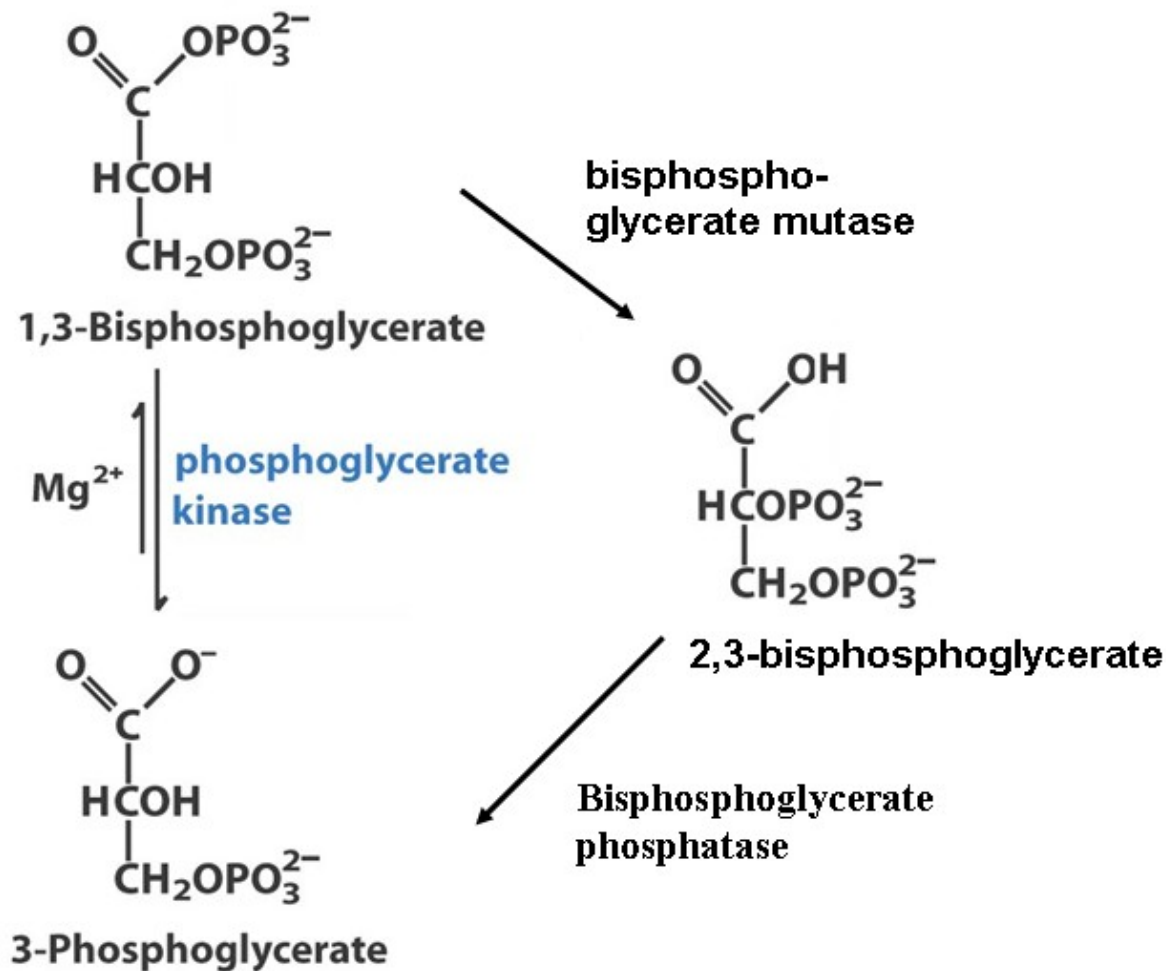
Oxidative-level 2 X 3 = 6 ATP

Total **4 ATP**

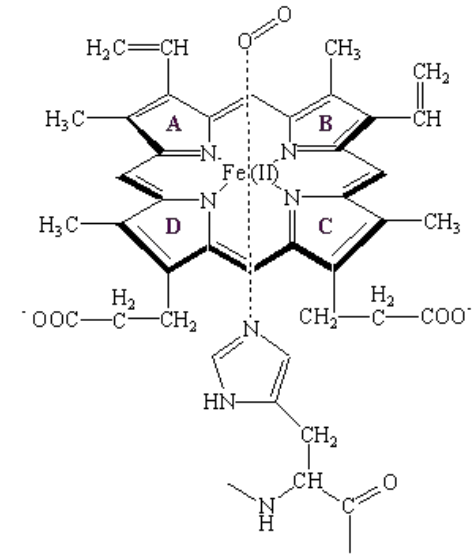
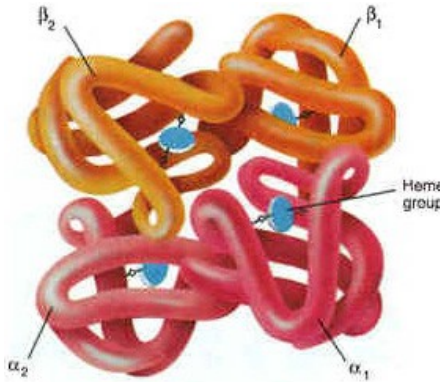
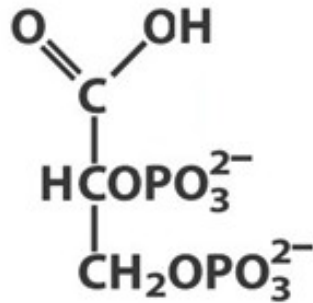
Net: $4 - 2 = 2$ ATP

Anaerobic Glycolysis: RBCs 2,3-BPG Shunt





2,3-bisphosphoglycerate pathway in RBC



- 2,3-Bisphosphoglycerate (2,3-BPG) is present in human red blood cells at approximately 5 mmol/L.
- It binds with greater affinity to deoxygenated hemoglobin (e.g. when the red cell is near respiring tissue) than it does to oxygenated hemoglobin (e.g. in the lungs).
- In bonding to partially deoxygenated hemoglobin it upregulates the release of the remaining oxygen molecules bound to the hemoglobin, thus enhancing the ability of RBCs to release oxygen near tissues that need it most.

Glycolysis in RBCs: ATP Production

ATP Consumed:

2 ATP

ATP Produced:

Substrate-level

OR

2 X 2 =

4

ATP

1 X 2 =

2

ATP

Oxidative-level

2 X 3 =

6

ATP

Total

4

ATP

Net:

OR

4 - 2 =

2

ATP

2 - 2 =

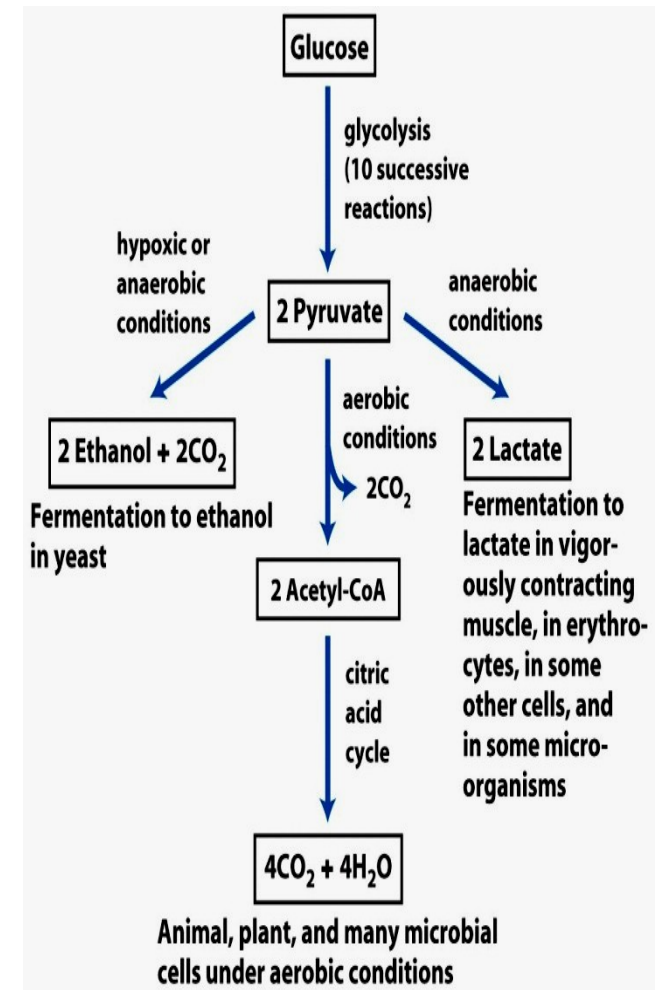
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ATP

Glycolysis is important for erythrocytes

Erythrocyte:

- Simplest cell in body.
- No subcellular organelles.
- No DNA-RNA-protein synthesis.
- No mitochondria—no oxidative phosphorylation
- Relies exclusively on glucose as fuel.
 - Glucose derived from blood.
 - Yields ATP from glycolysis.
 - End product is lactate.



Glycolysis in RBCs: Summary

End product:

Lactate

No net production or consumption of NADH

Energy yield:

No 2,3-BPG	2 ATP
2,3-BPG shunt	0 ATP

PKD hemolytic anemia depends on:

Degree of PKD

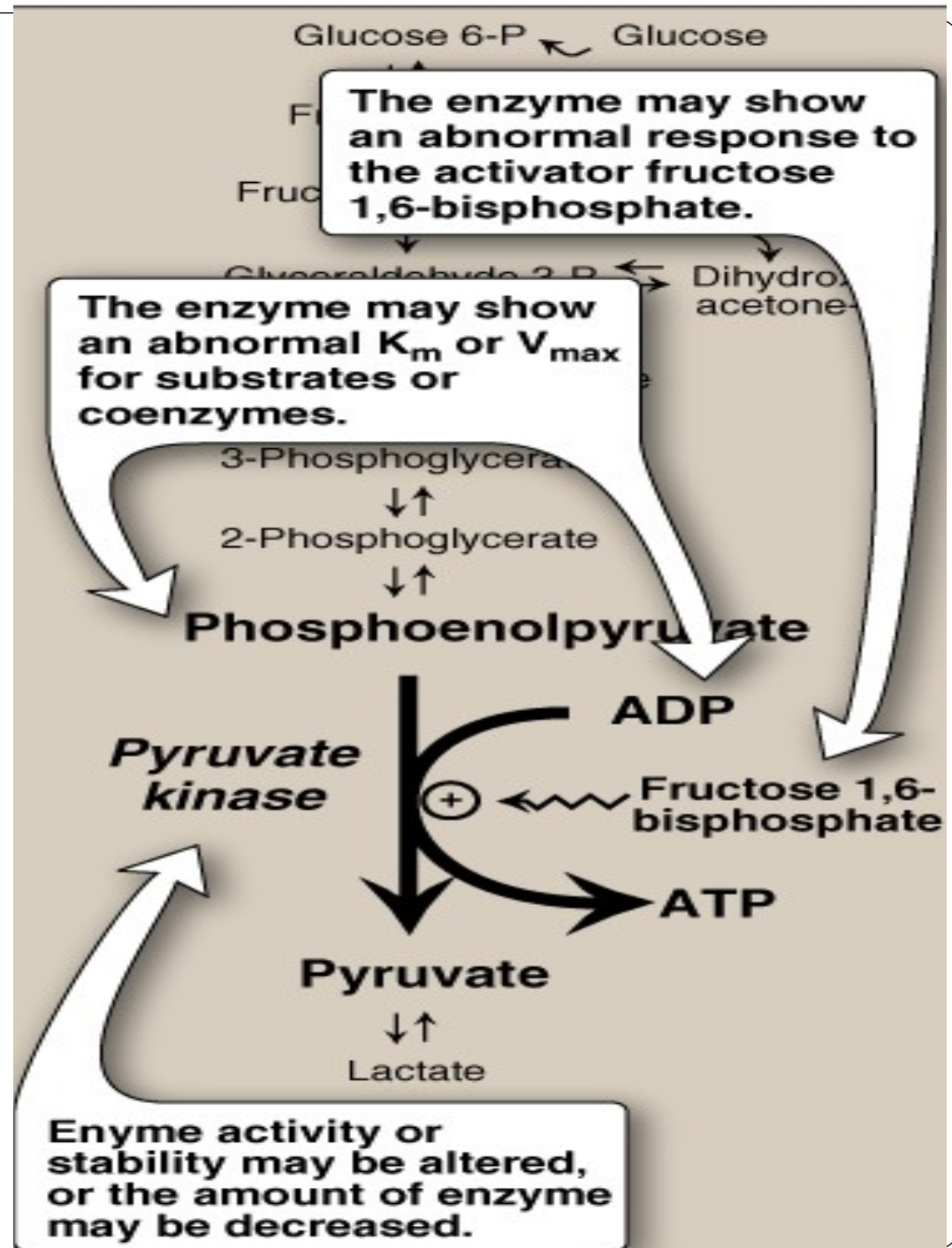
Compensation by 2,3-BPG

Pyruvate Kinase Deficiency

Hemolytic Anemia

PK Mutation

- Altered Enz. Kinetics
- Decreased Enz. Stability
- Altered response to activator

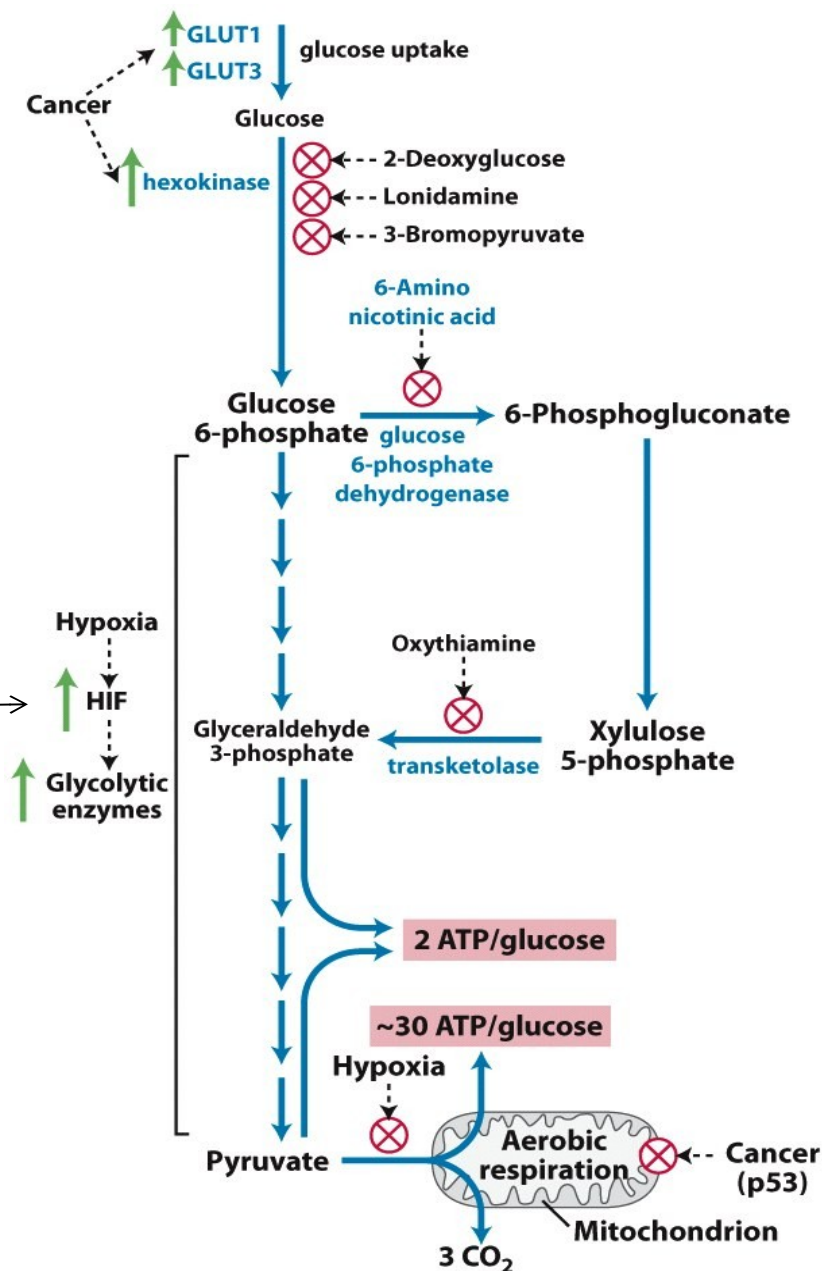


Most tumors: Glycolysis goes 10X faster than normal cells. And is mainly fermentative (producing lactic acid).

The strategy is to:

- 1. Use this in Detection**
- 2. To slow glycolysis down in cancer cells.**

Hypoxia Induced Transcription Factor



Box 14-1 figure 1

Lehninger Principles of Biochemistry, Fifth Edition

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Gluconeogenesis

- Pathways that responsible for converting non-carbohydrate precursors to glucose or glycogen.
- In mammals -occurs in **liver and kidney**

Major substrate :

1. Lactic acid - from muscle, erythrocyte
2. Glycerol - from TG hydrolysis
3. Glucogenic amino acid
4. propionyl-CoA

Biomedical Importance

- The biosynthesis of glucose is an absolute necessity of all mammals, because the brain and nervous system, as well as erythrocytes, testes, renal medulla, and embryonic tissue, require glucose from the **blood as their sole or major fuel source.**
- The human brain alone requires 120 g of glucose each day. Below a critical blood glucose concentration (normal = 65-110 mg/dL or 3.6-6 mM), **brain dysfunction can occur which can lead to coma and death.**
- Even when fat may be supplying most of the caloric requirements of an organism, there is always a certain basal requirement for glucose e.g. in skeletal muscle under anaerobic conditions.
- Glucose is precursor of lactose in the mammary gland.
- Gluconeogenic mechanisms are used to clear lactate (from muscle and erythrocytes) and glycerol (adipose tissue) from blood.

- Gluconeogenesis meets the needs of the body for glucose when carbohydrate is not available from the diet or from glycogenolysis.
- A supply of glucose is necessary especially for nervous system and erythrocytes.
- The enzymes :
 1. Pyruvate carboxylase
 2. Phosphoenolpyruvate carboxikinase
 3. Fructose 1,6-biphosphatase
 4. Glucose-6-phosphatase

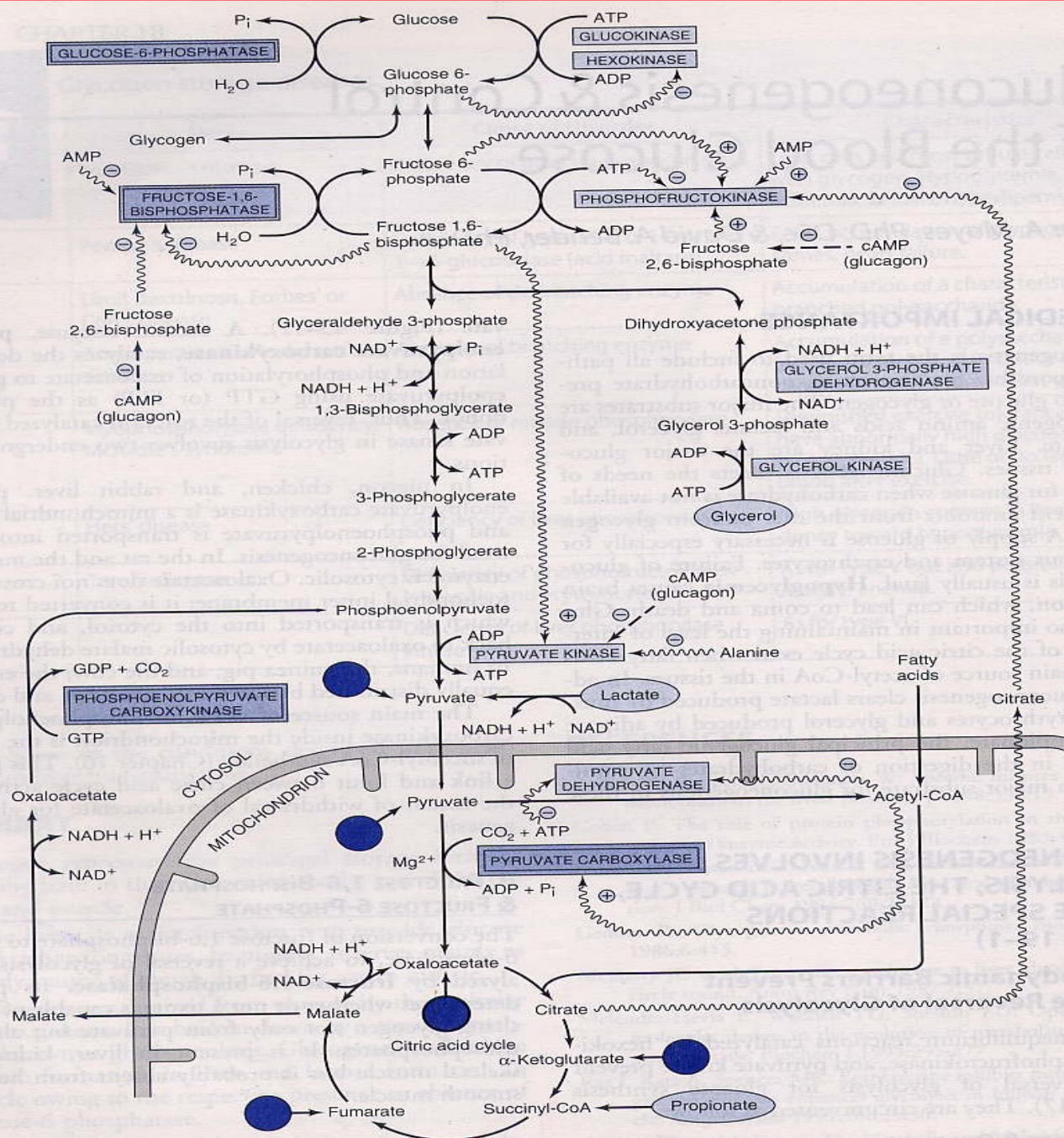
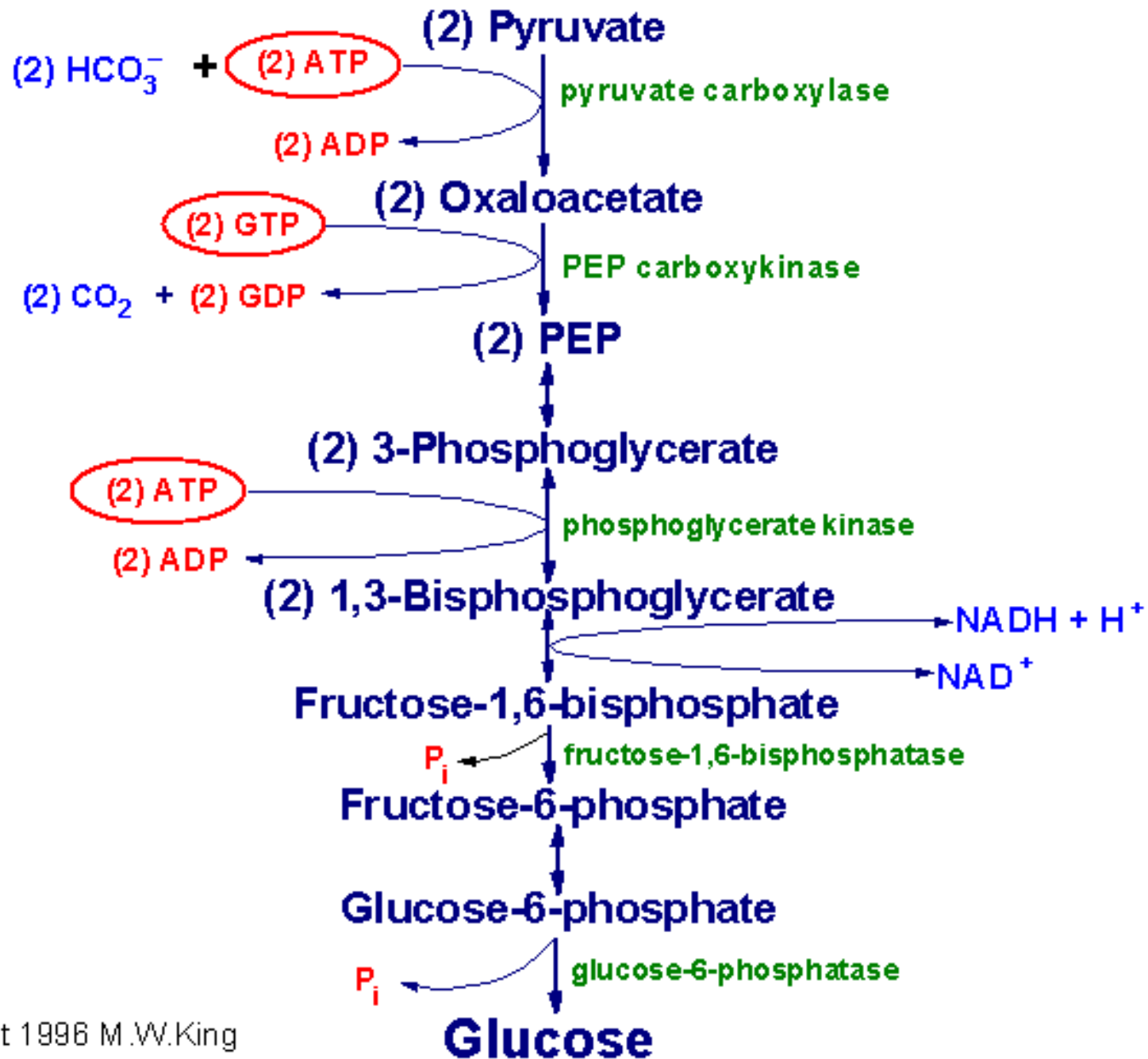
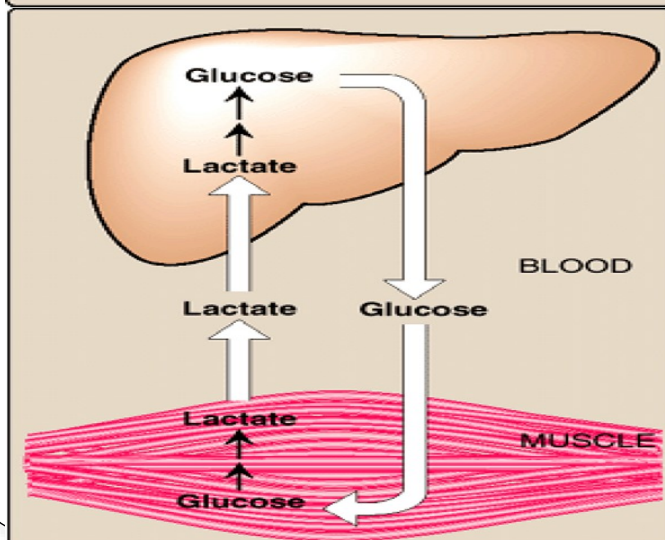
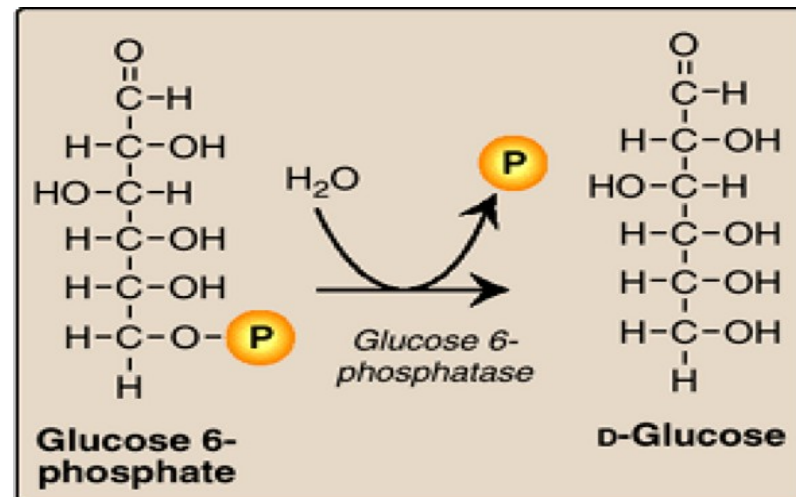
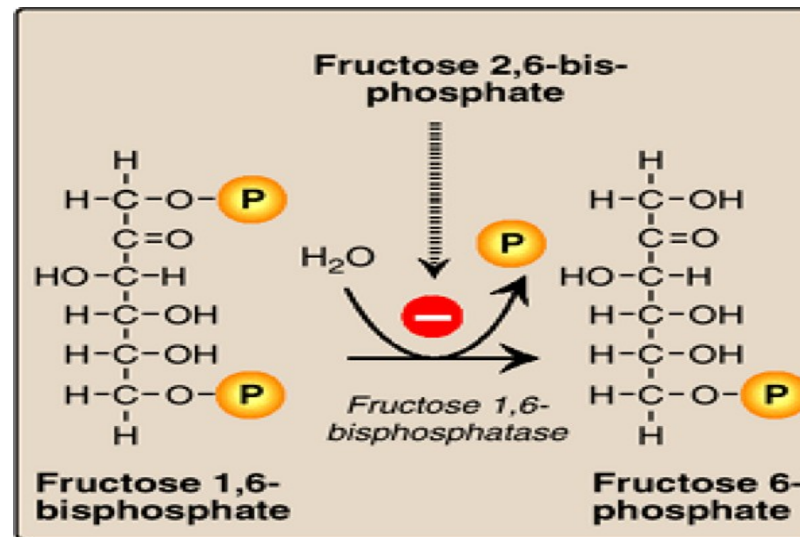
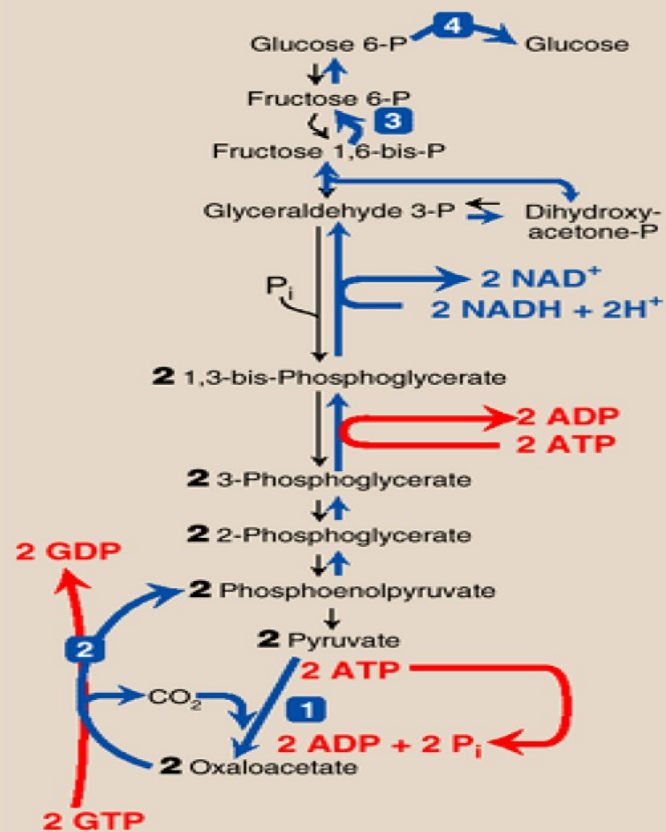


Figure 19-1. Major pathways and regulation of gluconeogenesis and glycolysis in the liver. Entry points of gluconeogenic amino acids after transamination are indicated by arrows extended from circles. The

Glucogenesis





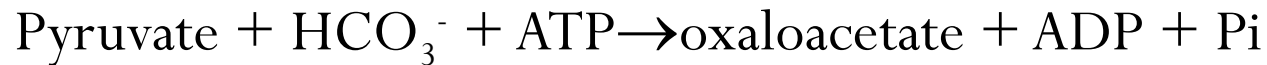
Gluconeogenesis Vs Glycolysis

- Thermodynamic barriers prevent a simple reversal of glycolysis in conversion of pyruvate to glucose.
- 7 of 10 reactions of gluconeogenesis are the reverse of glycolytic reactions.
- Three reactions of glycolysis are essentially irreversible in vivo and cannot be used in gluconeogenesis
 - ✓ the conversion of glucose to glucose 6-phosphate
 - ✓ the phosphorylation of fructose 6-phosphate to fructose 1,6-bisphosphate by phosphofructokinase-1
 - ✓ the conversion of phosphoenolpyruvate to pyruvate by pyruvate kinase.
- In cells, these three reactions are characterized by a large negative ΔG , whereas other glycolytic reactions have a ΔG near zero and can be reversed in vivo during gluconeogenesis.

BYPASS REACTIONS

1. Conversion of pyruvate to phosphoenolpyruvate

- Pyruvate is first transported into the mitochondria from the cytosol, or generated from alanine by transamination within mitochondria. Pyruvate is converted to oxaloacetate by **pyruvate carboxylase** which requires biotin:



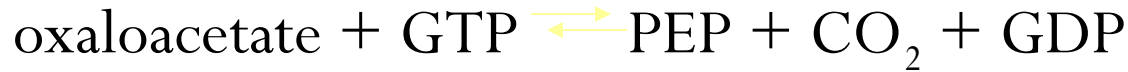
- Pyruvate carboxylase requires acetyl-CoA as a positive effector, and biotin acts as a carrier of activated HCO_3^-
- The oxaloacetate formed is reduced to malate by **mitochondrial malate dehydrogenase**:



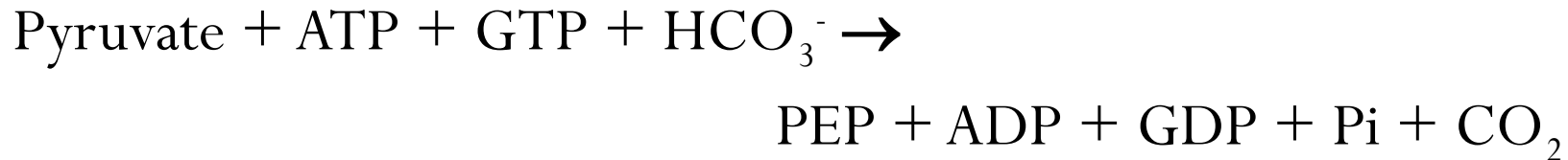
- Malate leaves the mitochondrion and is re-oxidized to oxaloacetate, with production of cytosolic NADH



- The oxaloacetate is then converted to phosphoenolpyruvate (PEP) by **phosphoenolpyruvate carboxykinase**, requiring Mg^{2+} and GTP:



Overall equation for this set of bypass reactions:



- Two high-energy phosphate equivalents must be expended to phosphorylate one molecule of pyruvate to PEP

2. Conversion of fructose 1,6-bisphosphate to fructose 6-phosphate:

- this is catalyzed by Mg^{2+} -dependent **fructose 1,6-bisphosphatase** which **hydrolyzes** the C-1 phosphate.



3. Conversion of glucose 6-phosphate to free glucose

- This is catalyzed by **glucose 6-phosphatase**, a Mg^{2+} -activated enzyme, and involves a simple **hydrolysis** of a phosphate ester:
$$\text{glucose 6-phosphate} + \text{H}_2\text{O} \rightarrow \text{glucose} + \text{P}_i$$
- The enzyme is found in hepatocytes and renal cells, **but not** in **muscle or brain**, and is located on the luminal side of the ER membrane

Citric acid cycle intermediates and many amino acids are glucogenic

- The biosynthetic pathway of gluconeogenesis allows the net synthesis of glucose from pyruvate and also from the citric acid cycle intermediates citrate, isocitrate, α -ketoglutarate, succinyl-CoA, succinate, fumarate, and malate, all of which can undergo oxidation to **oxaloacetate**.
- Some or all of the carbon atoms of many of the amino acids derived from protein are ultimately converted to pyruvate or to intermediates in the citric acid cycle. Such amino acids are said to be **glucogenic**.
- **Alanine** and **glutamine** are particularly important because they are the primary molecules that transport **amino groups** from extrahepatic tissue to the liver.

Glucogenic Amino Acids, Grouped by Site of Entry*

Pyruvate

Alanine

Cysteine

Glycine

Serine

Threonine

Tryptophan*

α -Ketoglutarate

Arginine

Glutamate

Glutamine

Histidine

Proline

Succinyl-CoA

Isoleucine*

Methionine

Threonine

Valine

Fumarate

Phenylalanine*

Tyrosine*

Oxaloacetate

Asparagine

Aspartate

Note: All these amino acids are precursors of blood glucose or liver glycogen, because they can be converted to pyruvate or citric acid cycle intermediates. Of the 20 common amino acids, only leucine and lysine are unable to furnish carbon for net glucose synthesis.

GLUCONEOGENESIS FROM AMINO ACID

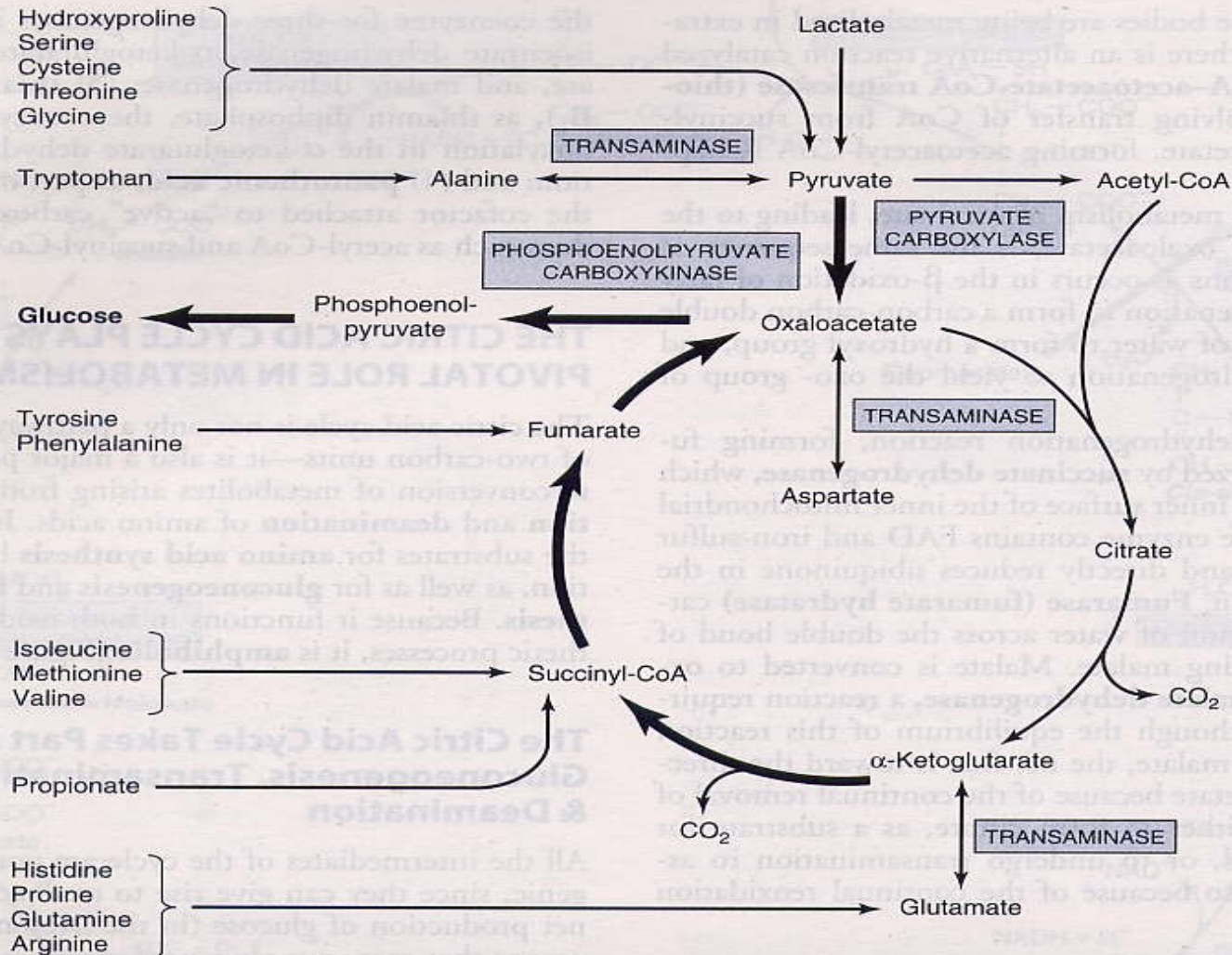
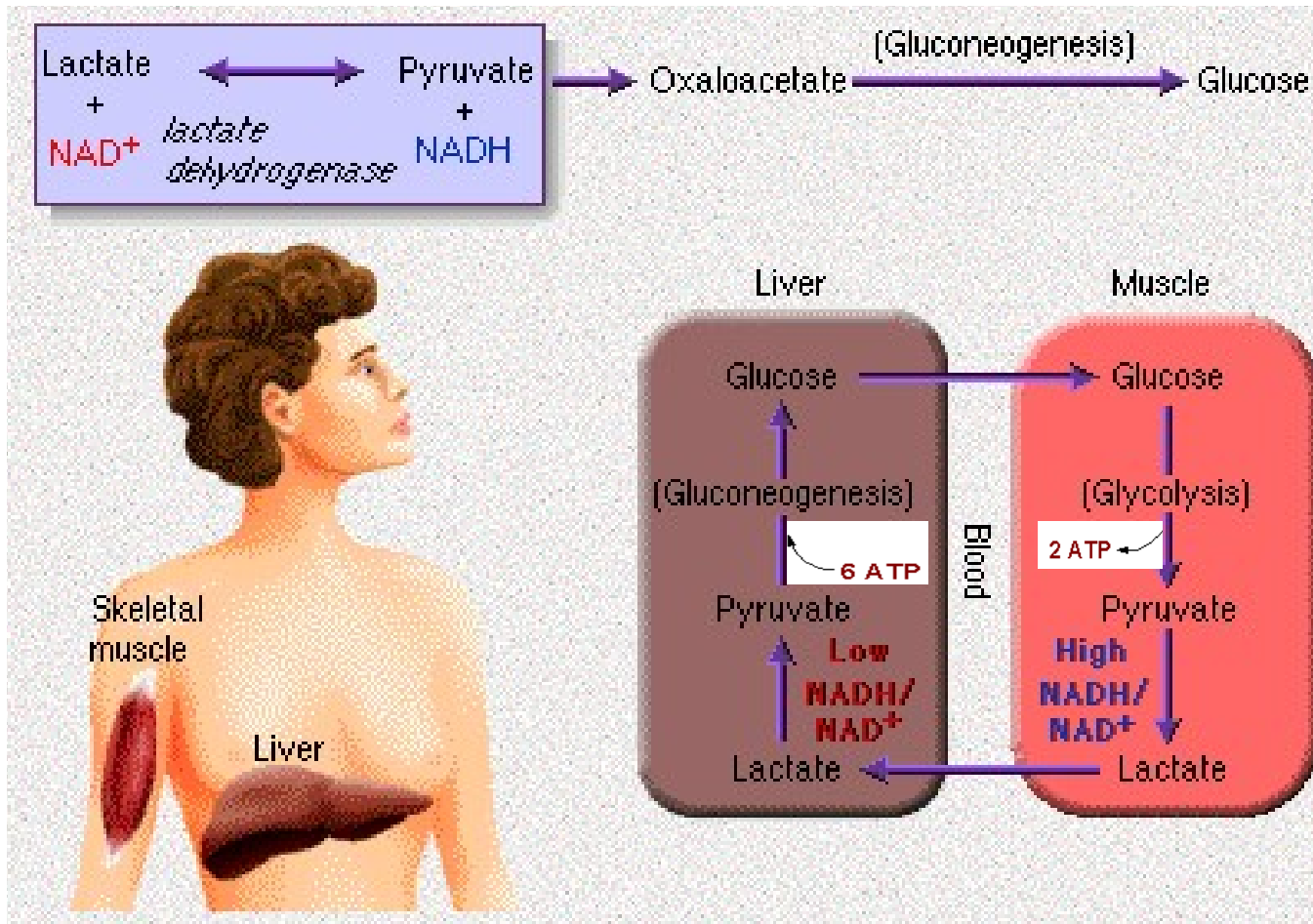


Figure 16-4. Involvement of the citric acid cycle in transamination and gluconeogenesis. The bold arrows indicate the main pathway of gluconeogenesis.

The Cori Cycle



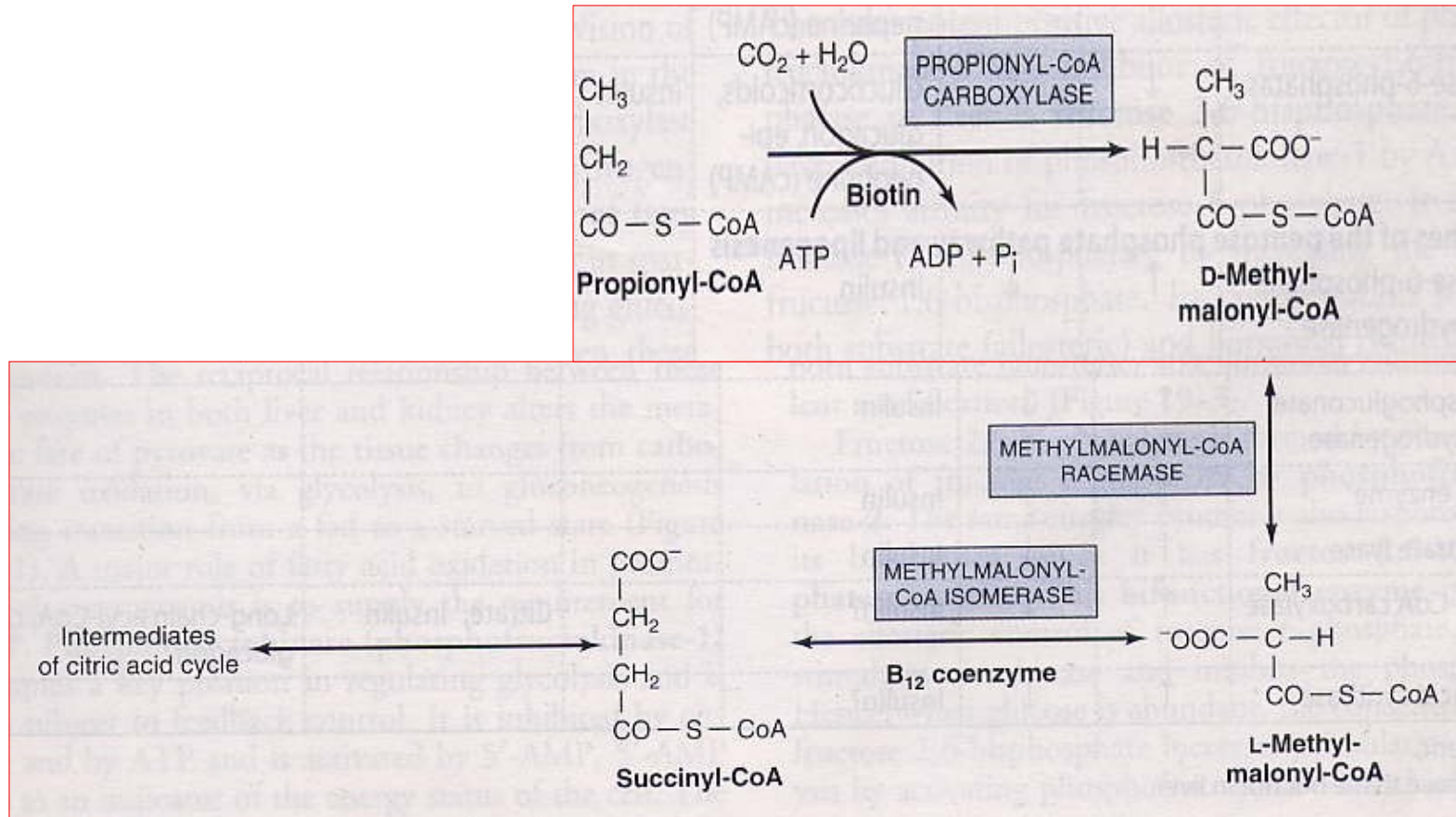
➤ Gluconeogenesis costing the body 4 moles of ATP more than are produced during glycolysis.

➤ Therefore, the cycle cannot be sustained indefinitely.

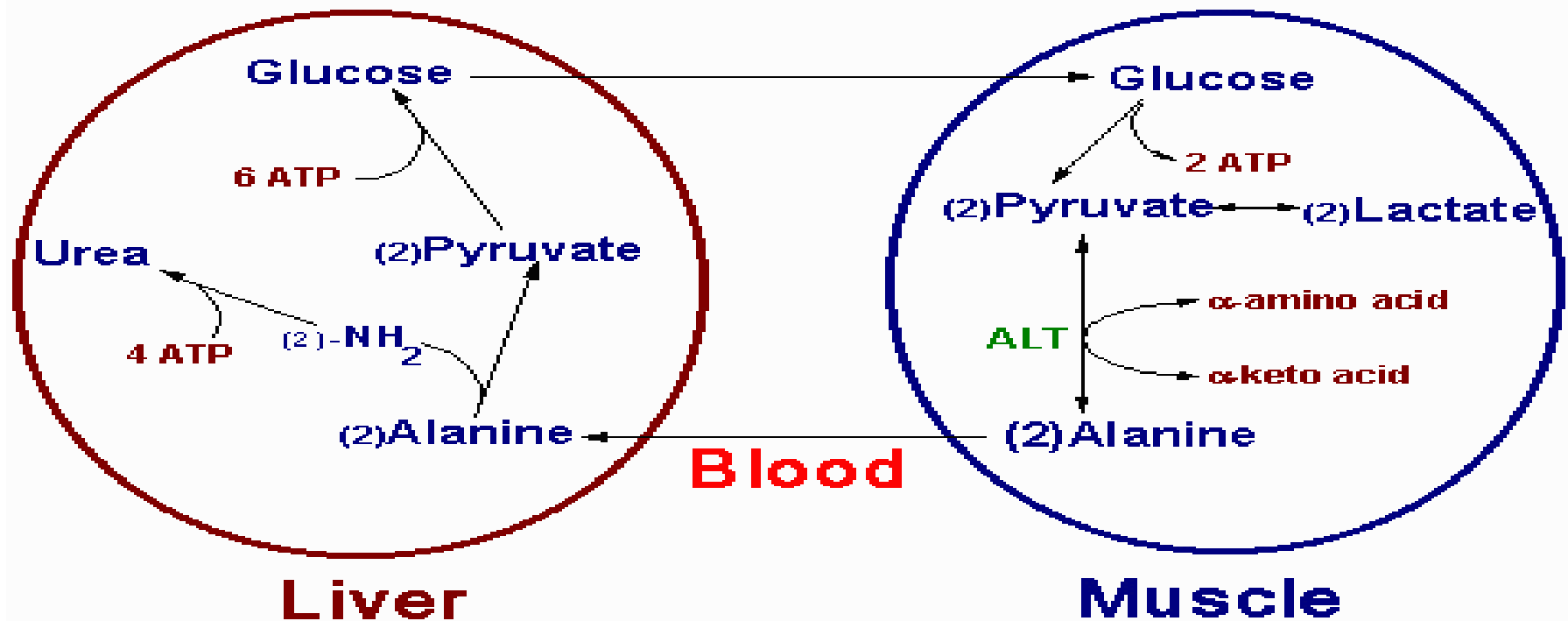
➤ The Cori cycle involves the utilization of lactate, produced by glycolysis in non-hepatic tissues, (such as muscle and erythrocytes) as a carbon source for hepatic gluconeogenesis.

➤ Liver can convert the anaerobic byproduct of glycolysis, lactate, back into more glucose for reuse by non-hepatic tissues.

GLUCONEOGENESIS FROM Propionyl-CoA



Glucose/Alanine Cycle



The glucose-alanine cycle is *used primarily as a mechanism for skeletal muscle to eliminate nitrogen* while replenishing its energy supply.

Glucose oxidation produces pyruvate which can undergo transamination to alanine.

- This reaction is catalyzed by *glutamate-pyruvate transaminase*, GPT (also called *alanine transaminase*, ALT).

Regulation of Glycolysis and Gluconeogenesis

- High glucose levels and insulin promote glycolysis
- Low glucose levels and glucagon promote gluconeogenesis

	Glycolysis	Gluconeogenesis
Enzyme	Hexokinase	Glucose-6-phosphatase
Activated by	High glucose levels Insulin, epinephrine	Low glucose levels Glucose-6-phosphate
Inhibited by	Glucose-6-phosphate	
Enzyme	Phosphofructokinase	Fructose-1,6-bisphosphatase
Activated by	AMP	Low glucose levels, glucagon
Inhibited by	ATP	AMP, insulin
Enzyme	Pyruvate kinase	Pyruvate carboxylase
Activated by	Fructose-1,6-bisphosphate	Low glucose level, glucagon
Inhibited by	ATP Acetyl CoA	Insulin

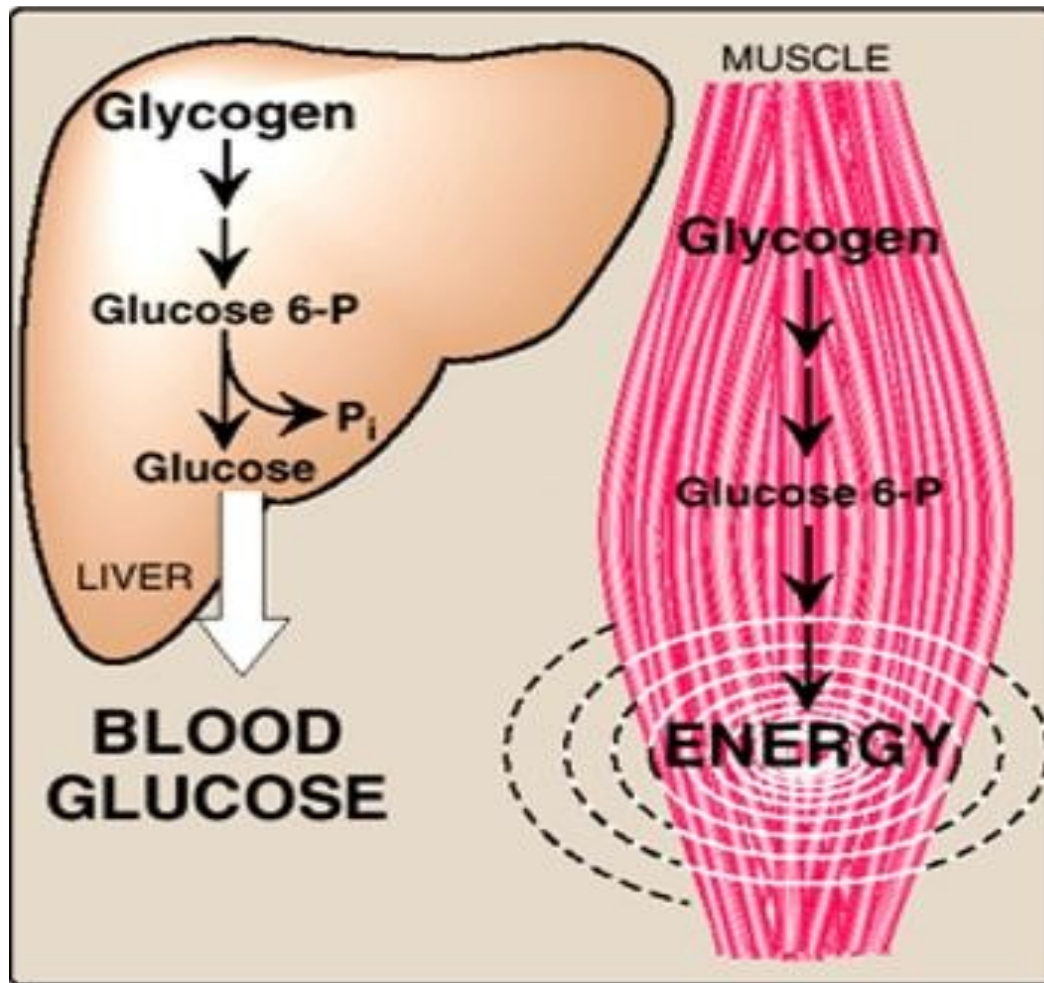
Glycogen Metabolism

- Blood glucose can be obtained from **three primary sources**:
 - ☛ the diet,
 - ☛ degradation of glycogen, and
 - ☛ gluconeogenesis.
- Dietary intake of glucose is **sporadic** and, depending on the diet, is not always a reliable source of blood glucose.
- Gluconeogenesis can provide sustained synthesis of glucose, but it is somewhat slow in responding to a falling blood glucose level.

- Therefore, the body has developed mechanisms for storing a supply of glucose in a rapidly mobilizable form, namely, **glycogen**.
- In the absence of a dietary source of glucose, this compound is rapidly released from **liver glycogen**.
- Similarly, **muscle glycogen** is extensively degraded in exercising muscle to provide that tissue with an important energy source.

Structure and Function of Glycogen

- The main stores of glycogen in the body are found in **skeletal muscle** and **liver**, although most other cells store small amounts of glycogen for their own use.
- The function of **muscle glycogen** is to serve as a **fuel reserve** for the synthesis of ATP during muscle contraction.
- **liver glycogen** is **to maintain the blood glucose concentration**, particularly during the early stages of a fast.



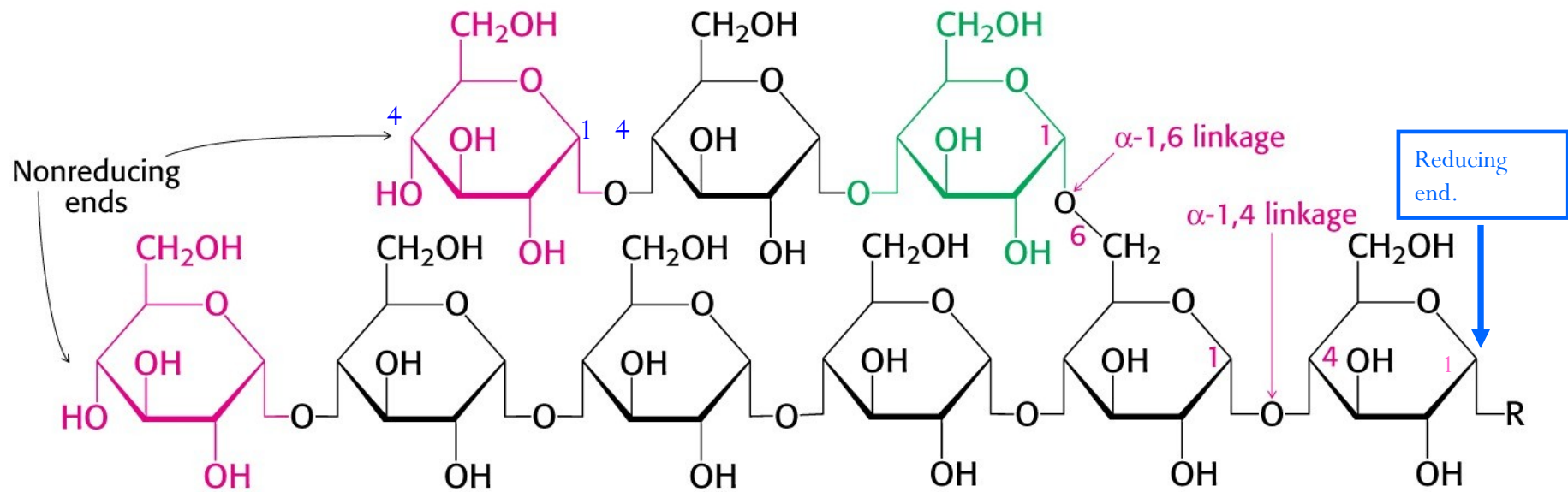
- Muscle glycogen does not directly yield free glucose (because muscle **lacks** *glucose 6-phosphatase*)

- pyruvate formed by glycolysis in muscle can undergo transamination to alanine, which is exported from muscle and used for gluconeogenesis in the liver.

Fig. function of muscle and liver glycogen.

Glycogenolysis: catabolism of glycogen.

Glycogenesis: anabolism or biosynthesis of glycogen.



Glycogen is a polymer of **glucose** residues linked by

- ♦ $\alpha(1 \rightarrow 4)$ glycosidic bonds, mainly
- ♦ $\alpha(1 \rightarrow 6)$ glycosidic bonds, at branch points

Glycogen chains & branches are longer than shown

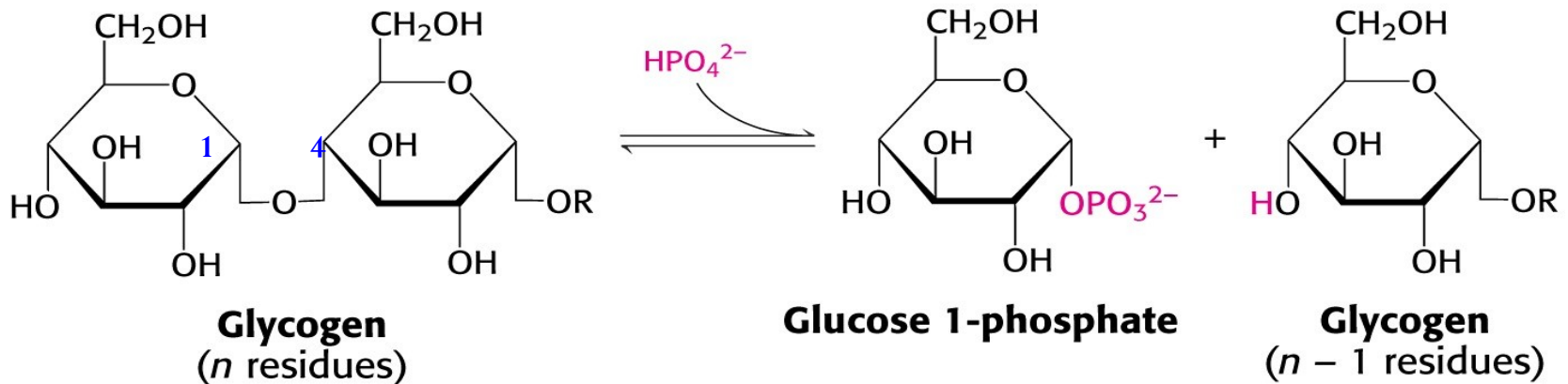
Glucose is stored as glycogen predominantly in **liver** and **muscle** cells.

Glycogenolysis

- To mobilizing glycogen, three enzymes are required:
 - glycogen phosphorylase,
 - debranching enzyme, and
 - phosphoglucomutase.
- The end product of glycogenolysis is glucose 6-phosphate.

Glycogen phosphorylase:

catalyzes an attack of an α -1,4 glycosidic bond linking 2 glucose residues by inorganic phosphate resulting in the removal of the terminal glucose residue as **glucose 1-phosphate**.

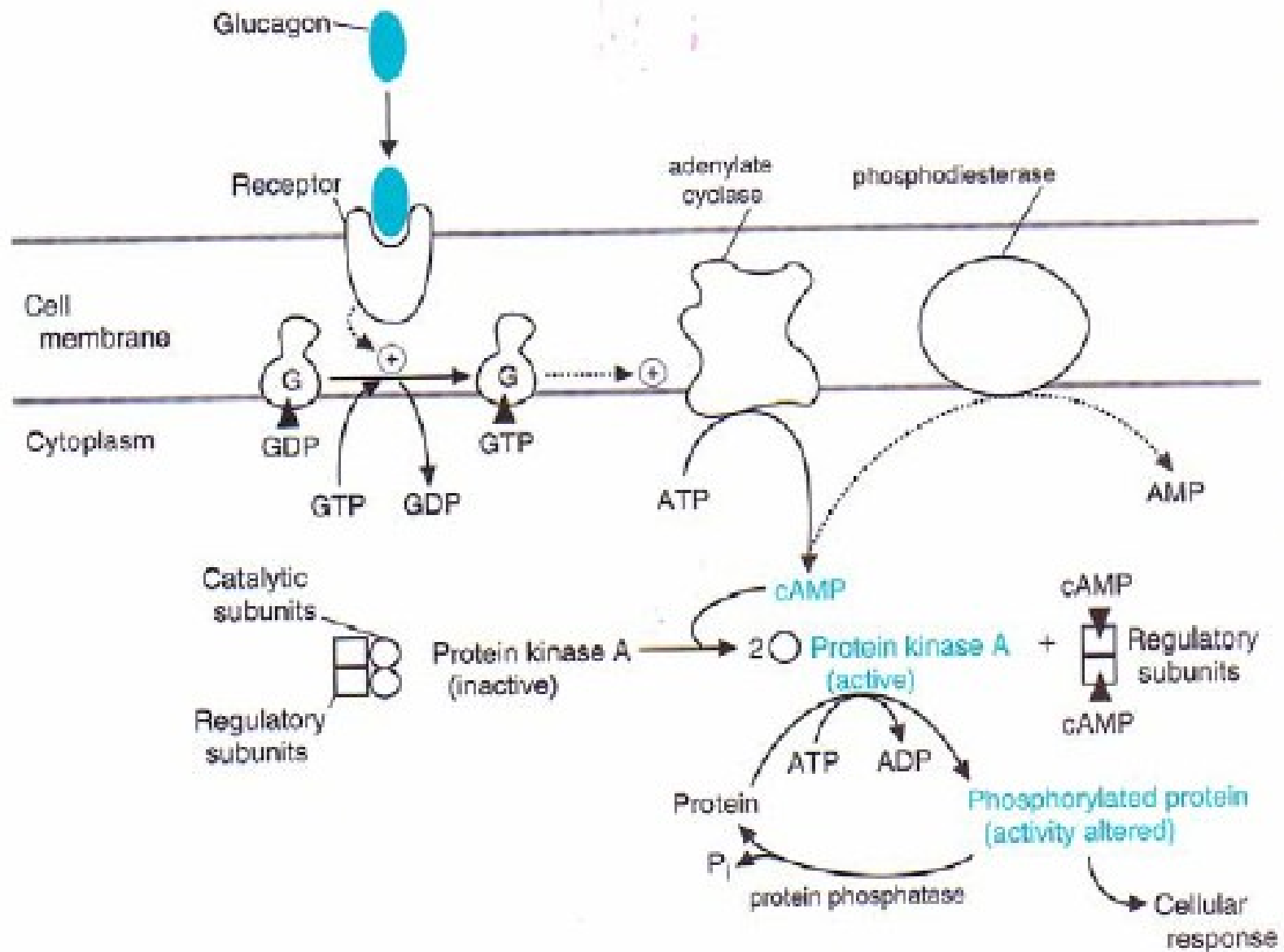


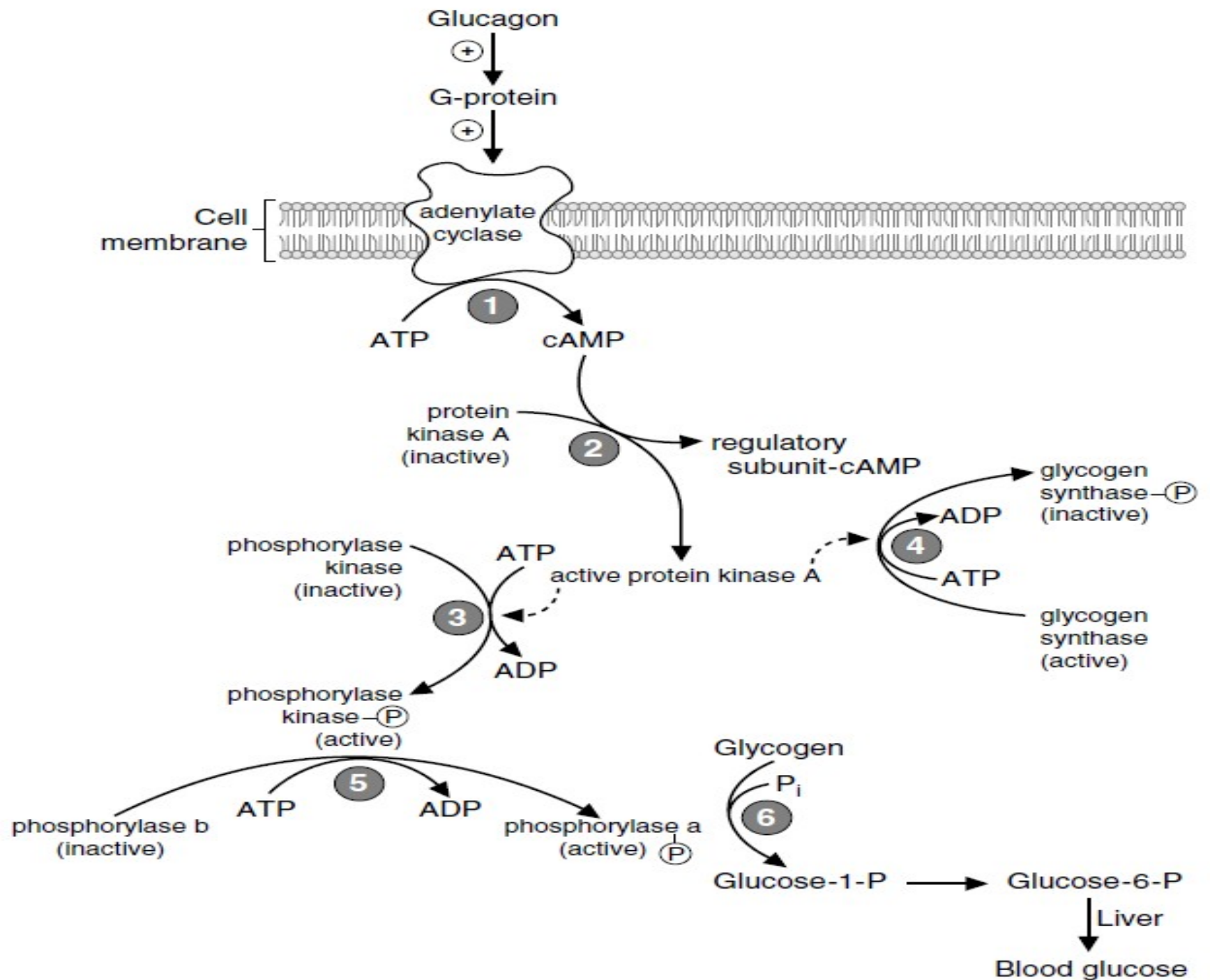
- **Phosphorylase** stops when it reaches a residue that is 4 positions away from a branch point.

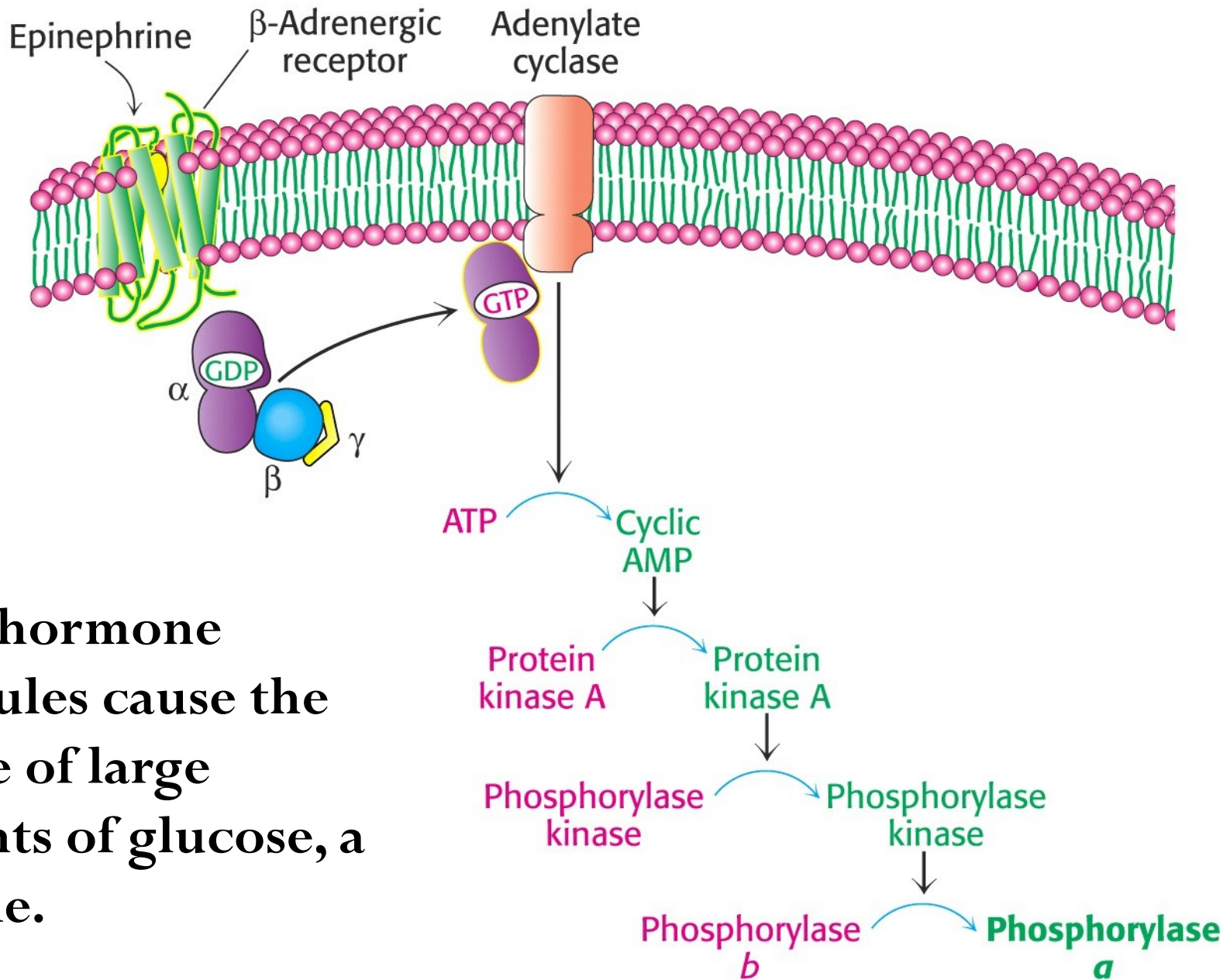
- A glycogen molecule that has been degraded by phosphorylase to the limit caused by the branches is called a **phosphorylase-limit dextrin**.

Epinephrine and Glucagon Stimulate Glycogen breakdown

- Muscle is responsive to epinephrine.
- Liver is responsive to glucagon and somewhat responsive to epinephrine.
- Both signal a cascade of molecular events leading to glycogen breakdown.
- Both utilize a G-protein-dependent signal-transduction pathway.





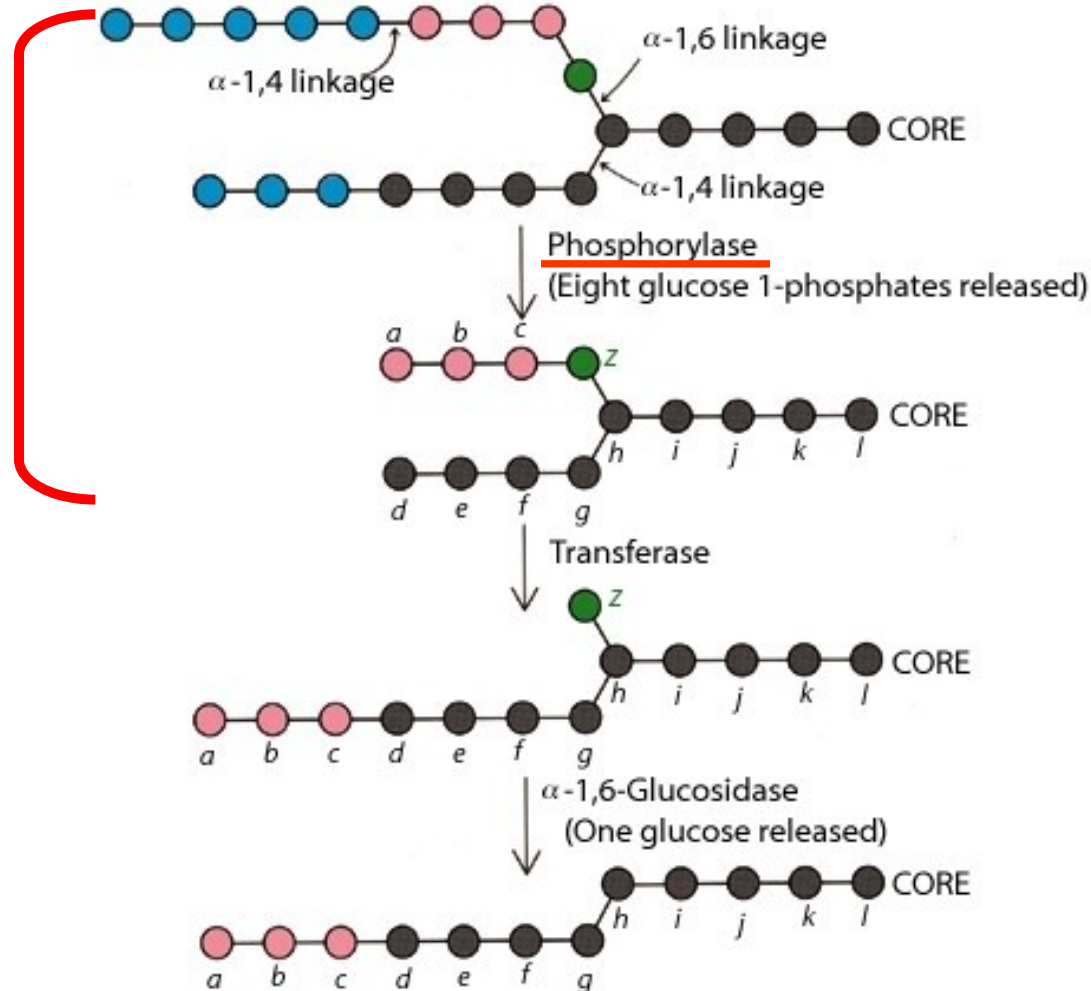


A few hormone molecules cause the release of large amounts of glucose, a cascade.

- ◆ Glucagon, which is synthesized by α -cells of the pancreas, activates cAMP formation in liver.
- ◆ Epinephrine activates cAMP formation in muscle.
 - Epinephrine is released from the adrenal glands in response to neural signals indicating an immediate need for enhanced glucose utilization in muscle, the so called **fight or flight response**.
 - **Muscle cells lack glucagon receptors.**
 - The presence of glucagon receptors on muscle cells would be futile anyway **since the role of glucagon release is to increase blood glucose concentrations** and muscle glycogen stores cannot contribute to blood glucose levels... **why?**

- α -1,6 glycosidic linkages at the branch points are **not** degraded by phosphorylase.

- When phosphorylase reaches a point 4 residues away from a branch point it stops.

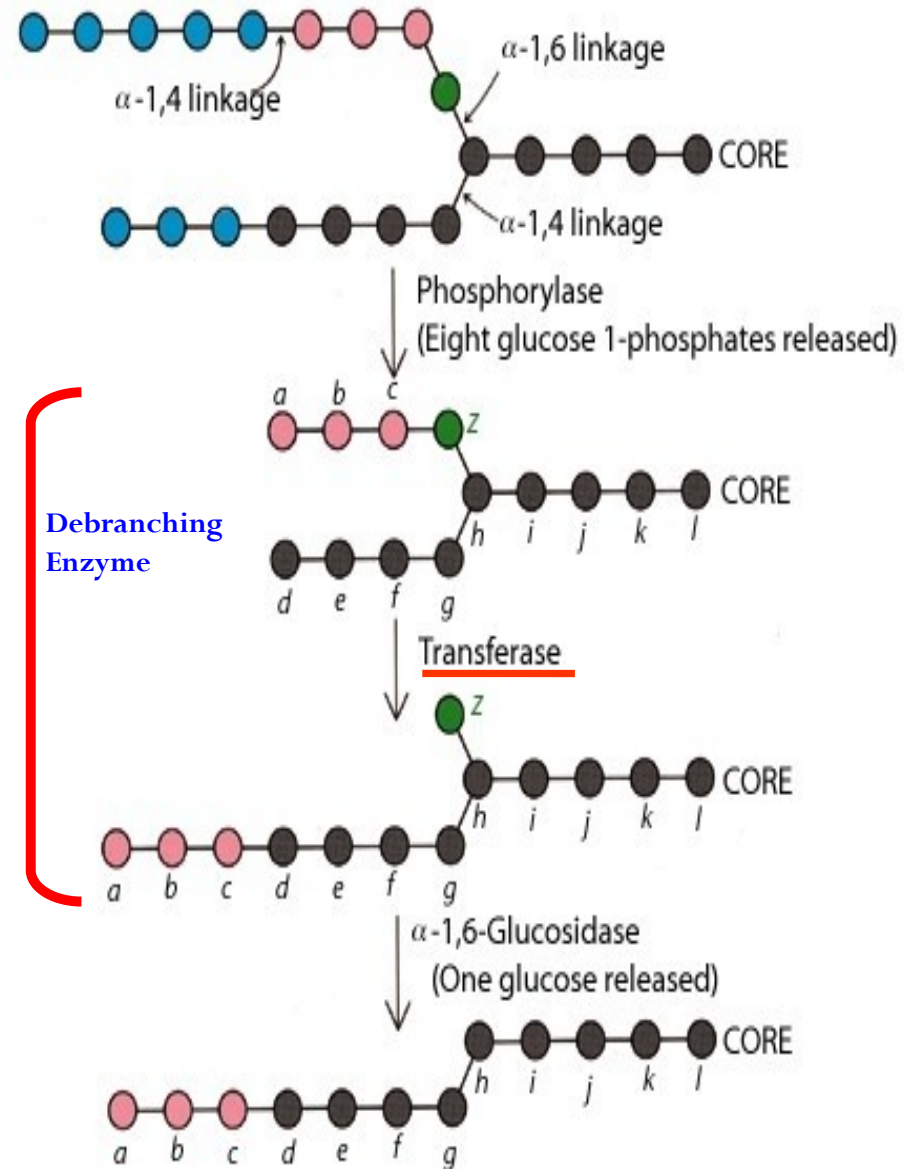


Further degradation occurs via the action of *debranching enzyme*.

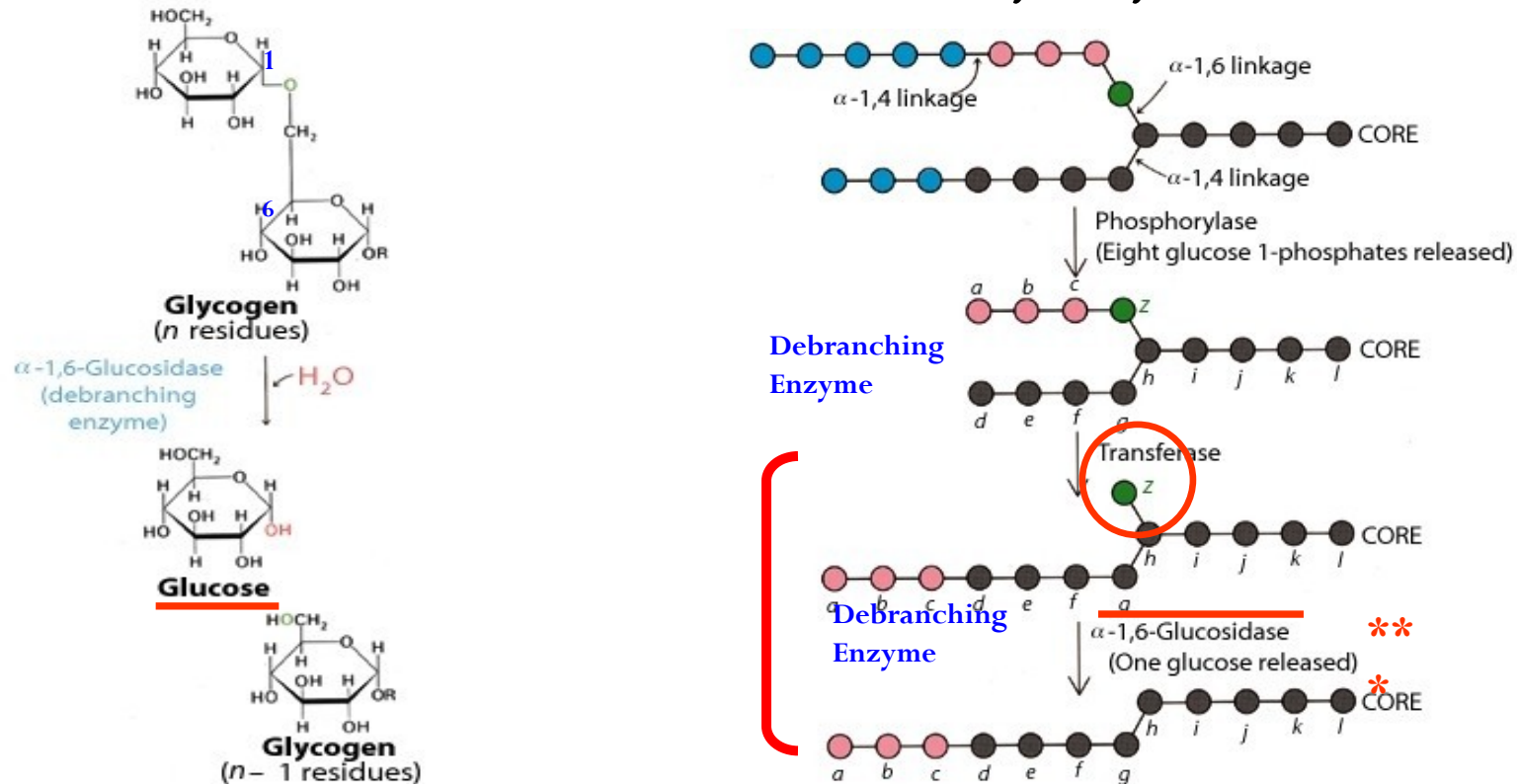
Debranching enzyme is bifunctional (it catalyzes 2 different reactions).

(1) A transferase activity in which a block of 3 glucosyl residues are transferred from one outer branch to the other.

The result is a longer amylose chain and only one glucosyl remaining in α -1,6 glycosidic linkage in the other chain.



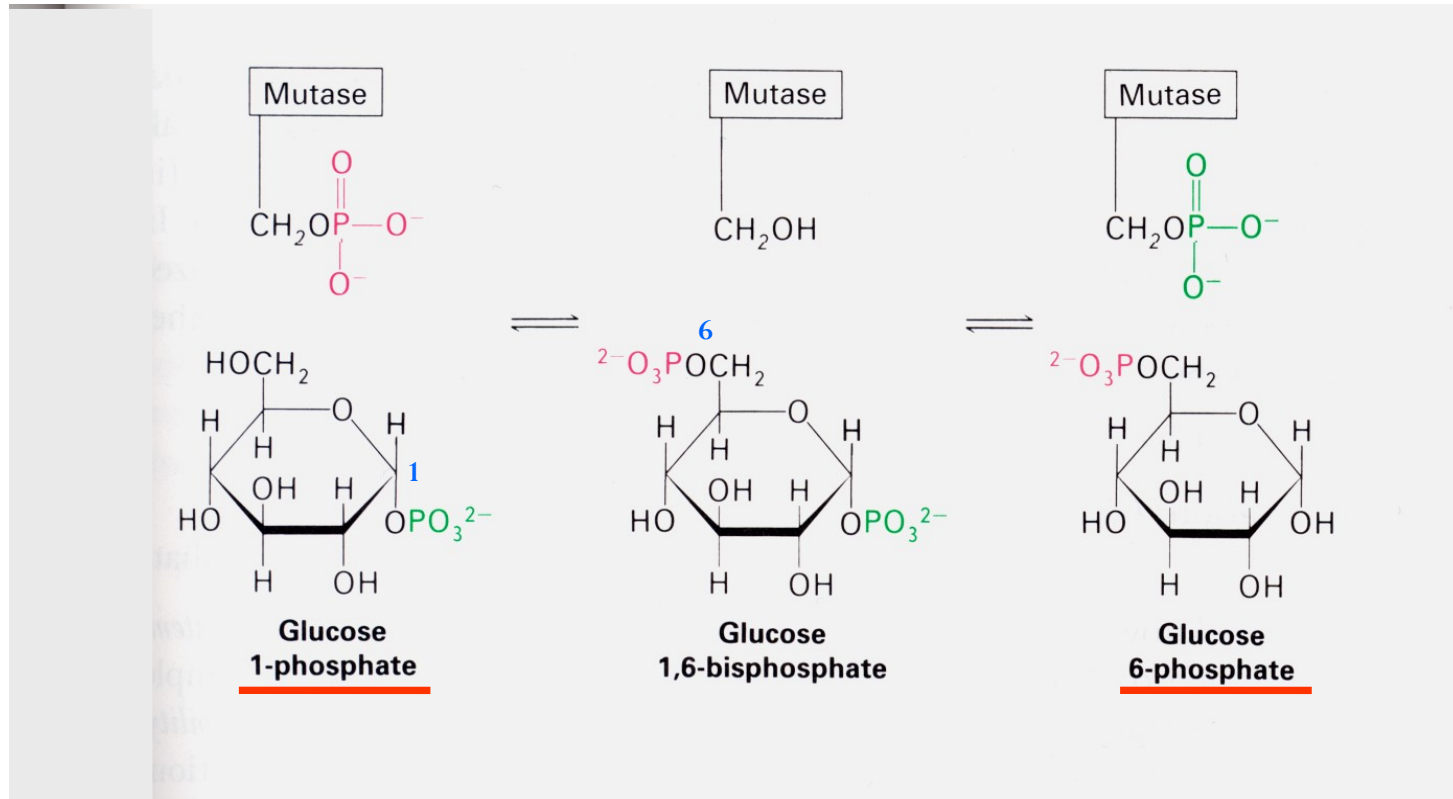
The remaining α -1,6 linkage is hydrolyzed by the **second enzymatic activity of debranching enzyme, namely its α -1,6-glucosidase activity** wherein an α -1,6 glycosidic bond between two residues is hydrolyzed.



Thus, the sum of the **transferase** and the **glucosidase** activities of **debranching enzyme** convert a branched structure into a linear one.

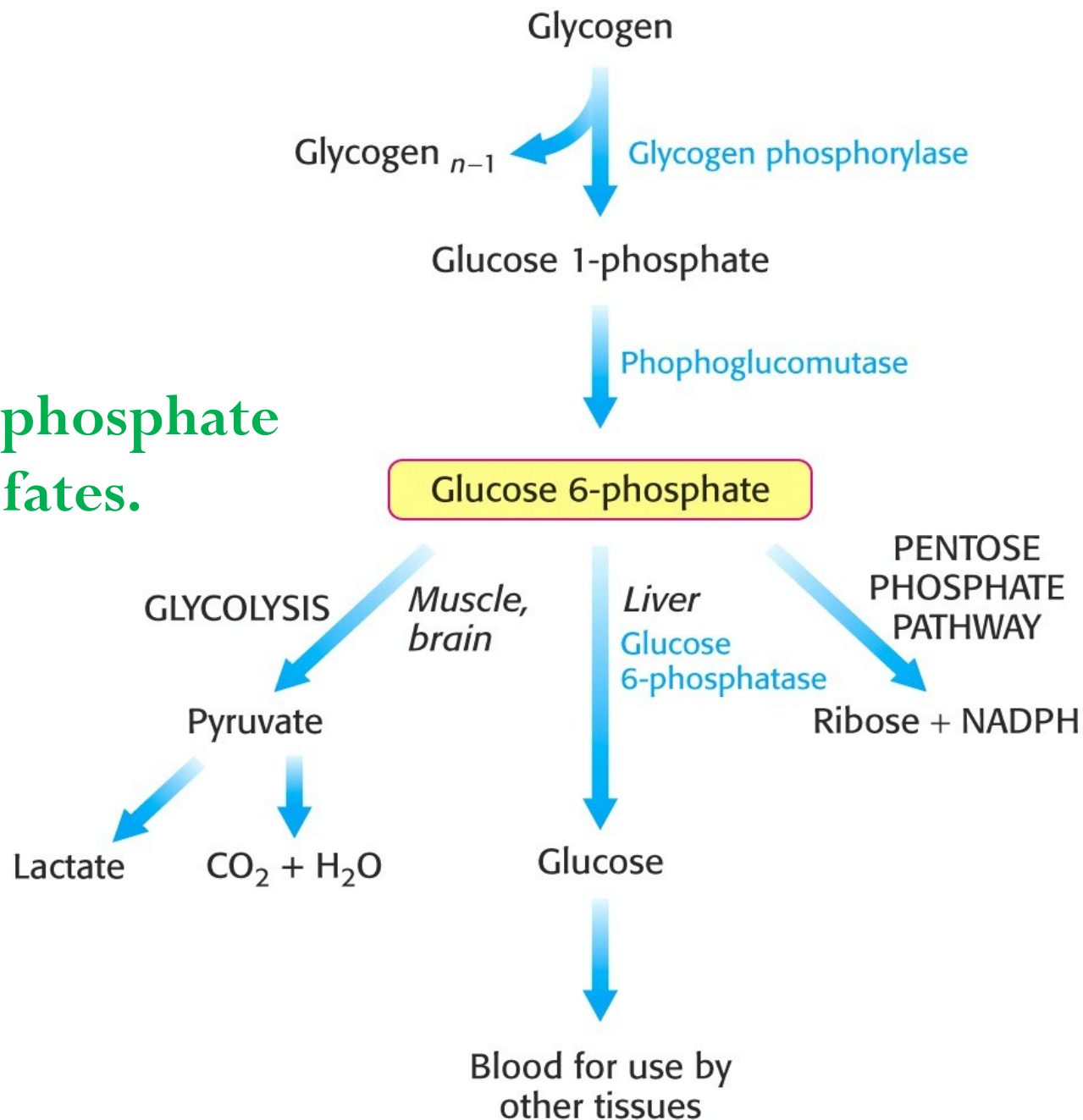
Net Result: mainly glucose 1-phosphate + some glucose (from the branch points).

Phosphoglucomutase: converts glucose 1-phosphate to glucose 6-phosphate via shifting a phosphoryl group.



The glucose 6-phosphate can then enter the metabolic mainstream.

Glucose 6-phosphate
has 3 fates.



Remember!

- Liver contains glucose 6-phosphatase.
- Muscle does not have this enzyme.

WHY?

- The liver releases glucose to the blood to be taken up by brain and active muscle. The liver regulates blood glucose levels.
- The muscle **retains** glucose 6-phosphate to be use for energy. **Phosphorylated glucose is not transported out of muscle cells.**

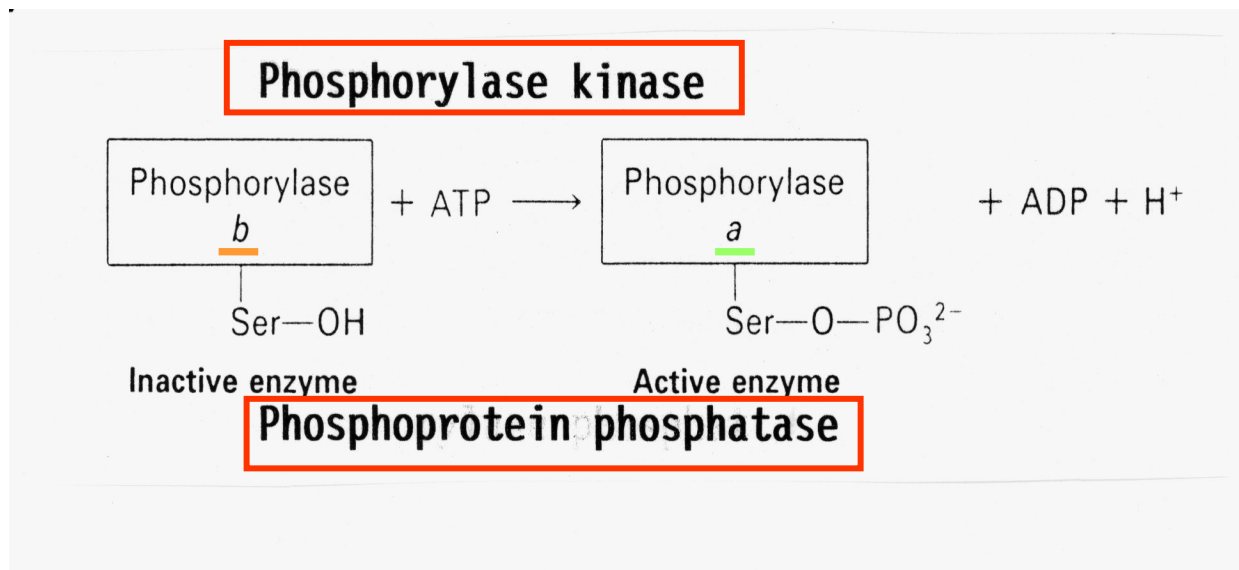
Regulation of Glycogen Phosphorylase

Regulated by both allosteric effectors and by reversible phosphorylation.

In skeletal muscle the phosphorylase exists in 2 inter-convertible forms:

Phosphorylase a: usually active form

Phosphorylase b: usually (but not always) inactive form.

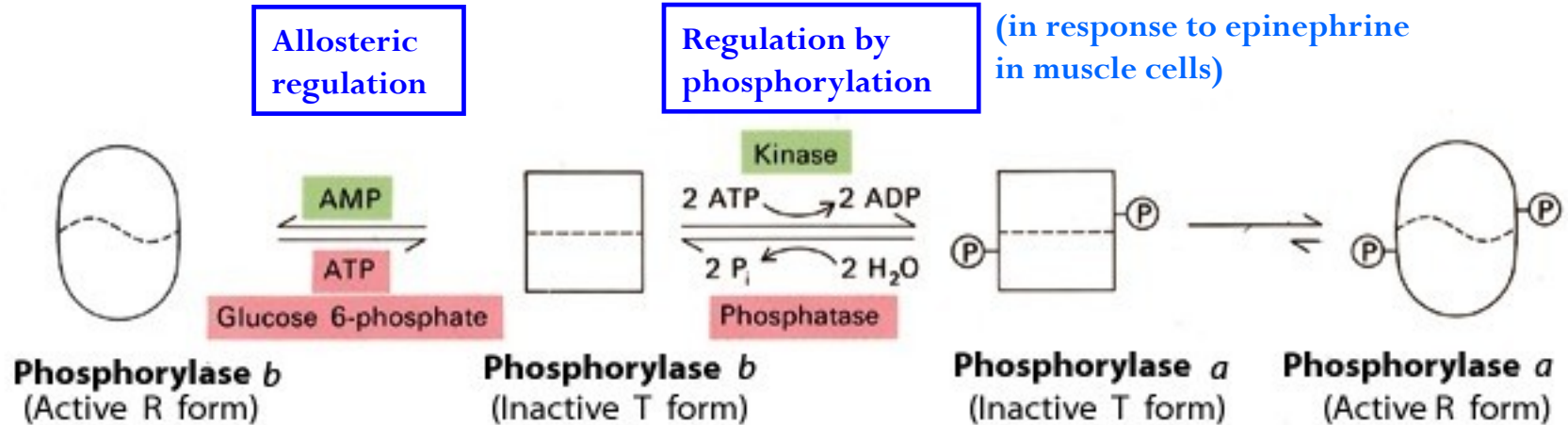


Phosphorylase b can be allosterically activated by AMP.

ATP and glucose 6-phosphate can prevent this activation.

Thus, the b form is usually inactive (because ATP and glucose 6-phosphate are plentiful).

The b form is active only when the energy charge of the cell is very low.



The a form is active regardless of the cell's energy charge.

At rest, most of the enzyme exists in the b form. During exercise, elevated AMP activates the b form.

Glycogenesis

- is the **Synthesis of Glycogen** from α -D Glucose.
- The process occurs in the **cytosol**, and requires energy supplied by **ATP** (for the phosphorylation of glucose) and uridine triphosphate (**UTP**).
 - *i.e. energy consuming process.*
- **Site-** Though every cell can form glycogen it chiefly occurs in **liver** and **skeletal muscle**.

- Even though-energy rich **fat** is abundant in the body, **skeletal muscle** prefers to store glucose (energy) as glycogen. **Because;**

1. Fat can not be oxidized under anaerobic condition.
2. Acetyl-CoA of fat oxidation can not be converted to glucose.
3. Skeletal muscle is unable to mobilize fat rapidly.

Glycogenesis: formation of glycogen

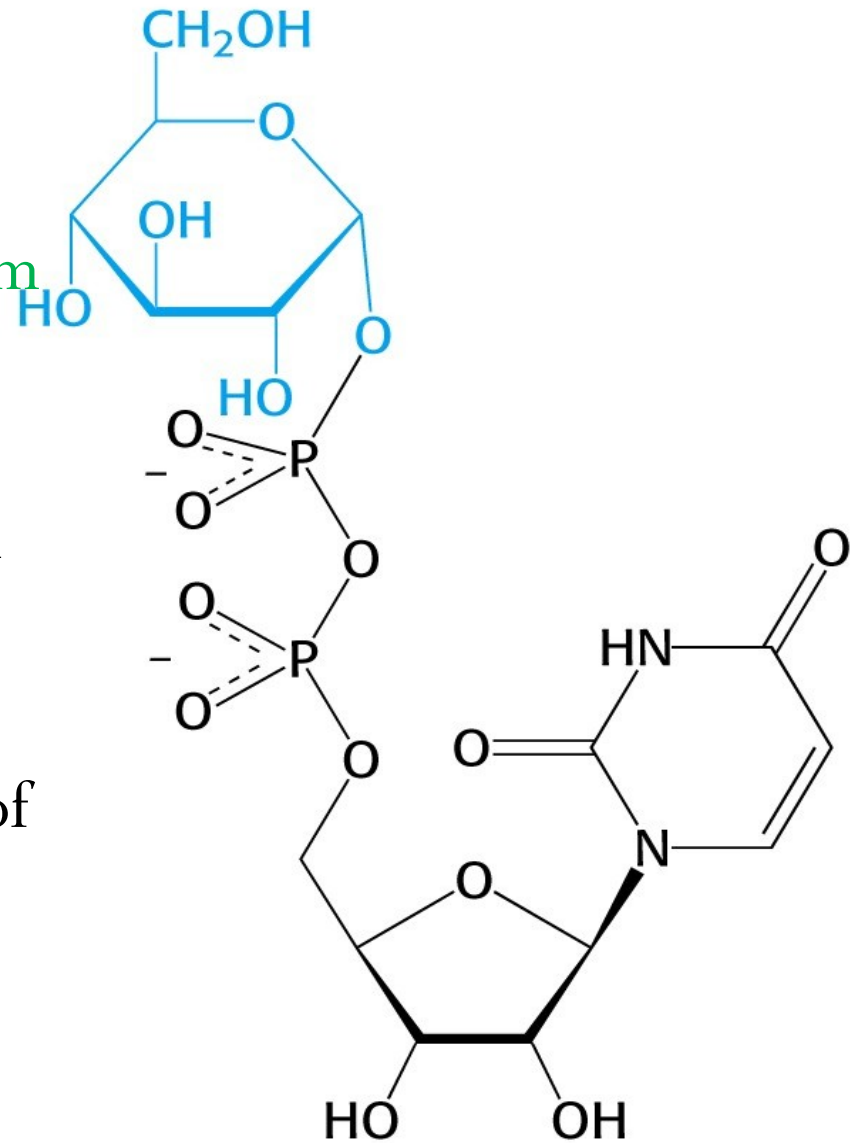
The following enzymes/proteins are required for the formation of glycogen:

- Glycogenin,
- *Glycogen synthase*,
- *Glycogen-branching enzyme*, and
- *UDP-glucose pyrophosphorylase*.

UDP glucose is the activated form of glucose.

Acetyl CoA is the activated form of acetate.

AA-tRNA is the activated form of amino acids.



**Uridine diphosphate glucose
(UDP-glucose)**

Glycogenesis

- Glycogen synthesis occurs in virtually all animal tissues, but is most prominent in the liver and skeletal muscle.

- 1) The first reaction is catalyzed by *hexokinase (or glucokinase in liver)*:

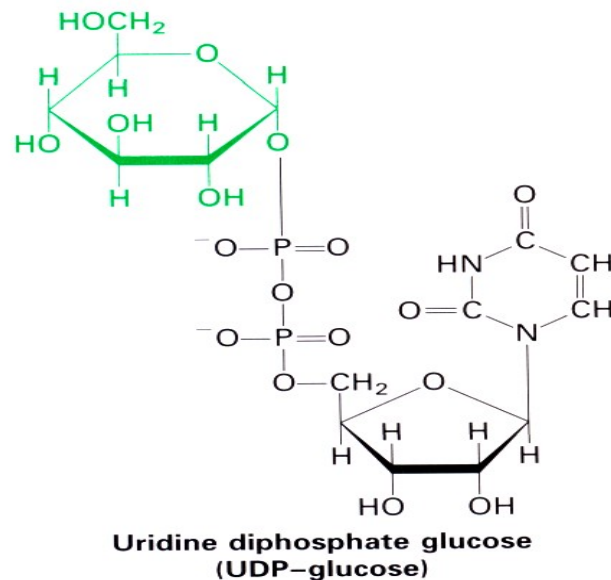
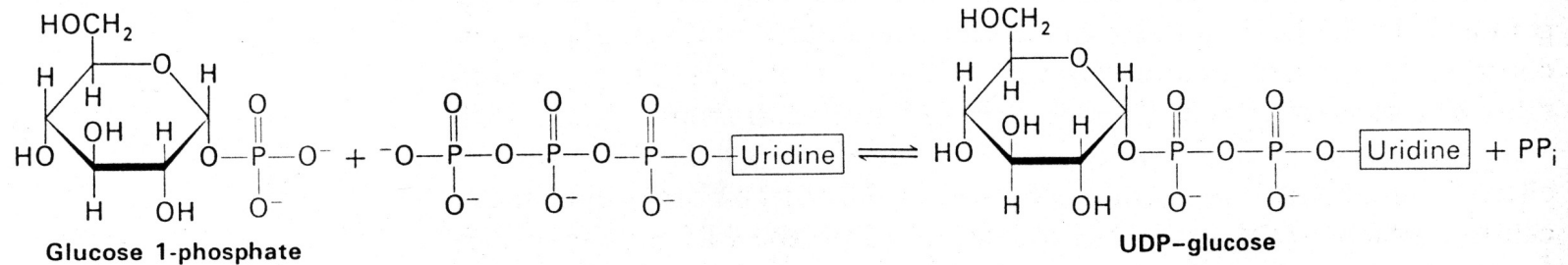


- 2) The second reaction is catalyzed by *phosphoglucomutase*:



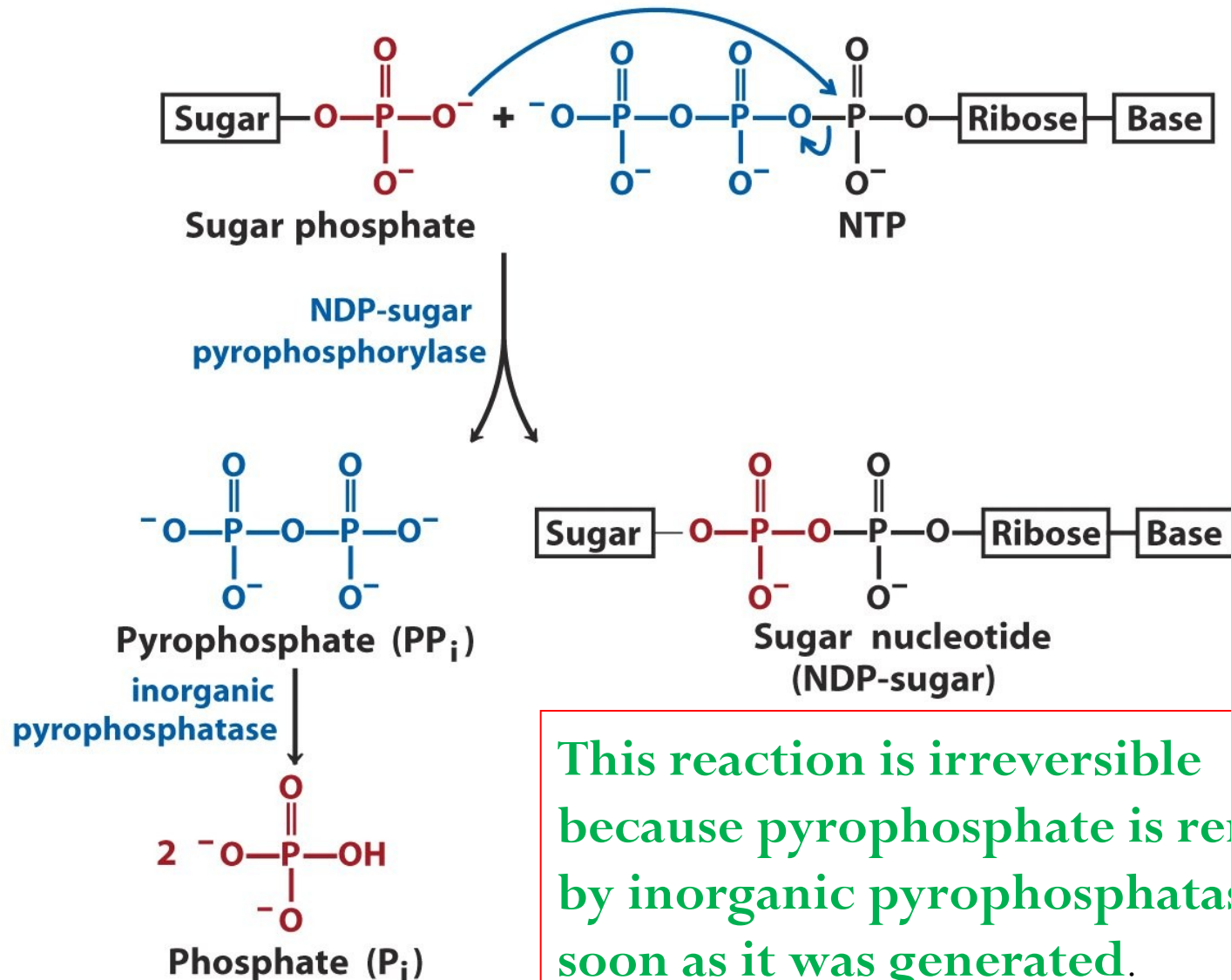
3) *UDP-glucose pyrophosphorylase*: catalyzes the formation of **UDP-glucose**, which will serve as **a glucose donor** in subsequent reactions.

Thus, **UDP glucose** can be considered as an **activated form of glucose**.



PP_i is rapidly hydrolyzed to 2 Pi driving the above reaction to completion.

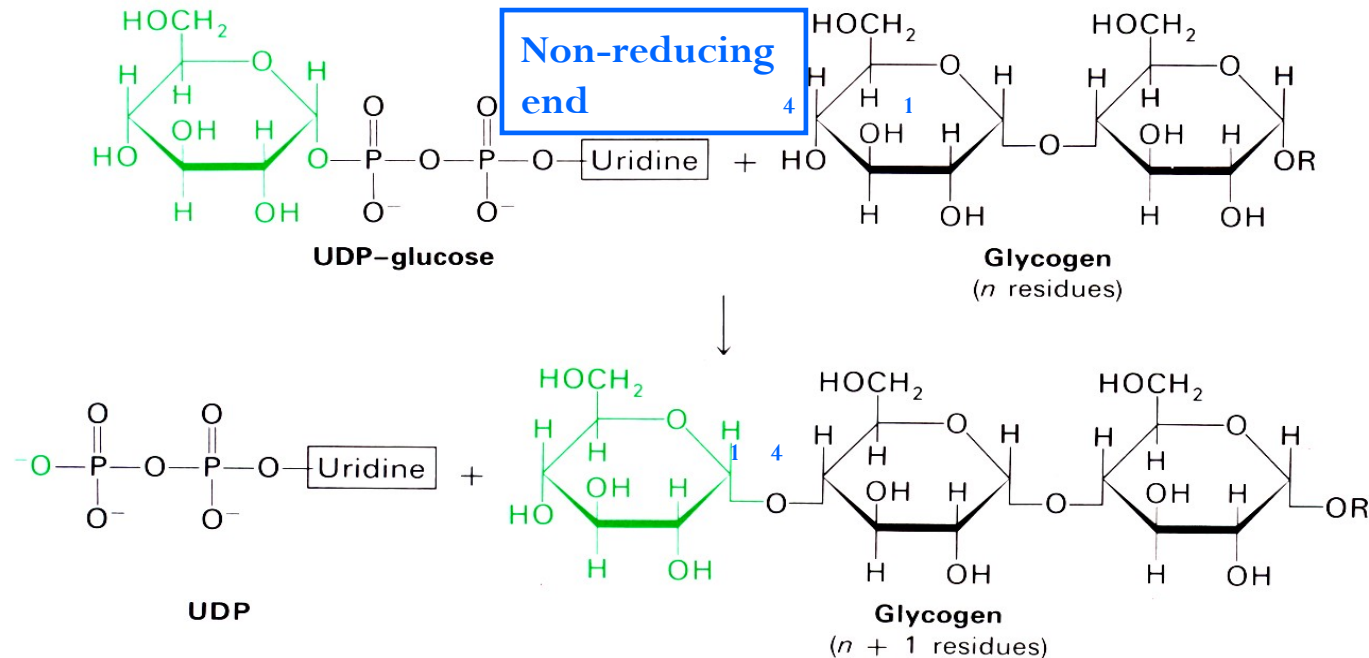
Formation of UDP-glucose



Net reaction: Sugar phosphate + NTP $\longrightarrow$ NDP-sugar + 2P_i

4) **Glycogen synthase**: catalyzes the transfer of glucose from UDP-glucose to **a growing chain**.

Glucosyl units are added to the **non-reducing terminal residue** of glycogen (i.e., at the C-4 terminus) to form an **α -1,4-glycosidic bond**.



- The reaction **requires at least 4 residues of glucose as a primer**.
- Thus a tetrasaccharide primer is elongated by one residue.
- This process is repeated many times to form a long, **unbranched** glycogen molecule.

5) *Glycogen synthase* requires a primer. This priming function is carried out by *glycogenin*, a 37 kDa protein to which the first glucose residue is attached.

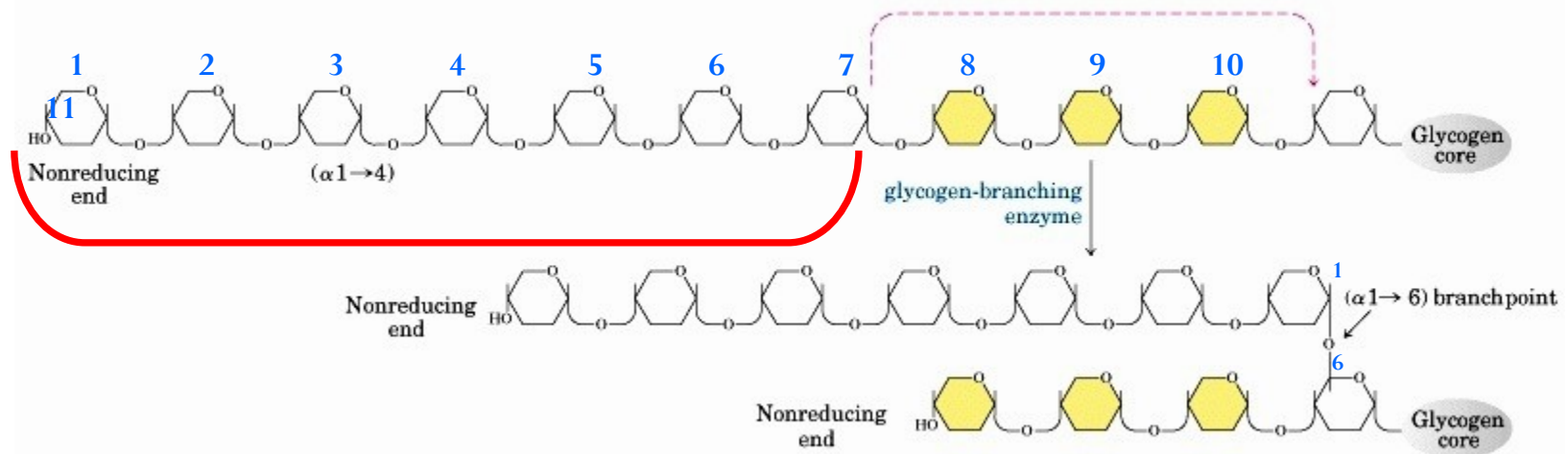
Glycogenin acts as a catalyst for synthesis of the nascent glycogen molecule by attaching a glucose to the hydroxyl group of a specific tyrosine residue within its structure.

Glycogenin then autocatalyzes the addition of up to 8 glucose units in α -1,4 glycosidic linkages with UDP-glucose being the glucose donor.

Glycogen synthase then takes over.

6) **Branching enzyme** forms the **α -1,6 linkages** that make glycogen a branched polymer.

Transfers a terminal fragment, 6-7 glucosyl residues in number, from the **non-reducing end of a glycogen branch having at least 11 residues, to the C-6 hydroxyl group of a glucose residue of the same or another glycogen chain at a more interior point, thereby creating a branch.**



Branching: increases the solubility of glycogen (by increasing the number of hydroxyl groups); AND

increases the rate of glycogen synthesis and degradation by creating a large number of terminal residues at which the appropriate enzymes can act.

Hormonal Control of Glycogenesis

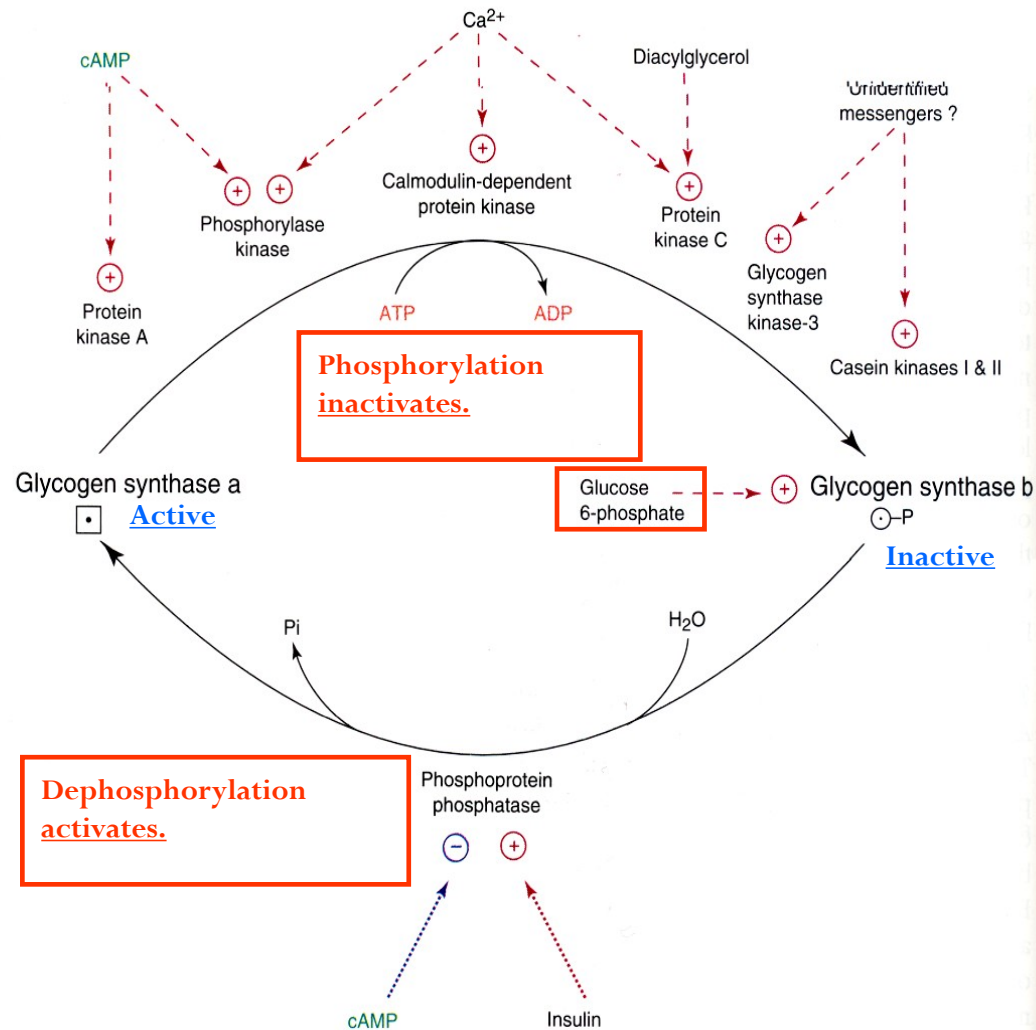
The activity of *glycogen synthase* is regulated by BOTH covalent modification and allosterically.

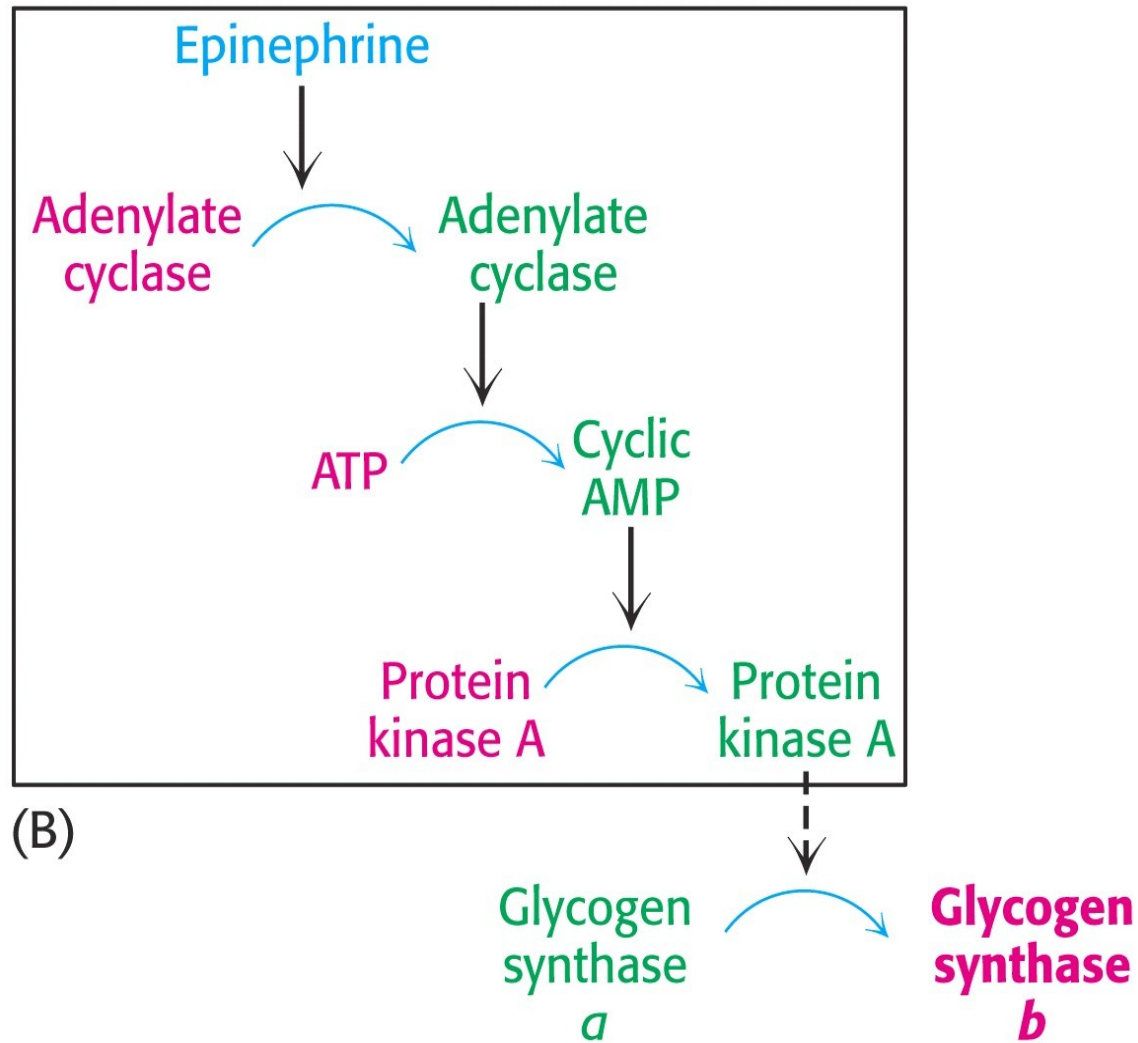
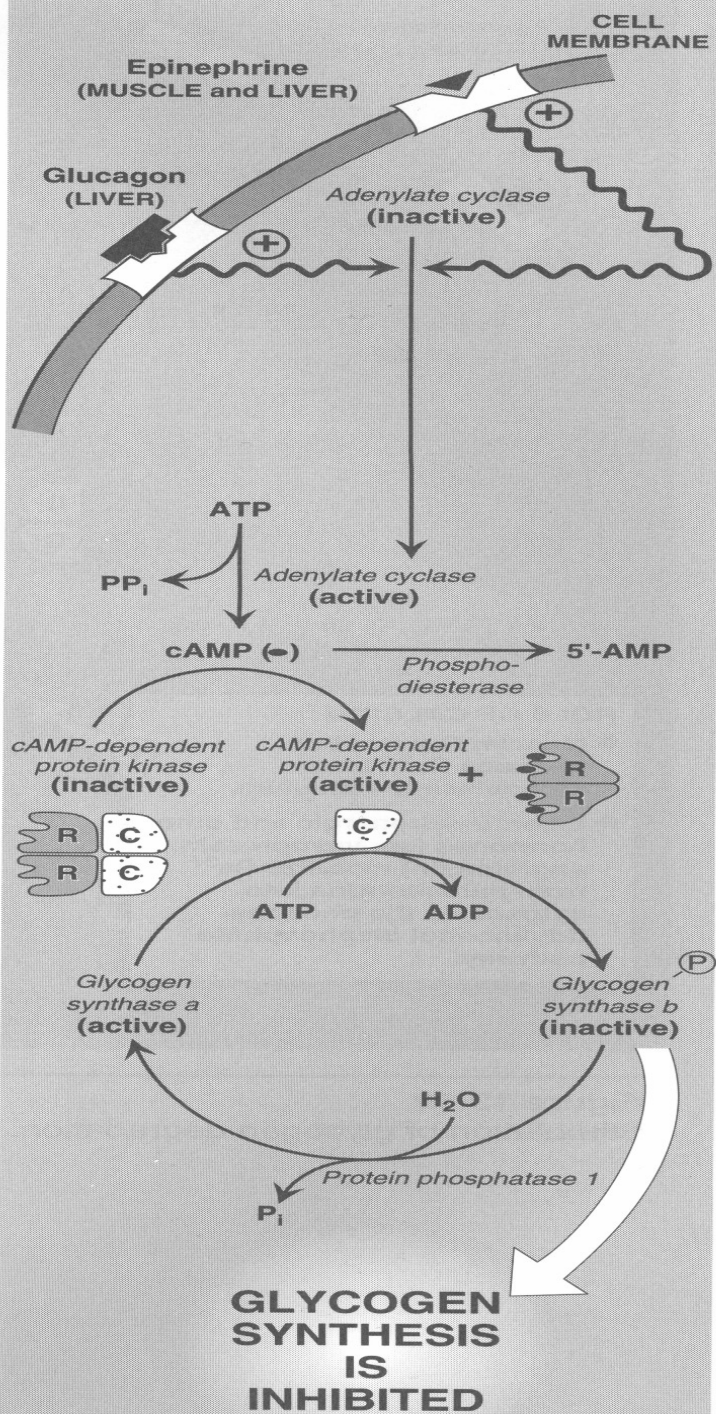
The active a form of glycogen synthase is converted to the inactive b form by phosphorylation via a variety of kinases.

The synthase is reactivated by dephosphorylation by *phospho protein phosphatase*.

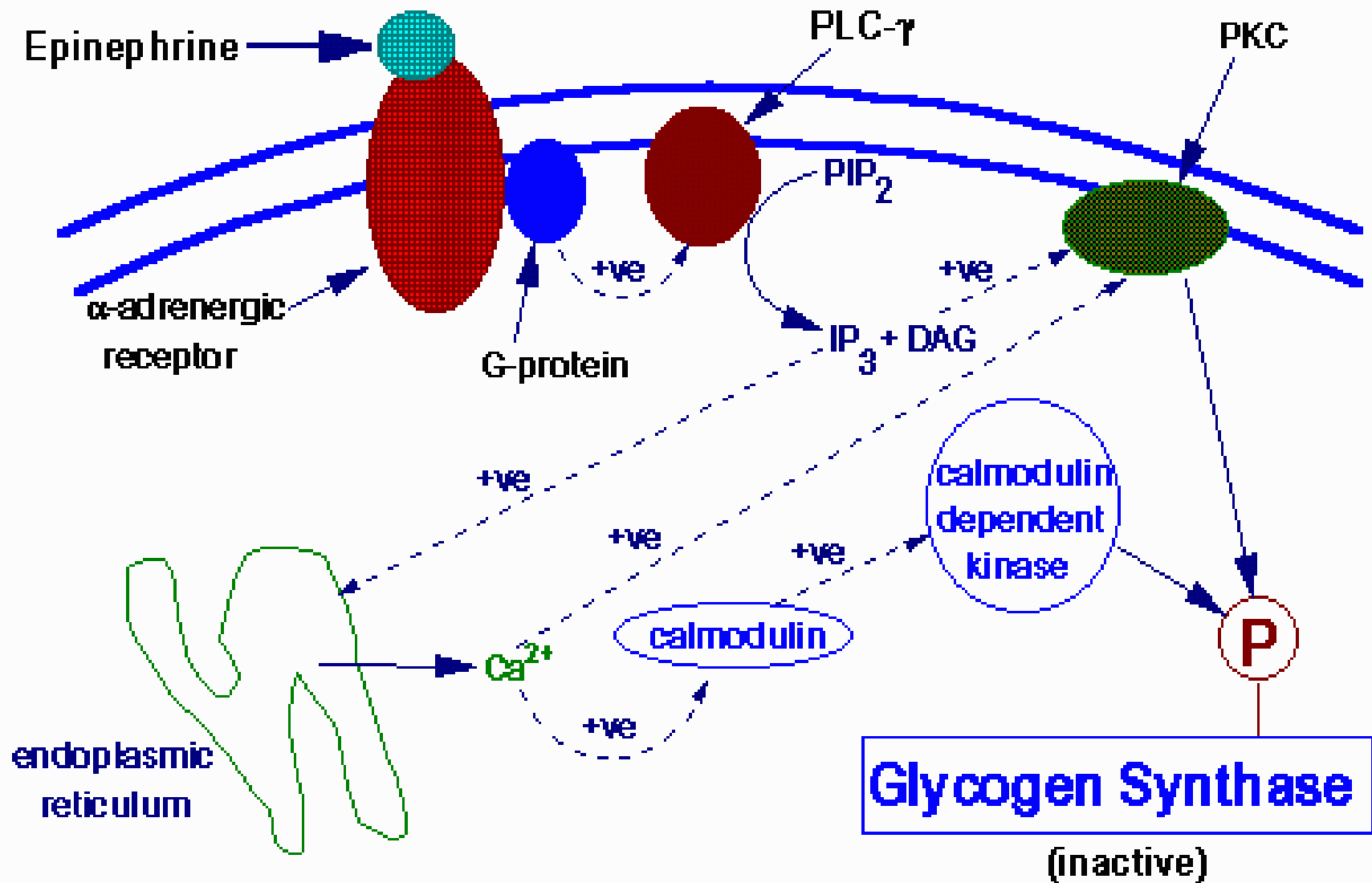
Hence the *phosphatase* accelerates glycogen synthesis.

Finally, the inactive b form can be allosterically activated by glucose 6-phosphate.





α -Receptor-Mediated Responses on Glycogen Synthase



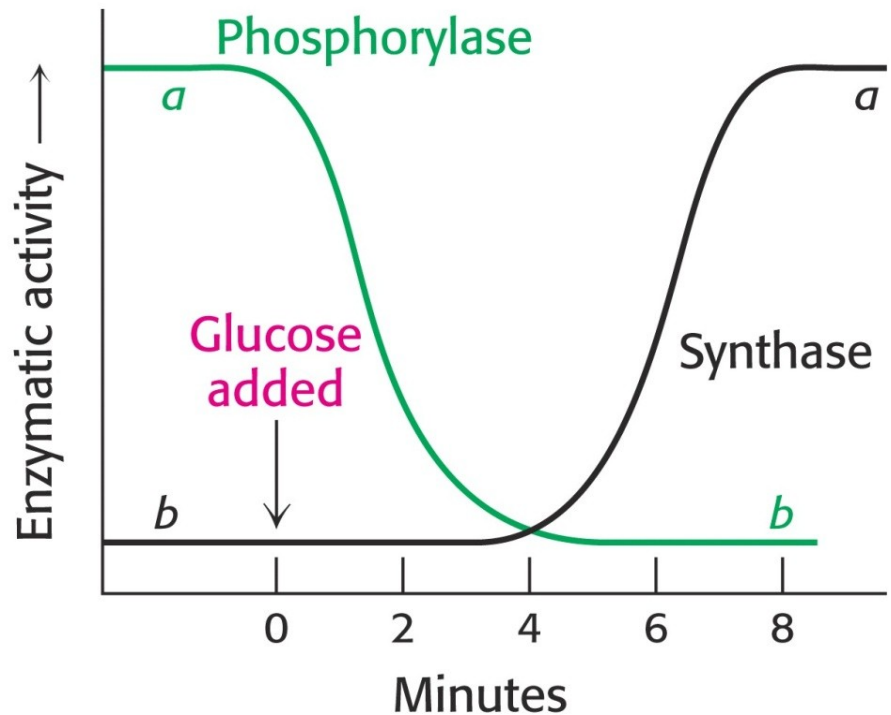
Co-ordinated Regulation of Glycogen Synthesis/ Breakdown

Low blood glucose:

↑ glucagon, ↓ insulin, ↑ PKA,
↑ phosphorylase kinase
↑ glycogen phosphorylase
↓ glycogen synthase
↑ glycogen breakdown, ↓ synthesis

High blood glucose:

↑ insulin, ↓ glucagon,
↑ phosphoprotein phosphatase
↓ phosphorylase kinase
↓ glycogen phosphorylase
↑ glycogen synthase
↓ glycogen breakdown, ↑ glycogen synthesis



Epinephrine has an effect similar to that of glucagon.

Glycogen Storage Diseases

Result from a **defect** in an enzyme required for either synthesis or degradation.

In general:

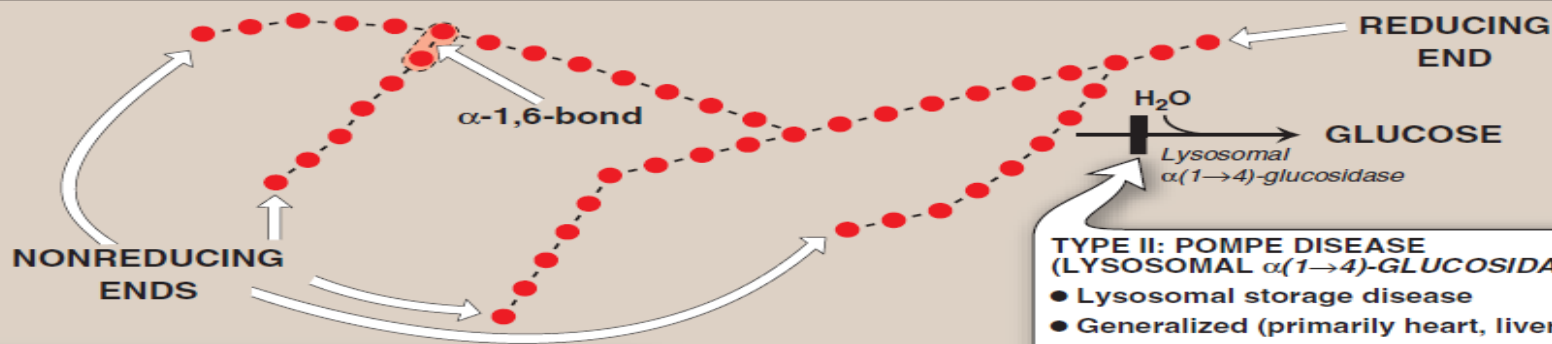
Affected organs either contain **more glycogen or have abnormally structured glycogen;**

Liver is usually the tissue most affected; heart and muscle can also be affected;

Abnormally structured glycogen due to defects in **either debranching enzyme or branching enzyme;**

Depending on the disease type, prognosis varies from poor to easily manageable;

Most of the defective enzymes are involved in glycogenolysis.



**TYPE V: McARDLE SYNDROME
(SKELETAL MUSCLE GLYCOGEN PHOSPHORYLASE
OR MYOPHOSPHORYLASE DEFICIENCY)**

- Skeletal muscle affected; liver enzyme normal
- Temporary weakness and cramping of skeletal muscle after exercise
- No rise in blood lactate during strenuous exercise
- Normal mental development
- Myoglobinemia and myoglobinuria
- Relatively benign, chronic condition
- High level of glycogen with normal structure in muscle
- Deficiency of the liver isozyme causes Type VI: Hers disease with mild fasting hypoglycemia.

**TYPE II: POMPE DISEASE
(LYSOSOMAL $\alpha(1 \rightarrow 4)$ -GLUCOSIDASE DEFICIENCY)**

- Lysosomal storage disease
- Generalized (primarily heart, liver, muscle)
- Excessive glycogen concentrations found in abnormal vacuoles in the lysosomes
- Normal blood sugar levels
- Massive cardiomegaly
- Enzyme replacement therapy available
- Infantile form: early death typically from heart failure
- Normal glycogen structure

P_i

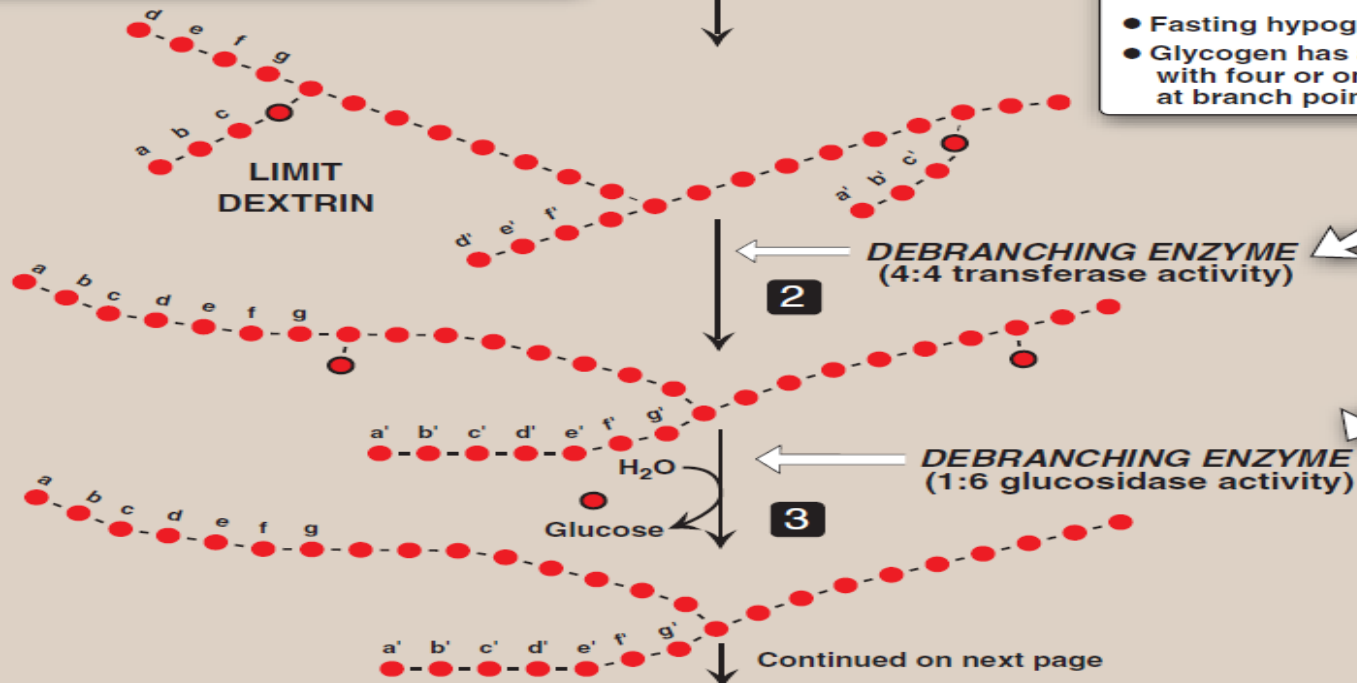
Glycogen phosphorylase

1

Glucose 1-P

**TYPE III: CORI DISEASE
(4:4 TRANSFERASE and/or 1:6
GLUCOSIDASE DEFICIENCY)**

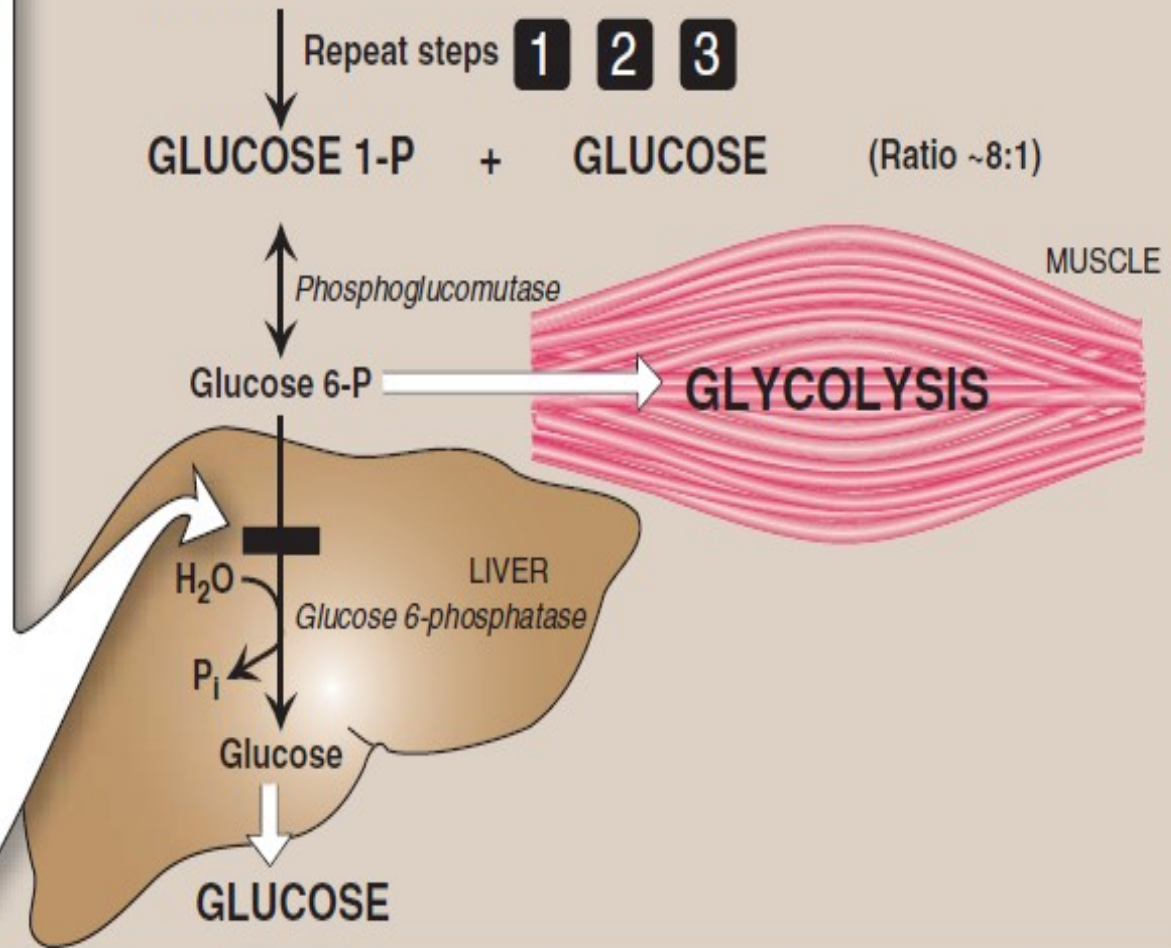
- Fasting hypoglycemia
- Glycogen has abnormal structure with four or one glucosyl residues at branch points



TYPE Ia: VON GIERKE DISEASE
(*GLUCOSE 6-PHOSPHATASE DEFICIENCY*)

TYPE Ib: *GLUCOSE 6-PHOSPHATE*
TRANSLOCASE DEFICIENCY

- Affects liver and kidney
- Fasting hypoglycemia—severe
- Fatty liver, hepato- and renomegaly
- Progressive renal disease
- Growth retardation and delayed puberty
- Hyperlacticacidemia, hyperlipidemia, and hyperuricemia
- Normal glycogen structure; increased glycogen stored
- Type Ib is characterized by neutropenia and recurrent infections
- Treatment: Nocturnal gastric infusions of glucose or regular administration of uncooked cornstarch



A summary of the glycogen storage diseases.

Types of Glycogenoses

Type	Glycogenosis	Deficient enzyme	Biochemical diagnosis	Clinical symptoms
1	Hepatorenal g., Gierke disease	glucose-6-phosphatase	Normal glycogen; excessive amounts in liver and kidneys	Hypoglycemia, hyperlipemia, ketosis, hyperuricemia, hepatomegaly, dwarfism
2	Generalized, malignant g.; Pompe disease; cardiomegalia glycogenica	α -1,4-glucosidase	Normal glycogen, excessive in all organs	Muscle hypotonia, heart failure, neurologic symptoms, infant death
3	Hepatomuscular, benign g.; Cori disease, Forbes disease (with subvariants 3b through f)	Amylo-1,6-glucosidase	Abnormal glycogen, with short outer chains, in liver and (more rarely) in muscles	Hepatomegaly, hypoglycemia; mild course of disease
4	Liver, cirrhotic, reticuloendothelial g.; Anderson disease; amylopectinosis	α -1,4-glucan: α -1,4-glucan-6-glycosyltransferase	Abnormal glycogen, with long outer chains, in liver, spleen, and lymph nodes	Cirrhosis of the liver; hepatosplenomegaly
5	Muscular g., Mcardle-Schmid-Pearson disease	α -glucanphosphorylase of the muscle	Normal glycogen, excessive amounts in muscle	Generalized myasthenia and myalgia, myoglobinuria
6	Hepatic g., Hers disease	α -glucanphosphorylase I of the liver	Normal glycogen, excessive amounts in liver	Hepatomegaly, relatively benign
7	Muscular g.; Tarui disease	Phosphofructokinase of the muscle	Normal glycogen, in the skeletal muscle	Muscle cramping, myoglobinuria
8	Hepatic g.; X-chromosome inheritance	Phosphorylase-b kinase of the liver	Normal glycogen, in the liver	Clinically mild manifestation, hepatomegaly, hypoglycemia

Know types I – V, defective enzyme and effect on glycogen.

Glycogen Storage Diseases

- Type I **Von-Gierke's**
- Glucose-6-Phosphatase Deficiency
 - Affects liver, kidney, and intestine
 - **Fatty liver, hepatomegaly**
 - Severe Fasting HYPOglycemia
 - Normal Glycogen structure
 - NOTE: In Von Geirke's the only source of glucose is food

Glycogen Storage Diseases

- **Type II-Pompe's Disease**
- (Lysosomal Alpha-Glucosidase Deficiency)
- Alpha 1-4 glucosidase (ACID MALTASE)
 - Inborn lysosomal enzyme defect
 - Generalized (LIVER, **HEART, MUSCLE**)
 - **EXCESSIVE GLYCOGEN CONCENTRATION-found in cytosomal vacuoles**
 - Normal blood sugar levels
 - **SEVERE CARDIOMEGALY**
 - Early death occurs
 - Normal Glycogen Structure

Glycogen Storage Diseases

- **Type V- McArdle's Syndrome**
 - **(Glycogen Phosphorylase Deficiency)**
 - Temporary weakness and cramping after exercise
 - No rise in blood lactate during exercise
 - Myoglobinuria in later years
 - Fair to good prognosis
 - **High level of glycogen in Muscle**
 - No ATP synthesis → no co A → no TCA cycle

Thank You!!